# EUCLIDEAN INVARIANCE AND CLUSTERING FOR THE CURVATURE FIELD OF THE THREE-DIMENSIONAL LATTICE ABELIAN HIGGS MODEL

Abstract. We consider the lattice abelian Higgs model in three dimensions, with noncompact gauge field and a photon mass, in boxes with King's gauge-fixed box action: free boundary conditions for the gauge field with an exact gauge-fixing term, and Dirichlet boundary conditions for the Higgs field. On gauge-invariant observables these box states are reflection positive at the site planes of all nine reflection families of the cubic lattice, in boxes invariant under the reflection; the curvature satisfies a Gaussian upper bound uniform in the box and in all parameters; and the expectations of $e^{iF(g)}$ and of $e^{F(g)}$, where $F(g)$ is the curvature smeared with a test cochain $g$, are monotone in the box. Hence the box states have an infinite-volume limit at fixed spacing, which coincides with King's limit with Neumann and Dirichlet boundary conditions. Every continuum limit point of the curvature field is the law of a closed tempered two-form which is Gaussian dominated, invariant under translations and the cubic group, and reflection positive at every translate of the nine families of planes. Combined with a rotation theorem from a recent preprint, this gives invariance under the full Euclidean group and reflection positivity at every affine plane. Almost every translation-ergodic component is Euclidean invariant, is reflection positive at every plane, has a unique vacuum for every choice of time direction, and has the cluster property. We show that the torus counterterm of a companion paper converges, at every fixed spacing, as the torus grows. For photon masses outside a countable set, the volume-first limits of the torus states of that paper have the same curvature two-point function as the corresponding continuum limit points of the box states (at the same photon mass and at the limiting torus counterterm), which is therefore Euclidean invariant. We also prove a rotation-invariance criterion for cubically symmetric scalar two-point functions with positive spectral representations in two frames, and show that its singularity threshold is optimal.

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## 1. Introduction

1.1. **Background.** The abelian Higgs model couples a $U(1)$ gauge field $A$ to a complex scalar field $\varphi$. In three Euclidean dimensions its lattice version with noncompact gauge field, a photon mass $\mu_0 > 0$ and a mass counterterm was constructed in the continuum limit in finite volume by King [K1], using ultraviolet stability bounds of Bałaban [Ba1, Ba2, Ba3, Ba4], and the infinite-volume limit was then taken in [K2], using the correlation inequalities of Brydges, Fröhlich and Seiler [BFS1]. The cubic lattice $\varepsilon\mathbb{Z}^3$ is invariant only under the group $\varepsilon\mathbb{Z}^3 \rtimes O_h$, where $O_h$ is the group of the 48 signed permutation matrices, and invariance of continuum limits under the full rotation group is not automatic.

A recent preprint [S] isolates a mechanism that produces continuous rotations for the field strength tensor of a three-dimensional gauge theory. If a law of a closed tempered two-form $F$ satisfies the Gaussian bound

$$\mu\big(e^{F(f)}\big) \le e^{\frac{1}{2}\|f\|_2^2} \qquad (f \text{ a real test two-form}), \tag{1.1}$$

is invariant under translations and under $O_h$, and is reflection positive at every plane of the nine families

$$x_j = c \quad (1 \le j \le 3), \qquad x_i - x_j = c, \qquad x_i + x_j = c \quad (1 \le i < j \le 3), \qquad c \in \mathbb{R}, \tag{1.2}$$

then it is invariant under all Euclidean motions [S, Thm. 18.21]. The diagonal reflections enter through a spectral bound $H_j \geq |P_i|$ on the reconstructed Hilbert space [S, Prop. 18.1]; see Section 6. The input required from a lattice construction is therefore reflection positivity of the limiting law at all nine families. On finite periodic lattices with a cubic period lattice this property is not available directly: a diagonal reflection of a cubic torus with an even number of sites per side maps a second layer of sites to itself without fixing it pointwise, and positivity fails for the algebra of the closed half-torus which contains that layer (Remark 3.6).

In this paper the volume is taken to infinity through boxes, with the gauge-fixed box action of King [K2, (3.38), p. 335], which we call the auxiliary action: plaquette term, photon mass and the exact gauge-fixing term $\frac{1}{2}\|d^*_\Omega A\|^2$ of the box $\Omega$ for the gauge field, Dirichlet boundary conditions for the Higgs field, and a box-independent mass counterterm $\delta m^2$, for instance that of [K1] (Section 2). For every reflection of the nine families there are boxes mapped onto themselves by it, and on gauge-invariant observables the auxiliary box state equals the Proca state of the box. A direct factorisation (Theorem 3.2) shows that these states are reflection positive at the site planes of all nine families. No comparison of tori of different shapes and no infrared estimate is needed. Throughout, the limits are taken in the *volume-first order*: the infinite-volume limit is taken at fixed lattice spacing, before the continuum limit.

1.2. **The main result.** The states and notation used in the following theorem are defined in Sections 2–5 and in Appendix B; we give a brief guide after the statement. Let $\mathcal{N} = \{e_1, e_2, e_3, e_1 \pm e_2, e_1 \pm e_3, e_2 \pm e_3\}$ be the nine normals of (1.2), and for $n \in \mathcal{N}$ and $c \in \mathbb{R}$ let $\pi_{n,c} = \{x \in \mathbb{R}^3 : n \cdot x = c\}$. Let $E(3) = \mathbb{R}^3 \rtimes O(3)$ be the Euclidean group.

**Theorem 1.1.** *Let $L \geq 2$ be an integer and $\varepsilon_K = L^{-K}$ for $K \in \mathbb{N}$. For each $K$ fix real parameters $\mu_{0,K} > 0$, $e_K$, $\lambda_K > 0$, $m_K^2$ and $\delta m_K^2$, and let $\langle\cdot\rangle_{\Omega,K}$ be the auxiliary box states of Definition 2.4(a) at spacing $\varepsilon_K$ with these parameters.*

(a) *(Infinite volume at fixed spacing.) For each $K$ the states $\langle\cdot\rangle_{\Omega,K}$ converge, as $\Omega$ runs through any exhausting sequence of boxes, on all exponentials $\exp\sum_a z_a F(g_a)$ of the lattice curvature, to a law $\omega_K$ of the curvature which does not depend on the sequence. On the classes $\{e^{iF(g)}\}$ and $\{e^{F(g)}\}$ it coincides with the infinite-volume limit of King's states with Neumann boundary conditions for $A$ and Dirichlet boundary conditions for $\varphi$ at the same spacing. The law $\omega_K$ is invariant under $\varepsilon_K\mathbb{Z}^3 \rtimes O_h$, satisfies $\omega_K(e^{F(g)}) \leq e^{\frac{1}{2}\|g\|^2_{2,\varepsilon_K}}$, and is reflection positive at the planes $\pi_{n,c}$, $n \in \mathcal{N}$, $c \in \varepsilon_K\mathbb{Z}$ (Theorem 4.3).*

(b) *(Continuum limit points.) Let $\mu_K$ be the law of the continuum curvature field $F_{\varepsilon_K}$ of $\omega_K$ (Section 5.2). For every $r > 7$ the sequence $(\mu_K)$ is tight on the Hermite–Sobolev space $H^{-r}(\mathbb{R}^3;\Lambda^2)$. Every weak limit point $\mu$ is the law of a closed tempered two-form, satisfies* (1.1) *and the moment bound*

$$|\mu\big(F(f_1)\cdots F(f_m)\big)| \leq 2\cdot 2^{m/2}(m!)^{1/2}\prod_{i=1}^m \|f_i\|_2, \tag{1.3}$$

*is invariant under $\mathbb{R}^3 \rtimes O_h$, and is reflection positive at every plane $\pi_{n,c}$, $n \in \mathcal{N}$, $c \in \mathbb{R}$ (Theorem 5.4).*

(c) *(Euclidean invariance.) Every such $\mu$ is invariant under $E(3)$, its moments obey the tensor transformation law under $E(3)$, and $\mu$ is reflection positive at every affine plane (Corollary 6.3).*

(d) *(Ergodic components.) Let $\mu = \int \mu_\xi\, \mu(d\xi)$ be the translation-ergodic decomposition of $\mu$ (Theorem A.4). For $\mu$-almost every $\xi$ the law $\mu_\xi$ is translation ergodic, is carried by closed two-forms, is invariant under $E(3)$ and reflection positive at every affine plane; for*

*every unit vector $n$, the Osterwalder–Schrader reconstruction of $\mu_\xi$ at the plane $\{n\cdot x = 0\}$ (Proposition B.3) has* $\operatorname{Ker} H_n = \mathbb{C}\psi_0^\xi$*, where $\psi_0^\xi$ is its vacuum vector; and*

$$\lim_{|a|\to\infty} \mu_\xi\big(\Phi\,\alpha_a\Psi\big) = \mu_\xi(\Phi)\,\mu_\xi(\Psi) \qquad \text{for all } \Phi,\Psi \in L^2(\mu_\xi), \tag{1.4}$$

*where $\alpha_a$ is the translation of observables by $a \in \mathbb{R}^3$ (Theorems 7.1 and 7.4).*

Parts (a) and (b) are proved in this paper, with the correlation inequalities of [BFS1] as the only input from the literature. Part (c) and the assertions of (d) that concern $O(3)$ and arbitrary planes use the rotation theorem [S, Thm. 18.21], which we quote as Theorem 6.2. The remaining assertions of (d), namely translation ergodicity, invariance under $\mathbb{R}^3 \rtimes O_h$, positivity at the nine families, $\operatorname{Ker} H_j = \mathbb{C}\psi_0^\xi$ for $j = 1, 2, 3$, and the cluster property (1.4), use no result of [S] other than the spectral bound [S, Prop. 18.1], quoted as Theorem 6.1. Part (d) is the uniqueness of the vacuum and the cluster property [OS73, (E4)] (axiom OS4 in [GJ, Sect. 6.1]) for almost every ergodic component, at the level of measures.

*Guide to the notation.* The lattice curvature is $F = dA$, and $F(g) = \varepsilon^3 \sum_p g(p)F(p)$ for a real two-cochain $g$ (Section 2.1). The norms $\|\cdot\|_{2,\varepsilon}$ and $\|\cdot\|_2$ are the $\ell^2$-norm with weight $\varepsilon^3$ and the $L^2$-norm of two-forms, with each unordered pair of indices counted once. The continuum field is $F_\varepsilon(f) = F(\iota_\varepsilon f)$, where $\iota_\varepsilon f$ samples a test two-form at plaquette centres (Section 5.2). Reflection positivity at a plane is Definition 5.1, and the reconstruction $(\mathcal{H}_n, H_n, \psi_0)$, with vacuum vector $\psi_0$, is Proposition B.3.

1.3. **Comparison with the torus states.** The companion paper [v1] constructs, for $m > 0$, $0 < \lambda < 0.1m$ and $0 < e^2 < 0.1m$, continuum states with cubic symmetry of a closely related lattice model on periodic cubic tori [v1, Thm. 3.1]. There the gauge field has the Feynman form, the counterterm is computed on each torus at a fixed reference photon mass $\mu_* = m$, so that it does not depend on the photon mass $\mu_0$, and the spacing and the torus go to their limits jointly. Theorem 1.1 does not apply to those states directly (Remark 3.6). In Section 9 we compare the two constructions at the level of two-point functions. First, at each fixed spacing $\varepsilon$ the torus counterterm $b_{T_j}(\varepsilon)$ of [v1] converges, as the torus grows, to the coefficient $a_\natural(\varepsilon)$ given by the same graphs on $\varepsilon\mathbb{Z}^3$ (Proposition 9.2): by [v1, Lemma G.1] each coefficient is a finite sum of products of torus Riemann sums of continuous functions on the Brillouin zone and of position sums of products of three positive kernels, and these converge. On the characteristic functions $e^{iA(f)}$ of the gauge field, the correlation inequalities of [BFS1] bound the box states at the coefficient $a_\natural(\varepsilon)$ by the limits of the torus states, so the two-point function of the torus limits is at most that of the box limit. Convexity of the pressure in $\mu_0^2$ and a comparison of torus and box pressures give equality of the diagonal terms at every $\mu_0$ at which the box second moment is continuous in $\mu_0^2$, and positivity of the difference then gives equality of the whole two-point function. Part (a) of the following statement is Theorem 9.11(ii), and part (b) is Corollary 9.14; the invariance in (b) uses Theorem 1.1(c), hence [S, Thm. 18.21]. The identification concerns the two-point function and the volume-first order only; the full law and the joint limits of [v1] are open (Section 10).

**Theorem 1.2.** *Let $L$, the spacings $\varepsilon_K$ and the tori $T_j$ be as in* [v1, Sect. 2.1]*, let $m > 0$, $\lambda > 0$, $e \neq 0$ and $\mu_* = m$, and let $\langle\cdot\rangle_{\varepsilon,j,\mu_0}$ be the torus states of* [v1, Def. 2.7]*. There is a countable set $\mathsf{E} \subset (0,\infty)$, defined by the box states alone, such that for every $\mu_0 \in (0,\mu_*] \setminus \mathsf{E}$:*

(a) *for every $K \geq 0$ and all plaquettes $p, q$ of $\varepsilon_K\mathbb{Z}^3$,*

$$\lim_{j\to\infty} \langle F(p)F(q)\rangle_{\varepsilon_K,j,\mu_0} = \omega_{\varepsilon_K,\mu_0}\big(F(p)F(q)\big),$$

*where $\omega_{\varepsilon_K,\mu_0}$ is the infinite-volume law of the auxiliary box states (Theorem 1.1(a)) at the spacing $\varepsilon_K$ with the parameters $\mu_0$, $e$, $\lambda$, $m^2$ and $\delta m^2 = a_\natural(\varepsilon_K) - m^2$;*

(b) *for every volume-first limit point $\nu$ of the torus states at photon mass $\mu_0$ (Definition 9.13) there is a law $\mu$ as in Theorem 1.1(b), a continuum limit point of the laws $\omega_{\varepsilon_K,\mu_0}$ of (a), such that $\nu(F(f)F(h)) = \mu(F(f)F(h))$ for all real $f, h \in L^2(\mathbb{R}^3;\Lambda^2)$; in particular $\nu(F(\gamma\cdot f)F(\gamma\cdot h)) = \nu(F(f)F(h))$ for every $\gamma \in E(3)$.*

1.4. **Comments.** *Positivity at the diagonal planes.* The factorisation of Theorem 3.2 treats all nine families at once. Bonds never cross a site plane of the nine families. For a diagonal plane $x_i \pm x_j = c$ some plaquettes do: two opposite corners of such a plaquette lie on the plane, and its curvature is $\varepsilon^{-1}(v_p - v_p \circ \hat\rho)$ for a sum $v_p$ of two bond variables on the positive side (Lemma 3.1). Hence the plaquette weight $e^{-\frac{\varepsilon}{2}(v_p - v_p\circ\hat\rho)^2}$ is a positive mixture of products $h \cdot \overline{h\circ\hat\rho}$, and positivity follows by Fubini. For coordinate planes the mechanism is the classical one [OSe, Sect. 2.3, Thm. 2.1]; King proves positivity of the auxiliary states at coordinate planes between lattice planes [K2, Prop. 3.3], using [BFS1, Cor. 5.4].

*Infinite volume.* On gauge-invariant observables the auxiliary state of a box is its Proca state (Lemma 2.6). Enlarging the box adds plaquette and hopping terms, so the correlation inequalities of [BFS1] make the states monotone in the box on the classes $\{e^{iF(g)}\}$ and $\{e^{F(g)}\}$ (Lemma 2.9). The infinite-volume limit exists along every exhausting sequence and is unique. The identification with King's Neumann/Dirichlet limit at the same spacing uses two comparison lemmas: an exact identity between the two quadratic forms on a box (Lemma 2.10), and an upper comparison with a margin proportional to $1/\mu_0$ (Lemma 2.12).

*Parameters and non-triviality.* Every bound used for the continuum limit is uniform in the box, in the spacing and in all parameters (Proposition 4.2). For this reason the parameters in Theorem 1.1 are arbitrary sequences, and no ultraviolet estimate enters. For the same reason Theorem 1.1 says nothing about non-triviality: the point mass at $F = 0$ has all the properties listed in (b)–(d).

*Ergodic components.* Invariance of $\mu$ under a connected group does not imply invariance of its ergodic components (Remark A.8). Invariance of the components comes from reflections: a reflection at a plane where $\mu$ is reflection positive acts trivially on the invariant $\sigma$-algebra (Lemma A.5). The statements in (d) are statements about measures. Gaussian domination and moment bounds are not asserted for the components, and a Wightman reconstruction for each component is not claimed.

*The Wick square.* Section 8 contains a criterion for rotation invariance of a scalar two-point function with cubic symmetry, a singularity bound $|G(x)| \le C|x|^{-a}$ with $a < 9$, and positive spectral representations in an axis frame and a diagonal frame. Example 8.4 shows that $a < 9$ cannot be replaced by $a \le 9$. The criterion applies to the two-point function of the Wick square $|\varphi|^2$ once that field is constructed in the volume-first order with these properties; see Section 10.

1.5. **Organisation.** Section 2 defines the auxiliary, Proca and Neumann/Dirichlet box states, proves the correlation and comparison inequalities, and constructs the infinite-volume limits on the classes $\{e^{iF(g)}\}$, $\{e^{F(g)}\}$. Section 3 proves reflection positivity at the nine families. Section 4 proves Gaussian domination and completes the infinite-volume construction at fixed spacing. Section 5 constructs the continuum limit points and proves their properties. Section 6 states the two results quoted from [S] and applies them. Section 7 treats the ergodic components and proves Theorem 1.1. Section 8 treats the two-point function of a scalar field. Section 9 proves that the torus counterterm of [v1] converges at fixed spacing and compares the two-point functions with those of the torus states of [v1], and Section 10 lists open problems. Appendix A proves the ergodic decomposition and the inheritance of invariance and positivity. Appendix B collects the facts about Gaussian-dominated and reflection-positive laws that we use.

## 2. The lattice model and the box states

Throughout Sections 2–4 the lattice spacing $\varepsilon > 0$ and the parameters $\mu_0 > 0$, $e \in \mathbb{R}$, $\lambda > 0$, $m^2 \in \mathbb{R}$, $\delta m^2 \in \mathbb{R}$ are fixed. We write $\kappa := \mu_0 \varepsilon$ for the photon mass in lattice units.

2.1. **Lattice, cochains and boxes.** Let $e_1, e_2, e_3$ be the standard basis of $\mathbb{R}^3$. The sites are the points of $\varepsilon\mathbb{Z}^3$. A bond is an ordered pair $b = \langle x, y\rangle$ of nearest-neighbour sites, with $b_- = x$, $b_+ = y$ and $-b = \langle y, x\rangle$; it is positively oriented if $y = x + \varepsilon e_\mu$ for some $\mu$, and we then also write $b = (x;\mu)$. For a site $x$ and $\mu < \nu$ the positively oriented plaquette $p = (x;\mu,\nu)$ has corners $x$, $x+\varepsilon e_\mu$, $x+\varepsilon e_\mu+\varepsilon e_\nu$, $x+\varepsilon e_\nu$, boundary

$$\partial p = \langle x, x+\varepsilon e_\mu\rangle + \langle x+\varepsilon e_\mu, x+\varepsilon e_\mu+\varepsilon e_\nu\rangle + \langle x+\varepsilon e_\mu+\varepsilon e_\nu, x+\varepsilon e_\nu\rangle + \langle x+\varepsilon e_\nu, x\rangle,$$

a formal sum of oriented bonds, centre $c_p = x + \frac{\varepsilon}{2}(e_\mu + e_\nu)$ and edge vectors $(u_p, v_p) = (e_\mu, e_\nu)$; the reversed plaquette $-p$ has edge vectors $(e_\nu, e_\mu)$. Oriented cubes and their boundaries are defined in the same way. A $k$-cochain ($k = 0,1,2,3$) is a real function on oriented $k$-cells which changes sign under reversal of orientation; we identify it with its restriction to positively oriented cells. The coboundary $d$ is

$$(d\chi)(b) = \varepsilon^{-1}\big(\chi(b_+) - \chi(b_-)\big), \qquad (dA)(p) = \varepsilon^{-1}\sum_{b\in\partial p} A(b), \qquad (dF)(c) = \varepsilon^{-1}\sum_{p\in\partial c} F(p),$$

so that $dd = 0$. On $k$-cochains we use $\langle u, v\rangle_\varepsilon := \varepsilon^3 \sum_\sigma u(\sigma)\overline{v(\sigma)}$, the sum running over positively oriented $k$-cells, and $\|u\|^2_{2,\varepsilon} = \langle u, u\rangle_\varepsilon$. These are the conventions of [K1, K2]; in particular $F(p) = (dA)(p)$ is King's field strength [K2, (2.11)].

A *box* is a set $\Omega = \prod_{\nu=1}^3 \big([a_\nu, b_\nu] \cap \varepsilon\mathbb{Z}\big)$ with $a_\nu \le b_\nu$ in $\varepsilon\mathbb{Z}$. We give it the induced complex: $\Omega^*$ is the set of positively oriented bonds with both endpoints in $\Omega$ [K1, p. 651], and $\Omega^{**}$ is the set of positively oriented plaquettes with all four corners in $\Omega$. The cochain spaces are $C^0(\Omega) = \mathbb{R}^\Omega$, $C^1(\Omega) = \mathbb{R}^{\Omega^*}$, $C^2(\Omega) = \mathbb{R}^{\Omega^{**}}$, $d_\Omega$ is the restriction of $d$, and $d^*_\Omega$ is its adjoint with respect to $\langle\cdot,\cdot\rangle_\varepsilon$. For a box $\Omega$ and $r \in \mathbb{N}$ we write $\Omega^{(r)}$ for the box of all sites at sup-distance at most $r\varepsilon$ from $\Omega$.

**Lemma 2.1.** *Let $\Omega$ be a box. (a) The kernel of $d_\Omega$ on $C^0(\Omega)$ consists of the constants. (b) If $A \in C^1(\Omega)$ and $d_\Omega A = 0$, then $A = d_\Omega\chi$ for some $\chi \in C^0(\Omega)$. (c) $C^1(\Omega) = \operatorname{Ran} d_\Omega \oplus \operatorname{Ker} d^*_\Omega$, orthogonally.*

*Proof.* (a) holds because $\Omega$ is connected by bonds of $\Omega^*$. For (b), fix the corner $o = (a_1, a_2, a_3)$ and define $\chi(x)$ as $\varepsilon$ times the sum of $A$ along the lattice path from $o$ to $x$ which first changes the first coordinate, then the second, then the third. All these paths lie in $\Omega$, because $\Omega$ is a product of intervals. For a bond $\langle x, x+\varepsilon e_\mu\rangle$ the paths to $x$ and to $x+\varepsilon e_\mu$, together with the bond, bound a union of plaquettes of $\Omega^{**}$, so $dA = 0$ gives $\chi(x+\varepsilon e_\mu) - \chi(x) = \varepsilon A(\langle x, x+\varepsilon e_\mu\rangle)$. Part (c) is linear algebra. □

We also use lattice units. For $A \in \mathbb{R}^{\Omega^*}$ let $|A|^2 = \sum_{b\in\Omega^*} A(b)^2$, let $d_0 = \varepsilon d_\Omega$ on $C^0(\Omega)$ and $d_1 = \varepsilon d_\Omega$ on $C^1(\Omega)$ be the integer incidence matrices, and let $d_0^{\mathsf T}$ be the transpose of $d_0$. Then

$$\|d_\Omega A\|^2_{2,\varepsilon} = \varepsilon|d_1 A|^2, \qquad \|d^*_\Omega A\|^2_{2,\varepsilon} = \varepsilon|d_0^{\mathsf T} A|^2, \qquad \mu_0^2\|A\|^2_{2,\varepsilon} = \varepsilon\kappa^2|A|^2. \tag{2.1}$$

For a real $g \in C^2(\Omega)$ the curvature observable is

$$F(g) := \langle g, d_\Omega A\rangle_\varepsilon = \varepsilon^3 \sum_{p\in\Omega^{**}} g(p)\,(dA)(p), \tag{2.2}$$

which is King's $F(g)$ [K2, (2.12)]. It is a linear function of $A$, and $F(g) = \langle d^*_\Omega g, A\rangle_\varepsilon$.

2.2. **The Higgs field and gauge invariance.** The Higgs field on a box $\Lambda$ is $\varphi \in \mathbb{C}^\Lambda$, with $\varphi_x = r_x e^{i\vartheta_x}$. All results of this paper hold for an arbitrary real counterterm $\delta m^2$ which does not depend on the box. The case of interest is King's mass counterterm, given by the graphical expansion of [K1, Sect. 3.3] computed with free-boundary propagators [K2, p. 330]; it depends on $\varepsilon$ and on the couplings but not on the box [K2, p. 334]. Put

$$v(r) := 3\varepsilon r^2 + \varepsilon^3\big(\tfrac{1}{2}(m^2+\delta m^2)r^2 + \lambda r^4\big) \qquad (r \ge 0). \tag{2.3}$$

Since $\lambda > 0$, $\int_0^\infty e^{\alpha r^2 - v(r)} r\,dr < \infty$ for every $\alpha \in \mathbb{R}$. For $c = (c_b)_{b\in\Lambda^*}$ with $c_b \ge 0$ the matter action is

$$S^{\mathrm{m}}_{\Lambda,c}(A,\varphi) := \sum_{x\in\Lambda} v(|\varphi_x|) - \sum_{b\in\Lambda^*} c_b\ \mathrm{Re}\,\big(\overline{\varphi(b_-)}\, e^{ie\varepsilon A(b)}\varphi(b_+)\big), \tag{2.4}$$

and $S^{\mathrm{m}}_\Lambda := S^{\mathrm{m}}_{\Lambda,c}$ with $c_b \equiv \varepsilon$. Note that $\mathrm{Re}(\overline{\varphi(b_-)}e^{ie\varepsilon A(b)}\varphi(b_+)) = r_{b_-}r_{b_+}\cos(\vartheta_{b_+} - \vartheta_{b_-} + e\varepsilon A(b))$.

*Remark* 2.2. With $D_A\psi(b) := \varepsilon^{-1}(e^{ie\varepsilon A(b)}\psi(b_+) - \psi(b_-))$ one has

$$S^{\mathrm{m}}_\Lambda(A,\varphi) = \frac{1}{2}\sum_b \varepsilon^3 |D_A(\mathbf{1}_\Lambda\varphi)(b)|^2 + \sum_{x\in\Lambda}\varepsilon^3\big(\tfrac{1}{2}(m^2+\delta m^2)|\varphi_x|^2 + \lambda|\varphi_x|^4\big),$$

where the first sum runs over all positively oriented bonds of $\varepsilon\mathbb{Z}^3$; on bonds not in $\Lambda^*$ the value of $A$ does not matter. This is the matter part of King's actions [K2, (2.5), (3.38)] with Dirichlet boundary conditions for $\varphi$. Each site of $\Lambda$ carries the diagonal terms $\frac{\varepsilon}{2}|\varphi_x|^2$ of all six bonds of $\varepsilon\mathbb{Z}^3$ at $x$, which gives the term $3\varepsilon r^2$ in (2.3). Consequently the single-site measure $e^{-v(|\varphi_x|)}d^2\varphi_x$ is the same at every site of every box.

A gauge transformation is $(A,\varphi) \mapsto (A + d_\Lambda\chi, e^{-ie\chi}\varphi)$ with $\chi \in \mathbb{R}^\Lambda$. The action $S^{\mathrm{m}}_{\Lambda,c}$ is invariant, and a function $G(A,\varphi)$ is called *gauge invariant* if $G(A + d_\Lambda\chi, e^{-ie\chi}\varphi) = G(A,\varphi)$ for all $\chi$. Every function of $d_\Lambda A$ and of $(|\varphi_x|)_x$ is gauge invariant.

2.3. **Box measures and the three box states.**

**Definition 2.3.** Let $\Lambda$ be a box, $\mathfrak{B} \supset \Lambda^*$ a finite set of positively oriented bonds, $\gamma$ a centred Gaussian probability measure on $\mathbb{R}^{\mathfrak{B}}$ with nondegenerate covariance, and $c \in [0,\infty)^{\Lambda^*}$. For Borel $G$ with $\int\gamma(dA)\int d\varphi\,|G|e^{-S^{\mathrm{m}}_{\Lambda,c}} < \infty$ put

$$\langle G\rangle_{\gamma,\Lambda,c} := \frac{\int\gamma(dA)\int_{\mathbb{C}^\Lambda} d\varphi\ G(A,\varphi)\, e^{-S^{\mathrm{m}}_{\Lambda,c}(A,\varphi)}}{\int\gamma(dA)\int_{\mathbb{C}^\Lambda} d\varphi\ e^{-S^{\mathrm{m}}_{\Lambda,c}(A,\varphi)}},$$

where $d\varphi$ is Lebesgue measure. We write $\langle\cdot\rangle_{\gamma,\Lambda}$ when $c_b \equiv \varepsilon$. For a positive definite quadratic form $Q$ on $\mathbb{R}^{\mathfrak{B}}$ let $\gamma_Q$ be the centred Gaussian measure with density proportional to $\exp(-\frac{\varepsilon}{2}Q(A))$.

The denominator is positive and finite: since $c_b rr' \le \frac{c_b}{2}(r^2+r'^2)$ and every site has at most six bonds, $e^{-S^{\mathrm{m}}_{\Lambda,c}} \le \prod_x e^{-v(r_x)+3c_{\max}r_x^2}$ with $c_{\max} = \max_b c_b$, which is integrable. In particular, if $c_b \le \varepsilon$ for all $b$, then $\int d\varphi\, e^{-S^{\mathrm{m}}_{\Lambda,c}(A,\varphi)} \le \big(\int_{\mathbb{C}} e^{-v(|z|)+3\varepsilon|z|^2}d^2z\big)^{|\Lambda|}$ uniformly in $A$.

**Definition 2.4.** Let $\Omega$ and $C$ be boxes.

(a) The *auxiliary box state* is $\langle\cdot\rangle_\Omega := \langle\cdot\rangle_{\gamma_{Q_\Omega},\Omega}$ with $\mathfrak{B} = \Omega^*$ and

$$Q_\Omega(A) := |d_1A|^2 + |d_0^{\mathsf{T}}A|^2 + \kappa^2|A|^2.$$

By (2.1) its density is proportional to $e^{-S_\Omega}$ with $S_\Omega = \frac{1}{2}\|d_\Omega A\|^2_{2,\varepsilon} + \frac{1}{2}\|d^*_\Omega A\|^2_{2,\varepsilon} + \frac{1}{2}\mu_0^2\|A\|^2_{2,\varepsilon} + S^{\mathrm{m}}_\Omega$.

(b) The *Proca box state* is $\langle\cdot\rangle_\Omega^{\mathrm{Pr}} := \langle\cdot\rangle_{\gamma_{Q_\Omega^{\mathrm{Pr}}},\Omega}$ with $Q_\Omega^{\mathrm{Pr}}(A) := |d_1 A|^2 + \kappa^2|A|^2$.

(c) The *Neumann/Dirichlet state* is $\langle\cdot\rangle_C^{\mathrm{ND}} := \langle\cdot\rangle_{\gamma_{Q_C^{\mathrm{ND}}},C}$ with $\mathfrak{B} = C^*$, $Q_C^{\mathrm{ND}}(A) := K_C(A) + \kappa^2|A|^2$ and

$$K_C(A) := \sum_{\mu,\nu=1}^{3} \sum_{x} \big(A(x+\varepsilon e_\nu;\mu) - A(x;\mu)\big)^2,$$

where for given $\mu,\nu$ the sum runs over the sites $x$ for which both bonds $(x;\mu)$ and $(x+\varepsilon e_\nu;\mu)$ belong to $C^*$.

The auxiliary state is defined by the gauge-fixed box action of [K2, (3.38), p. 335], which we call the auxiliary action, on the induced complex of $\Omega$. King's gauge-fixing term $\frac{1}{2}\sum_{x\in\Omega}\varepsilon^3|\sum_{y\in x^*}\varepsilon^{-1}\Omega(y)A(\langle x,y\rangle)|^2$, where $x^*$ is the set of nearest neighbours of $x$ and $\Omega(\cdot)$ is the indicator of $\Omega$, equals $\frac{1}{2}\|d_\Omega^* A\|_{2,\varepsilon}^2$. Thus the divergence is the exact adjoint of $d_\Omega$. The mass term runs over $\Omega^*$, the plaquette term over $\Omega^{**}$, and the matter part is that of Remark 2.2. The state of (c) is King's action [K2, (2.5), p. 324] with Neumann boundary conditions for $A$ and Dirichlet boundary conditions for $\varphi$ [K2, (2.9)–(2.10), p. 325], with the vector potential on the internal bonds $C^*$.

*Remark* 2.5. King identifies maps $\Omega^* \to \mathbb{R}$ with maps $\Omega \to \mathbb{R}^3$ through $A_\mu(x) = A(\langle x, x+\varepsilon e_\mu\rangle)$ [K2, (2.2)]. If the Neumann/Dirichlet state is defined with the variables $A_\mu(x)$, $x \in C$, $\mu = 1,2,3$, which include the bonds leaving $C$ through its faces $\{x_\mu = b_\mu\}$, and with $K_C$ summed over all $x, x+\varepsilon e_\nu \in C$, then all statements of this section remain true with the same proofs. The hypotheses $C^{(1)} \subset \Omega$ in Lemma 2.11 and $\Omega^{(N+1)} \subset B$ in Lemma 2.12 are chosen so as to cover both readings, and in Lemma 2.9(a) the variables of $C$ are among those of $C'$ whenever $C \subset C'$.

For a box $\Lambda$ we consider the classes

$$\mathcal{I}_F(\Lambda) := \{e^{iF(g)} : g \in C^2(\Lambda) \text{ real}\}, \qquad \mathcal{D}_F(\Lambda) := \{e^{F(g)} : g \in C^2(\Lambda) \text{ real}\}.$$

For every box measure the expectations of elements of $\mathcal{I}_F(\Lambda)$ are real. Indeed, $(A,\varphi) \mapsto (-A,\overline{\varphi})$ preserves $\gamma$ and $S^{\mathrm{m}}_{\Lambda,c}$ and maps $F(g)$ to $-F(g)$.

### 2.4. **The Proca form and the dependence on the curvature.**

**Lemma 2.6.** *Let $\Lambda$ be a box and $G$ a gauge-invariant Borel function with $|G(A,\varphi)| \le c\,e^{a|A|}$ for some constants $a, c$.*

(a) $\langle G\rangle_\Lambda = \langle G\rangle_\Lambda^{\mathrm{Pr}}$.

(b) *For every box measure $\langle\cdot\rangle_{\gamma,\Lambda}$ with $\mathfrak{B} = \Lambda^*$, the expectation $\langle G\rangle_{\gamma,\Lambda}$ depends on $\gamma$ only through the law of $d_1 A$ under $\gamma$.*

*Proof.* Put $\Phi_G(A) := \int d\varphi\, G(A,\varphi) e^{-S^{\mathrm{m}}_\Lambda(A,\varphi)}$. For $\chi \in \mathbb{R}^\Lambda$, the substitution $\varphi \mapsto e^{-ie\chi}\varphi$ preserves $d\varphi$, so gauge invariance of $G$ and of $S^{\mathrm{m}}_\Lambda$ gives $\Phi_G(A + d_\Lambda\chi) = \Phi_G(A)$. By the remark after Definition 2.3, $|\Phi_G(A)| \le c' e^{a|A|}$.

(a) By Lemma 2.1, every $A \in \mathbb{R}^{\Lambda^*}$ is uniquely $A = Z + d_0\chi$ with $d_0^{\mathsf{T}} Z = 0$ and $\chi \perp 1$, and $(Z,\chi) \mapsto A$ is a linear bijection with constant Jacobian. Since $d_1 d_0 = 0$ and $Z \perp \operatorname{Ran} d_0$,

$$Q_\Lambda(A) = |d_1 Z|^2 + \kappa^2|Z|^2 + |d_0^{\mathsf{T}} d_0\chi|^2 + \kappa^2|d_0\chi|^2, \qquad Q_\Lambda^{\mathrm{Pr}}(A) = |d_1 Z|^2 + \kappa^2|Z|^2 + \kappa^2|d_0\chi|^2.$$

Since $d_0\chi = \varepsilon\, d_\Lambda\chi$, we have $\Phi_G(Z + d_0\chi) = \Phi_G(Z)$. Hence in each of the two states the integral over $(Z,\chi)$ factorises into the integral of $\Phi_G(Z) e^{-\frac{\varepsilon}{2}(|d_1 Z|^2 + \kappa^2|Z|^2)}$ over $Z$, which is the same for both states, times a Gaussian integral over $\chi$. The latter depends on the state but is the same for $G$ and for $G \equiv 1$, so it cancels in the normalised expectation. Fubini's theorem applies because of the bound on $\Phi_G$ and $\kappa > 0$.

(b) If $d_1 A = d_1 A'$, then $A' - A = d_\Lambda \chi$ for some $\chi$, by Lemma 2.1(b), and so $\Phi_G(A') = \Phi_G(A)$. Hence $\Phi_G = h_G \circ d_1$ with $h_G := \Phi_G \circ T$ on $\operatorname{Ran} d_1$, where $T$ is a linear right inverse of $d_1$ on $\operatorname{Ran} d_1$. Then $\langle G \rangle_{\gamma,\Lambda} = \int h_G(d_1 A)\gamma(dA) \big/ \int h_1(d_1 A)\gamma(dA)$. □

2.5. **Correlation inequalities.** The following inequalities are due to Brydges, Fröhlich and Seiler [BFS1, Sect. 6].

**Lemma 2.7.** *Let $\Lambda$ be a box, $\mathfrak{B} = \Lambda^*$, and let $\ell$ be a real linear form on $\mathbb{R}^{\Lambda^*}$.*

(a) *For a fixed Gaussian $\gamma$, the function $c \mapsto \langle e^{i\ell(A)} \rangle_{\gamma,\Lambda,c}$ is nondecreasing, and $c \mapsto \langle e^{\ell(A)} \rangle_{\gamma,\Lambda,c}$ is nonincreasing, in each coordinate $c_b \in [0,\infty)$.*

(b) *For fixed $c$, if $\gamma_1, \gamma_2$ are centred Gaussians with nondegenerate covariances $C_1 \le C_2$, then $\langle e^{i\ell(A)} \rangle_{\gamma_2,\Lambda,c} \le \langle e^{i\ell(A)} \rangle_{\gamma_1,\Lambda,c}$ and $\langle e^{\ell(A)} \rangle_{\gamma_2,\Lambda,c} \ge \langle e^{\ell(A)} \rangle_{\gamma_1,\Lambda,c}$.*

*Proof.* If $e = 0$, the matter action does not depend on $A$. The box measure is then the product of $\gamma$ and a probability measure in $\varphi$, and $\langle e^{i\ell(A)} \rangle_{\gamma,\Lambda,c} = e^{-\frac{1}{2}\ell^{\mathsf{T}} C \ell}$, $\langle e^{\ell(A)} \rangle_{\gamma,\Lambda,c} = e^{\frac{1}{2}\ell^{\mathsf{T}} C \ell}$, where $C$ is the covariance of $\gamma$ and $\ell$ is identified with a vector of $\mathbb{R}^{\Lambda^*}$. These expressions do not depend on $c$, and they are nonincreasing, respectively nondecreasing, in $C$. This proves (a) and (b) for $e = 0$.

Let now $e \neq 0$. The measures of Definition 2.3 are of the type considered in [BFS1, Sect. 6.1, pp. 275–276]. The lattice is the finite set $\Lambda$, with nonnegative hopping coefficients $c_b$ on the pairs $b \in \Lambda^*$ and gauge variables $e\varepsilon A(b)$ (no ultraviolet cutoff). The vector potential has an arbitrary centred Gaussian law, and the Higgs field has Dirichlet boundary conditions. The radial single-site distribution $e^{-v(r)} r\, dr$ is a finite measure with $\int e^{\alpha r^2 - v(r)} r\, dr < \infty$ for all $\alpha$, as required in [BFS1, Sect. 4.1, p. 262]. In the notation of [BFS1, Sect. 6.3], the linear form of the gauge variables with coefficients $f = \ell/(e\varepsilon)$ is $A(f) = \ell(A)$, and similarly every real linear form $k$ of $A$ is $A(h)$ with $h = k/(e\varepsilon)$. With $g = 0$, [BFS1, Thm. 6.2, p. 280] gives, for every real linear form $k$ and every $b \in \Lambda^*$,

$$\langle e^{i\ell}; k^2 \rangle \le 0, \qquad \langle e^{i\ell}; r_{b_-} r_{b_+} \cos(\cdot) \rangle \ge 0, \qquad \langle e^{\ell}; k^2 \rangle \ge 0, \qquad \langle e^{\ell}; r_{b_-} r_{b_+} \cos(\cdot) \rangle \le 0,$$

where $\langle X; Y \rangle = \langle XY \rangle - \langle X \rangle \langle Y \rangle$ and $\cos(\cdot) = \cos(\vartheta_{b_+} - \vartheta_{b_-} + e\varepsilon A(b))$. (a) follows from $\partial_{c_b} \langle Y \rangle_{\gamma,\Lambda,c} = \langle Y; r_{b_-} r_{b_+} \cos(\cdot) \rangle$, which may be differentiated under the integral because $\int e^{\alpha r^2 - v(r)} r\, dr < \infty$. For (b), let $C_t = (1-t)C_1 + tC_2$ and write $C_2 - C_1 = \sum_\alpha d_\alpha w_\alpha w_\alpha^{\mathsf{T}}$ with $d_\alpha \ge 0$. Differentiating the Gaussian density gives

$$\partial_t \langle Y \rangle_{\gamma_t} = \frac{1}{2} \sum_\alpha d_\alpha \langle Y; k_\alpha(A)^2 \rangle_{\gamma_t}, \qquad k_\alpha(A) = w_\alpha^{\mathsf{T}} C_t^{-1} A,$$

where $\gamma_t$ has covariance $C_t$; the derivative of the normalisation cancels. Integrate over $t \in [0,1]$. This is [BFS1, Cor. 6.3(1), p. 281]. □

**Lemma 2.8.** *Let $\Lambda \subset \Lambda'$ be boxes and $\mathfrak{B} \subset \mathfrak{B}'$ finite sets of positively oriented bonds with $\Lambda^* \subset \mathfrak{B}$ and $\Lambda'^* \subset \mathfrak{B}'$. Let $Q$ and $Q'$ be positive definite forms on $\mathbb{R}^{\mathfrak{B}}$ and $\mathbb{R}^{\mathfrak{B}'}$ such that*

$$Q'(A) \ge Q(A|_{\mathfrak{B}}) + \kappa^2 \sum_{b \in \mathfrak{B}' \setminus \mathfrak{B}} A(b)^2 \qquad (A \in \mathbb{R}^{\mathfrak{B}'}). \tag{2.5}$$

*Then for every real $g \in C^2(\Lambda)$,*

$$\langle e^{iF(g)} \rangle_{\gamma_Q,\Lambda} \le \langle e^{iF(g)} \rangle_{\gamma_{Q'},\Lambda'}, \qquad \langle e^{F(g)} \rangle_{\gamma_Q,\Lambda} \ge \langle e^{F(g)} \rangle_{\gamma_{Q'},\Lambda'}.$$

*Proof.* We treat $Y = e^{iF(g)}$; the case $e^{F(g)}$ is the same with all inequalities reversed. The observable $Y$ and the matter action of $\Lambda$ depend on $A$ only through $A|_{\Lambda^*}$. In a box measure the variables in $\mathfrak{B} \setminus \Lambda^*$ can therefore be integrated out first, which replaces the Gaussian by its

marginal on $\mathbb{R}^{\Lambda^*}$ and does not change the expectation. Covariance inequalities pass to these marginals. With this understanding we apply Lemma 2.7 to measures with $\mathfrak{B} \neq \Lambda^*$.

Start from $\langle\cdot\rangle_{\gamma_{Q'},\Lambda'}$ and put $c_b = 0$ for $b \in \Lambda'^* \setminus \Lambda^*$. By Lemma 2.7(a) this does not increase $\langle Y\rangle$. In the resulting measure the sites of $\Lambda' \setminus \Lambda$ carry only their single-site factors, which cancel, so the resulting measure is $\langle\cdot\rangle_{\gamma_{Q'},\Lambda}$, with Gaussian variables $\mathfrak{B}'$. By (2.5) the covariance $(\varepsilon Q')^{-1}$ is at most the covariance of $\gamma_Q \otimes \gamma'$, where $\gamma'$ is the Gaussian on $\mathbb{R}^{\mathfrak{B}'\setminus\mathfrak{B}}$ with form $\kappa^2|\cdot|^2$. By Lemma 2.7(b), $\langle Y\rangle_{\gamma_{Q'},\Lambda} \geq \langle Y\rangle_{\gamma_Q\otimes\gamma',\Lambda} = \langle Y\rangle_{\gamma_Q,\Lambda}$. □

### 2.6. **Monotonicity in the box.**

**Lemma 2.9.** *Let $g \in C^2(\Lambda)$ be real, $Y = e^{iF(g)}$ and $Y' = e^{F(g)}$.*

(a) *If $\Lambda \subset C \subset C'$ are boxes, then $\langle Y\rangle_C^{\mathrm{ND}} \leq \langle Y\rangle_{C'}^{\mathrm{ND}}$ and $\langle Y'\rangle_C^{\mathrm{ND}} \geq \langle Y'\rangle_{C'}^{\mathrm{ND}}$.*

(b) *If $\Lambda \subset \Omega \subset \Omega'$ are boxes, then $\langle Y\rangle_\Omega \leq \langle Y\rangle_{\Omega'}$ and $\langle Y'\rangle_\Omega \geq \langle Y'\rangle_{\Omega'}$.*

*Proof.* (a) Every pair of bonds summed in $K_C$ is summed in $K_{C'}$, and all terms are nonnegative, so $Q_{C'}^{\mathrm{ND}}(A) \geq Q_C^{\mathrm{ND}}(A|_{C^*}) + \kappa^2 \sum_{b\in C'^*\setminus C^*} A(b)^2$. Apply Lemma 2.8. (b) By Lemma 2.6(a) we may replace both states by Proca states. Every plaquette of $\Omega^{**}$ has its bonds in $\Omega^*$, so $Q_{\Omega'}^{\mathrm{Pr}}(A) \geq Q_\Omega^{\mathrm{Pr}}(A|_{\Omega^*}) + \kappa^2 \sum_{b\in\Omega'^*\setminus\Omega^*} A(b)^2$. Apply Lemma 2.8. □

Part (a) is King's monotonicity argument [K2, Thm. 3.2 and its proof, pp. 333–334]. Part (b) holds because on gauge-invariant observables enlarging the box only adds plaquette and hopping terms to the Proca action, while the gauge-fixing term, which changes at the boundary, does not matter.

### 2.7. **Comparison with the Neumann/Dirichlet states.**

**Lemma 2.10.** *For every box $\Omega$ and $A \in \mathbb{R}^{\Omega^*}$,*

$$|d_1 A|^2 + |d_0^{\mathsf{T}} A|^2 = K_\Omega(A) + N_\partial(A), \qquad N_\partial(A) := \sum_{b\in\Omega^*} n_\partial(b) A(b)^2,$$

*where $n_\partial(b) \in \{0,1,2\}$ is the number of the two bonds $\langle x - \varepsilon e_\mu, x\rangle$, $\langle x + \varepsilon e_\mu, x + 2\varepsilon e_\mu\rangle$ collinear with $b = (x;\mu)$ which are not in $\Omega^*$. In particular $Q_\Omega = Q_\Omega^{\mathrm{ND}} + N_\partial \geq Q_\Omega^{\mathrm{ND}}$.*

*Proof.* We work in lattice units on $\mathbb{Z}^3$. Extend $A$ by zero to a finitely supported 1-cochain $a$ on $\mathbb{Z}^3$. For finitely supported 1-cochains on $\mathbb{Z}^3$, summation by parts gives

$$\sum_p |(d_1 a)(p)|^2 + \sum_x |(d_0^{\mathsf{T}} a)(x)|^2 = \sum_{\mu,\nu=1}^{3} \sum_x |a(x + e_\nu;\mu) - a(x;\mu)|^2, \tag{2.6}$$

which expresses the fact that $d^*d + dd^*$ is the componentwise lattice Laplacian on 1-cochains of $\mathbb{Z}^3$. For $x \in \Omega$ the divergence $(d_0^{\mathsf{T}} a)(x)$ equals $(d_0^{\mathsf{T}} A)(x)$, and for $x \notin \Omega$ it vanishes, since no bond of $\Omega^*$ has an endpoint outside $\Omega$. A plaquette $p \notin \Omega^{**}$ meets $\Omega^*$ in at most one bond. Indeed, two parallel sides in $\Omega^*$ would put all four corners in $\Omega$, and two adjacent sides in $\Omega^*$ would put three corners in $\Omega$, hence the fourth as well, because $\Omega$ is a product of intervals. Thus the left side of (2.6) equals $|d_1 A|^2 + |d_0^{\mathsf{T}} A|^2 + \sum' A(b)^2$, where $\sum'$ runs over the pairs $(p,b)$ with $p \notin \Omega^{**}$ and $b$ the unique bond of $p$ in $\Omega^*$.

On the right side, the pairs with both bonds in $\Omega^*$ give $K_\Omega(A)$, and the pairs with no bond in $\Omega^*$ give 0. The pairs with exactly one bond $b = (x;\mu)$ in $\Omega^*$ are of two kinds. If $\nu \neq \mu$, the two bonds of the pair are opposite sides of a plaquette $p$ spanned by $e_\mu, e_\nu$. Its two sides parallel to $e_\nu$ have an endpoint outside $\Omega$, since the $\nu$-coordinate of one of the two bonds leaves $[a_\nu, b_\nu]$, so $p \notin \Omega^{**}$ and $b$ is its only bond in $\Omega^*$. Conversely every such pair $(p,b)$ arises in this way from exactly one pair. These terms therefore cancel $\sum'$. If $\nu = \mu$, the pairs are those counted by $n_\partial(b)$, and they give $N_\partial(A)$. □

**Lemma 2.11.** *Let $C, \Omega$ be boxes with $C^{(1)} \subset \Omega$, and let $g \in C^2(C)$ be real. Then*

$$\langle e^{iF(g)}\rangle_C^{\mathrm{ND}} \le \langle e^{iF(g)}\rangle_\Omega, \qquad \langle e^{F(g)}\rangle_C^{\mathrm{ND}} \ge \langle e^{F(g)}\rangle_\Omega.$$

*Proof.* By Lemma 2.10, $Q_\Omega(A) \ge K_\Omega(A) + \kappa^2|A|^2$. Every pair of bonds summed in $K_C$ is summed in $K_\Omega$, so (2.5) holds with $Q = Q_C^{\mathrm{ND}}$, $\mathfrak{B} = C^*$, $Q' = Q_\Omega$, $\mathfrak{B}' = \Omega^*$. Apply Lemma 2.8. □

**Lemma 2.12.** *Let $\Omega \subset B$ be boxes and $N \ge 1$ an integer such that $\Omega^{(N+1)} \subset B$ and*

$$3\Big(\frac{\pi}{2N}\Big)^2 \le \kappa^2, \qquad \textit{that is,} \qquad \mu_0 N\varepsilon \ge \frac{\sqrt{3}\,\pi}{2}. \tag{2.7}$$

*Then for every real $g \in C^2(\Omega)$,*

$$\langle e^{iF(g)}\rangle_\Omega \le \langle e^{iF(g)}\rangle_B^{\mathrm{ND}}, \qquad \langle e^{F(g)}\rangle_\Omega \ge \langle e^{F(g)}\rangle_B^{\mathrm{ND}}.$$

*Proof. Step 1: comparison of curvature covariances.* For a positive definite form $Q$ on $\mathbb{R}^{\mathfrak{B}}$ with $\mathfrak{B} \supset \Omega^*$, let $\Sigma_Q$ be the covariance of $(d_1A)|_{\Omega^{**}}$ under $\gamma_Q$, a positive semidefinite form on $\mathbb{R}^{\Omega^{**}}$. For $h \in \mathbb{R}^{\Omega^{**}}$ put $\ell_h(A) = \sum_{p\in\Omega^{**}} h(p)(d_1A)(p)$. Since $\gamma_Q$ has density proportional to $e^{-\frac{\varepsilon}{2}Q}$,

$$\langle h, \Sigma_Q h\rangle = \mathrm{Var}_{\gamma_Q}(\ell_h) = \sup_{A\in\mathbb{R}^{\mathfrak{B}}} \big(2\ell_h(A) - \varepsilon Q(A)\big). \tag{2.8}$$

We claim that $Q_\Omega^{\mathrm{Pr}}(A|_{\Omega^*}) \le Q_B^{\mathrm{ND}}(A)$ for all $A \in \mathbb{R}^{B^*}$. Given this, (2.8) gives $\Sigma_{Q_B^{\mathrm{ND}}} \le \Sigma_{Q_\Omega^{\mathrm{Pr}}}$, since $\ell_h(A)$ depends only on $A|_{\Omega^*}$.

To prove the claim, work in lattice units, with $\Omega = \prod_\nu [a_\nu, b_\nu]$ in integer coordinates, and let $\theta = \pi/(2N)$. For each $\nu$ let $\phi_\nu : \mathbb{Z} \to [0,1]$ be 1 on $[a_\nu - 1, b_\nu + 1]$, $\phi_\nu(b_\nu + s) = \phi_\nu(a_\nu - s) = \cos(\theta(s-1))$ for $1 \le s \le N+1$, and 0 outside $[a_\nu - N, b_\nu + N]$. Put $\psi(x) = \prod_\nu \phi_\nu(x_\nu)$. Then $\psi = 1$ on $\Omega$, and $\psi(x) > 0$ only if $x \in \Omega^{(N)}$, in which case all bonds $(x;\mu)$ belong to $B^*$, because $\Omega^{(N+1)} \subset B$. Let $a$ be the 1-cochain on $\mathbb{Z}^3$ with $a(x;\mu) = \psi(x)A(x;\mu)$ if $\psi(x) > 0$ and $a = 0$ otherwise. Every plaquette of $\Omega^{**}$ has its bonds in $\Omega^*$, where $a = A$. By (2.6),

$$|d_1A|^2_{\Omega^{**}} \le \sum_p |(d_1a)(p)|^2 + \sum_x |(d_0^{\mathsf{T}}a)(x)|^2 = \sum_{\mu,\nu}\sum_x |\psi(x+e_\nu)A_\mu(x+e_\nu) - \psi(x)A_\mu(x)|^2,$$

with $A_\mu(x) = A(x;\mu)$ where $\psi(x) > 0$. The discrete IMS formula

$$\begin{aligned}\sum_x &|\psi(x+e)f(x+e) - \psi(x)f(x)|^2 \\ &= \sum_x \psi(x)\psi(x+e)|f(x+e) - f(x)|^2 + \sum_x \psi(x)\big(2\psi(x) - \psi(x+e) - \psi(x-e)\big)f(x)^2\end{aligned}$$

holds for every finitely supported $\psi$ and every $f$; it follows by expanding both squares and shifting the summation index in one term. Apply it with $f = A_\mu$ and $e = e_\nu$. In the first sum $\psi(x)\psi(x+e_\nu) \le 1$, and $\psi(x)\psi(x+e_\nu) > 0$ only if both bonds $(x;\mu)$ and $(x+e_\nu;\mu)$ belong to $B^*$, so the first sums add up to at most $K_B(A)$. The second sums add up to $\sum_x V(x)\sum_\mu A_\mu(x)^2$ with

$$V(x) = \sum_\nu \Big(\prod_{\nu'\ne\nu} \phi_{\nu'}(x_{\nu'})^2\Big)\,(\phi_\nu\Delta^-\phi_\nu)(x_\nu), \qquad (\phi\Delta^-\phi)(t) := \phi(t)\big(2\phi(t) - \phi(t+1) - \phi(t-1)\big).$$

A direct computation gives, for the one-dimensional profile at the layer $t = b_\nu + s$ (and symmetrically at $a_\nu - s$): $(\phi\Delta^-\phi)(t)$ equals 0 for $s \le 0$, $1 - \cos\theta$ for $s = 1$, $2\cos^2(\theta(s-1))(1-\cos\theta)$ for $2 \le s \le N$, and 0 for $s \ge N+1$. Since $2(1-\cos\theta) \le \theta^2$, we get $0 \le V \le 3\theta^2 \le \kappa^2$, and $V = 0$ on $\Omega$. The bonds $(x;\mu)$ with $x \notin \Omega$ are not in $\Omega^*$. Hence

$$Q_\Omega^{\mathrm{Pr}}(A|_{\Omega^*}) = |d_1A|^2_{\Omega^{**}} + \kappa^2 \sum_{b\in\Omega^*} A(b)^2 \le K_B(A) + \kappa^2 \sum_{b\in B^*} A(b)^2 = Q_B^{\mathrm{ND}}(A).$$

*Step 2: correlation inequalities.* Let $Y = e^{iF(g)}$; the case $e^{F(g)}$ is the same with the inequalities reversed. In $\langle\cdot\rangle_B^{\mathrm{ND}}$ put $c_b = 0$ for $b \in B^* \setminus \Omega^*$. By Lemma 2.7(a) this does not increase $\langle Y\rangle$, and the resulting measure is $\langle\cdot\rangle_{\gamma_1,\Omega}$, where $\gamma_1$ is the marginal of $\gamma_{Q_B^{\mathrm{ND}}}$ on $\mathbb{R}^{\Omega^*}$ (the sites of $B\setminus\Omega$ decouple). The law of $d_1A$ under $\gamma_1$ is centred Gaussian with covariance $\Sigma_1 := \Sigma_{Q_B^{\mathrm{ND}}}$, and under $\gamma_{Q_\Omega^{\mathrm{Pr}}}$ it is centred Gaussian with covariance $\Sigma_2 := \Sigma_{Q_\Omega^{\mathrm{Pr}}}$. Both covariances are carried by $\operatorname{Ran} d_1$. By Step 1, $\Delta := \Sigma_2 - \Sigma_1 \geq 0$. Let $T$ be a linear right inverse of $d_1$ on $\operatorname{Ran} d_1$. Let $\gamma_2$ be the law of $A + T\eta$, where $A \sim \gamma_1$ and $\eta \sim N(0,\Delta)$ are independent. Then $\gamma_2$ is a centred Gaussian whose covariance $C_1 + T\Delta T^{\mathsf{T}}$ dominates the covariance $C_1$ of $\gamma_1$, and the law of $d_1(A+T\eta) = d_1A + \eta$ is $N(0,\Sigma_2)$. By Lemma 2.7(b), $\langle Y\rangle_{\gamma_1,\Omega} \geq \langle Y\rangle_{\gamma_2,\Omega}$. By Lemma 2.6(b) and (a), $\langle Y\rangle_{\gamma_2,\Omega} = \langle Y\rangle_\Omega^{\mathrm{Pr}} = \langle Y\rangle_\Omega$. Altogether $\langle Y\rangle_B^{\mathrm{ND}} \geq \langle Y\rangle_\Omega$. □

*Remark* 2.13. The upper comparison is made with a Neumann/Dirichlet box $B$ which contains $\Omega$ with a margin of $N+1$ lattice layers, where $N$ is of order $(\mu_0\varepsilon)^{-1}$. In physical units the margin is of order $1/\mu_0$, independently of $\varepsilon$. In Step 1 the photon mass absorbs the localisation error $V$ of the cutoff $\psi$.

2.8. **Limits on the two classes.** A sequence of boxes $(\Omega_n)$ is *exhausting* if every finite subset of $\varepsilon\mathbb{Z}^3$ is contained in $\Omega_n$ for all large $n$. It is nested if $\Omega_n \subset \Omega_{n+1}$. King's cubes $C_J = \{x \in \varepsilon\mathbb{Z}^3 : -MJ \leq x_\mu < MJ\}$ [K2, (2.7)] form a nested exhausting sequence.

**Proposition 2.14.** *Let $g$ be a finitely supported real 2-cochain on $\varepsilon\mathbb{Z}^3$, and let $Y = e^{iF(g)}$, $Y' = e^{F(g)}$, regarded as elements of $\mathcal{I}_F(\Lambda)$, $\mathcal{D}_F(\Lambda)$ for every box $\Lambda$ with $\operatorname{supp} g \subset \Lambda^{**}$.*

(a) *For every nested exhausting sequence $(C_J)$ the limits $s^{\mathrm{ND}}(Y) = \lim_J \langle Y\rangle_{C_J}^{\mathrm{ND}}$ and $s^{\mathrm{ND}}(Y') = \lim_J \langle Y'\rangle_{C_J}^{\mathrm{ND}}$ exist; the first is a nondecreasing limit and the second a nonincreasing limit.*

(b) *For every exhausting sequence $(\Omega_n)$ the limits $s(Y) = \lim_n \langle Y\rangle_{\Omega_n}$ and $s(Y') = \lim_n \langle Y'\rangle_{\Omega_n}$ exist and do not depend on the sequence. In fact $s(Y) = \sup_\Omega \langle Y\rangle_\Omega$ and $s(Y') = \inf_\Omega \langle Y'\rangle_\Omega$, the supremum and infimum running over all boxes $\Omega$ with $\operatorname{supp} g \subset \Omega^{**}$.*

(c) *$s(Y) = s^{\mathrm{ND}}(Y)$ and $s(Y') = s^{\mathrm{ND}}(Y')$. In particular $s^{\mathrm{ND}}$ does not depend on the nested sequence.*

*Proof.* (a) By Lemma 2.9(a), $\langle Y\rangle_{C_J}^{\mathrm{ND}}$ is nondecreasing and bounded by 1, and $\langle Y'\rangle_{C_J}^{\mathrm{ND}}$ is nonincreasing and positive.

(b) Let $s(Y) = \sup_\Omega \langle Y\rangle_\Omega \leq 1$ and $\delta > 0$, and choose $\Omega_\delta$ with $\langle Y\rangle_{\Omega_\delta} > s(Y) - \delta$. For large $n$, $\Omega_n \supset \Omega_\delta$, so $s(Y) - \delta < \langle Y\rangle_{\Omega_\delta} \leq \langle Y\rangle_{\Omega_n} \leq s(Y)$ by Lemma 2.9(b). The argument for $Y'$ is the same.

(c) Given $J$ with $\operatorname{supp} g \subset C_J^{**}$, choose $n$ with $C_J^{(1)} \subset \Omega_n$. Lemma 2.11 gives $\langle Y\rangle_{C_J}^{\mathrm{ND}} \leq \langle Y\rangle_{\Omega_n} \leq s(Y)$, hence $s^{\mathrm{ND}}(Y) \leq s(Y)$. Conversely, given a box $\Omega$ with $\operatorname{supp} g \subset \Omega^{**}$, choose $N$ as in (2.7) and $J$ with $\Omega^{(N+1)} \subset C_J$. Lemma 2.12 gives $\langle Y\rangle_\Omega \leq \langle Y\rangle_{C_J}^{\mathrm{ND}} \leq s^{\mathrm{ND}}(Y)$, hence $s(Y) \leq s^{\mathrm{ND}}(Y)$. The same argument with reversed inequalities applies to $Y'$.

The two comparison lemmas also give the convergence in (b) without Lemma 2.9(b). Indeed, for every $J$ and all large $n$ they give $\langle Y\rangle_{C_J}^{\mathrm{ND}} \leq \langle Y\rangle_{\Omega_n} \leq s^{\mathrm{ND}}(Y)$, and the left side tends to $s^{\mathrm{ND}}(Y)$ as $J \to \infty$; similarly for $Y'$. □

## 3. Reflection positivity at the nine reflection families

3.1. **Reflections and layers.** Let $\mathcal{N} = \{e_1, e_2, e_3, e_1 \pm e_2, e_1 \pm e_3, e_2 \pm e_3\}$. For $n \in \mathcal{N}$ and $c \in \varepsilon\mathbb{Z}$ let

$$\rho = \rho_{n,c} : \; x \mapsto x - \frac{2}{|n|^2}(n\cdot x - c)\, n,$$

the reflection in the plane $\pi_{n,c} = \{n \cdot x = c\}$. Since the entries of $n$ lie in $\{0, \pm 1\}$ and $c \in \varepsilon\mathbb{Z}$, $\rho$ maps $\varepsilon\mathbb{Z}^3$ onto itself and nearest neighbours to nearest neighbours; its linear part is a signed permutation matrix. Hence $\rho \in \varepsilon\mathbb{Z}^3 \rtimes O_h$ acts on oriented cells by $\rho_* \langle x, y\rangle = \langle \rho x, \rho y\rangle$ and correspondingly on plaquettes and cubes, with $\rho_* \partial\sigma = \partial \rho_* \sigma$. Planes $\pi_{n,c}$ with $c \in \varepsilon\mathbb{Z}$ are called *site planes*. The layer function $\ell(x) := (n \cdot x - c)/\varepsilon$ is integer valued and satisfies

$$\ell \circ \rho = -\ell, \qquad \ell(x + \varepsilon e_\mu) - \ell(x) = n_\mu \in \{0, \pm 1\}.$$

On fields we let $(\hat\rho A)(b) := A(\rho_* b)$ and $(\hat\rho \varphi)(x) := \varphi(\rho x)$, and on functions $\theta G := \overline{G \circ \hat\rho}$. The map $\hat\rho$ is an involution, acts on the real coordinates of $(A, \varphi)$ by a signed permutation, and commutes with $d$: $d(\hat\rho A) = \hat\rho(dA)$ with $(\hat\rho F)(p) := F(\rho_* p)$.

Let $\Lambda \subset \varepsilon\mathbb{Z}^3$ be finite with $\rho\Lambda = \Lambda$, with the induced complex $\Lambda^*$, $\Lambda^{**}$ (bonds with both endpoints in $\Lambda$, plaquettes with all corners in $\Lambda$). Put $\Lambda_\pm := \Lambda \cap \{\pm\ell \geq 0\}$ and $\Lambda_0 := \Lambda \cap \{\ell = 0\}$. A cell *lies in* a set of sites if all its vertices do.

**Lemma 3.1.** *Let $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, $\rho = \rho_{n,c}$, and let $\Lambda$ be as above.*

(a) *Every bond of $\Lambda^*$ lies in $\Lambda_+$ or in $\Lambda_-$. A bond lying in both lies in $\Lambda_0$ and satisfies $\rho_* b = b$. No bond satisfies $\rho_* b = -b$.*

(b) *Let $X$ be the set of plaquettes of $\Lambda^{**}$ lying neither in $\Lambda_+$ nor in $\Lambda_-$. If $n$ is a coordinate vector, $X = \emptyset$. If $n = e_i \pm e_j$, every $p \in X$ is spanned by $e_i, e_j$, two opposite corners of $p$ lie in $\Lambda_0$, the other two corners have $\ell = 1$ and $\ell = -1$, and $\rho_* p = -p$.*

(c) *For $p \in X$ write $\partial p = b_1 + b_2 + b_3 + b_4$, where $b_1, b_2$ are the two oriented bonds of $\partial p$ with an endpoint at the corner with $\ell = 1$; they lie in $\Lambda_+$. Put $v_p(A) := A(b_1) + A(b_2)$. Then*

$$\varepsilon\,(dA)(p) = v_p(A) - v_p(\hat\rho A).$$

*Proof.* (a) The values of $\ell$ at the two endpoints of a bond differ by at most 1, so they are not of opposite strict signs. A bond lying in $\Lambda_+$ and in $\Lambda_-$ has $\ell = 0$ at both endpoints; $\rho$ fixes the plane pointwise, so $\rho_* b = b$. If $\rho_* \langle x, y\rangle = \langle y, x\rangle$, then $\rho x = y = x + \varepsilon e_\mu$ for some $\mu$ (up to sign), and $-\ell(x) = \ell(\rho x) = \ell(x) \pm n_\mu$. If $n_\mu = 0$ this forces $\ell(x) = 0$ and then $\rho x = x \neq y$; if $n_\mu = \pm 1$ it forces $2\ell(x) = \pm 1$, which is impossible.

(b) The corners of a plaquette spanned by $e_\mu, e_\nu$ have the values $t, t + n_\mu, t + n_\nu, t + n_\mu + n_\nu$ of $\ell$. If $n_\mu n_\nu = 0$ these lie in two consecutive integers, so the plaquette lies in $\Lambda_+$ or $\Lambda_-$. If $n_\mu n_\nu \neq 0$, then $n = e_\mu \pm e_\nu$ up to sign and $\{\mu, \nu\} = \{i, j\}$. The four values are then $s - 1, s, s, s + 1$ for some integer $s$, the two middle values being taken at opposite corners. If $s \neq 0$, all four values have the same weak sign. If $s = 0$, the two middle corners lie in $\Lambda_0$ and are fixed by $\rho$, and the others have $\ell = \pm 1$ and are exchanged. The reflection $\rho$ fixes the coordinates other than $x_i, x_j$, so $\rho_* p$ is the plaquette spanned by $e_i, e_j$ with the same two middle corners, which is $p$ up to orientation. On the affine plane through $p$ spanned by $e_i, e_j$, $\rho$ acts as a reflection in a line and reverses orientation. Hence $\rho_* p = -p$.

(c) The map $\rho$ exchanges the corners with $\ell = \pm 1$ and fixes the two others. Hence $\rho_*(b_1 + b_2)$ is a combination of $b_3, b_4$ only, and $\rho_*(b_3 + b_4)$ of $b_1, b_2$ only. Since $\rho_* \partial p = \partial \rho_* p = -\partial p$, comparing coefficients gives $\rho_*(b_1 + b_2) = -(b_3 + b_4)$. Therefore $\varepsilon(dA)(p) = A(b_1) + A(b_2) - A(\rho_* b_1) - A(\rho_* b_2) = v_p(A) - v_p(\hat\rho A)$. □

### 3.2. **The positivity theorem.**

**Theorem 3.2.** *Let $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, $\rho = \rho_{n,c}$, and let $\Lambda$ be finite with $\rho\Lambda = \Lambda$. For $x \in \Lambda$ let $v_x : [0, \infty) \to \mathbb{R}$ be Borel with $v_{\rho x} = v_x$ and $\int_0^\infty e^{3\varepsilon r^2 - v_x(r)} r\, dr < \infty$. On $\mathbb{R}^{\Lambda^*} \times \mathbb{C}^\Lambda$ let*

$$S(A, \varphi) = \tfrac{1}{2}\|dA\|_{2,\varepsilon}^2 + \tfrac{1}{2}\mu_0^2 \|A\|_{2,\varepsilon}^2 + \sum_{x \in \Lambda} v_x(|\varphi_x|) - \varepsilon \sum_{b \in \Lambda^*} \mathrm{Re}\left(\overline{\varphi(b_-)}\, e^{ie\varepsilon A(b)} \varphi(b_+)\right),$$

*where $\|dA\|_{2,\varepsilon}^2$ is the sum over $\Lambda^{**}$. Let $\mathfrak{A}_+$ be the set of bounded Borel functions of $\varphi|_{\Lambda_+}$ and of the values of $A$ on the bonds lying in $\Lambda_+$. Then*

$$\int \theta(G)\, G\, e^{-S}\, dA\, d\varphi \;\geq 0 \qquad (G \in \mathfrak{A}_+).$$

*Proof.* Write $S = \sum_\sigma T_\sigma$ as a sum over the cells $\sigma$ of the induced complex. The site terms are $T_x = v_x(|\varphi_x|)$. The bond terms are $T_b = \frac{1}{2}\mu_0^2\varepsilon^3 A(b)^2 - \varepsilon\,\mathrm{Re}(\overline{\varphi(b_-)}e^{ie\varepsilon A(b)}\varphi(b_+))$, which is unchanged when $b$ is replaced by $-b$. The plaquette terms are $T_p = \frac{1}{2}\varepsilon^3 (dA)(p)^2$. Each term satisfies $T_\sigma \circ \hat\rho = T_{\rho\sigma}$: for sites by $v_{\rho x} = v_x$, for bonds because $(\hat\rho A)(b) = A(\rho_* b)$ and $\rho_* b = \langle \rho b_-, \rho b_+\rangle$, and for plaquettes because $d$ commutes with $\hat\rho$.

By Lemma 3.1, every cell either lies in $\Lambda_0$, or lies in $\Lambda_+$ but not in $\Lambda_0$, or lies in $\Lambda_-$ but not in $\Lambda_0$, or belongs to $X$. Let $C_0$, $C_+$, $C_-$ be the first three sets. Then $\rho C_+ = C_-$. The cells of $C_0$ are fixed by $\rho$ with their orientation, since $\rho$ fixes the plane pointwise, so $T_\sigma \circ \hat\rho = T_\sigma$ for $\sigma \in C_0$. Put

$$S_+ := \sum_{\sigma\in C_+} T_\sigma + \frac{1}{2}\sum_{\sigma\in C_0} T_\sigma.$$

Then $S_+ \circ \hat\rho = \sum_{\sigma\in C_-} T_\sigma + \frac{1}{2}\sum_{C_0} T_\sigma$, and by Lemma 3.1(c)

$$S = S_+ + S_+ \circ \hat\rho + \sum_{p\in X} \frac{\varepsilon}{2}\big(v_p - v_p\circ\hat\rho\big)^2.$$

Split the variables into $\omega_0$ (the values of $\varphi$ on $\Lambda_0$ and of $A$ on bonds lying in $\Lambda_0$), $\omega_+$ (the remaining variables attached to sites of $\Lambda_+$ and bonds lying in $\Lambda_+$) and $\omega_-$ (the rest). Then $S_+$, $v_p$ ($p \in X$) and every $G \in \mathfrak{A}_+$ are functions of $(\omega_0, \omega_+)$, and $\hat\rho$ fixes $\omega_0$ and exchanges $\omega_+$ and $\omega_-$ by a signed permutation, which preserves Lebesgue measure.

For $a > 0$, $e^{-ay^2} = \int_{\mathbb{R}} e^{iky}\,\nu_a(dk)$ with $\nu_a$ the centred normal law of variance $2a$. With $\eta_k := e^{ikv_p}$ we have $e^{ik(v_p - v_p\circ\hat\rho)} = \eta_k\,\overline{\eta_k\circ\hat\rho}$. Hence, by Fubini's theorem,

$$\int \theta(G)\, G\, e^{-S} = \int \nu_{\varepsilon/2}^{\otimes X}(dk) \int d\omega_0 \,\Big|\int d\omega_+ \; G\, e^{-S_+} \prod_{p\in X} e^{ik_p v_p}\Big|^2 \;\geq 0.$$

Indeed, for fixed $k$ and $\omega_0$ the integral over $\omega_-$ of $\overline{G\circ\hat\rho}\, e^{-S_+\circ\hat\rho}\prod_p \overline{\eta_{k_p}\circ\hat\rho}$ equals, after the change of variables $\omega_- \mapsto \omega_+$, the complex conjugate of the integral over $\omega_+$ of $G\, e^{-S_+}\prod_p \eta_{k_p}$. Fubini's theorem applies because $|\eta_k| = 1$, $\nu_{\varepsilon/2}$ is a probability measure, and

$$\int e^{-S_+ - S_+\circ\hat\rho}\, dA\, d\varphi \leq \int \prod_{b\in\Lambda^*} e^{-\frac{1}{2}\mu_0^2\varepsilon^3 A(b)^2} \prod_{x\in\Lambda} e^{-v_x(|\varphi_x|)+3\varepsilon|\varphi_x|^2}\, dA\, d\varphi < \infty.$$

Here we dropped the nonnegative plaquette terms and used $\varepsilon\, r_{b_-} r_{b_+} \leq \frac{\varepsilon}{2}(r_{b_-}^2 + r_{b_+}^2)$ together with the fact that every site has at most six bonds. □

**Corollary 3.3.** *Let $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, and let $\Omega$ be a box with $\rho_{n,c}\Omega = \Omega$. For every gauge-invariant $G \in \mathfrak{A}_+$ (with $\Lambda = \Omega$), $\langle\theta(G)G\rangle_\Omega \geq 0$. This applies in particular to every bounded Borel function $G$ of finitely many curvatures $(dA)(p)$ with $p \in \Omega^{**}$ lying in $\{n\cdot x \geq c\}$.*

*Proof.* The map $\hat\rho$ conjugates gauge transformations into gauge transformations:

$$\hat\rho\big(A + d\chi, e^{-ie\chi}\varphi\big) = \big(\hat\rho A + d(\chi\circ\rho),\, e^{-ie\chi\circ\rho}\hat\rho\varphi\big).$$

Hence $\theta(G)G$ is bounded and gauge invariant, and $\langle\theta(G)G\rangle_\Omega = \langle\theta(G)G\rangle_\Omega^{\mathrm{Pr}}$ by Lemma 2.6(a). The Proca box state is the normalised measure $e^{-S}dA\,d\varphi$ of Theorem 3.2 with $\Lambda = \Omega$ and $v_x = v$ for all $x$. The last assertion holds because the bonds of such a plaquette lie in $\Omega_+$. □

**Lemma 3.4.** *Let $n \in \mathcal{N}$ and $c \in \varepsilon\mathbb{Z}$. Every finite subset of $\varepsilon\mathbb{Z}^3$ is contained in a box $\Omega$ with $\rho_{n,c}\Omega = \Omega$.*

*Proof.* Let $R \in \varepsilon\mathbb{Z}$ be large. For $n = e_i$ take $[c-R, c+R]$ in the $i$-th coordinate and $[-R, R]$ in the others. For $n = e_i - e_j$ the reflection is $(x_i, x_j) \mapsto (x_j + c, x_i - c)$; take $[-R, R] \times [-R - c, R - c]$ in the coordinates $(x_i, x_j)$. For $n = e_i + e_j$ the reflection is $(x_i, x_j) \mapsto (c - x_j, c - x_i)$; take $[-R, R] \times [c - R, c + R]$. In the remaining coordinate take $[-R, R]$. □

*Remark* 3.5. Corollary 3.3 concerns gauge-invariant observables. Positivity is proved for the Proca measure and transferred by Lemma 2.6. We make no assertion about functions of $A$ that are not gauge invariant under the gauge-fixed measure $\gamma_{Q_\Omega}$.

*Remark* 3.6. On cubic tori the diagonal reflections do not admit the factorisation of Theorem 3.2. Let $T = \varepsilon\mathbb{Z}^3/(N\varepsilon\mathbb{Z})^3$ and let $\rho$ be the reflection $(x_1, x_2, x_3) \mapsto (x_2, x_1, x_3)$. The layer index $\ell = (x_1 - x_2)/\varepsilon$ is defined modulo $N$, and $\rho$ maps the layer $\ell$ onto the layer $-\ell$. Let first $N$ be even. Then $\rho$ maps the layers $\ell \equiv 0$ and $\ell \equiv N/2$ to themselves. It fixes the first pointwise, but on the second it acts as the translation by $\frac{N\varepsilon}{2}(e_2 - e_1)$ modulo the period lattice, without fixed points. If $y$ lies in the layer $\ell \equiv N/2$, the function $f = e^{-|\varphi_y|^2} - e^{-|\varphi_{\rho y}|^2}$ depends only on the variables of the closed half $\{0 \le \ell \le N/2\}$ and satisfies $\theta f = -f$. Hence $\langle \theta(f) f \rangle = -\langle f^2 \rangle < 0$ for every measure under which $|\varphi_y| \neq |\varphi_{\rho y}|$ with positive probability. Thus positivity fails for the algebra of the closed half-torus $\{0 \le \ell \le N/2\}$, which contains the layer $\ell \equiv N/2$; we do not discuss other algebras, such as that of the open half-torus. If $N$ is odd, the layer $\ell \equiv 0$ is the only layer mapped to itself, and the bonds between the layers $\ell \equiv (N-1)/2$ and $\ell \equiv (N+1)/2$ join the two halves $\{0 \le \ell \le (N-1)/2\}$ and $\{(N+1)/2 \le \ell \le N\}$ without being mapped to their reverses by $\rho$. So the conclusion of Lemma 3.1(a) fails, and the proof of Theorem 3.2 does not apply. This is the reason for working with boxes.

## 4. Gaussian domination and the infinite-volume limit

4.1. **Current expansion.** For a box $\Lambda$ let $\mathcal{J}_\Lambda$ be the set of divergence-free integer currents, that is, of $n \in \mathbb{Z}^{\Lambda^*}$ with $d_0^{\mathsf{T}} n = 0$.

**Lemma 4.1.** *Let $\Lambda$ be a box and $c \in [0, \infty)^{\Lambda^*}$. Then for $A \in \mathbb{R}^{\Lambda^*}$,*

$$W_c(A) := \int_{\mathbb{C}^\Lambda} e^{-S^{\mathrm{m}}_{\Lambda,c}(A,\varphi)}\, d\varphi = \sum_{n \in \mathcal{J}_\Lambda} w(n) \exp\Big( ie\varepsilon \sum_{b \in \Lambda^*} n(b) A(b) \Big),$$

*where $w(n) = w(-n) \ge 0$ and $\sum_n w(n) = W_c(0) < \infty$. The series converges absolutely and uniformly in $A$.*

*Proof.* Write $d\varphi = \prod_x r_x\, dr_x\, d\vartheta_x$ and use $e^{t\cos u} = \sum_{k \in \mathbb{Z}} I_k(t) e^{iku}$, where the modified Bessel functions satisfy $I_k = I_{-k} \ge 0$ on $[0, \infty)$ and $\sum_k I_k(t) = e^t$. For every bond $b \in \Lambda^*$ put $u_b = \vartheta_{b_+} - \vartheta_{b_-} + e\varepsilon A(b)$ and expand $e^{c_b r_{b_-} r_{b_+} \cos u_b}$. The product over $b$ of these series converges absolutely, with sum of absolute values $\prod_b e^{c_b r_{b_-} r_{b_+}} \le \prod_x e^{3 c_{\max} r_x^2}$. This is integrable against $\prod_x e^{-v(r_x)} r_x\, dr_x\, d\vartheta_x$, so Fubini's theorem applies. Since $\sum_b n(b)(\vartheta_{b_+} - \vartheta_{b_-}) = \sum_x (d_0^{\mathsf{T}} n)(x) \vartheta_x$, the angular integral of $\exp(i \sum_b n(b)(\vartheta_{b_+} - \vartheta_{b_-}))$ equals $(2\pi)^{|\Lambda|}$ if $d_0^{\mathsf{T}} n = 0$ and 0 otherwise. This gives the expansion with

$$w(n) = (2\pi)^{|\Lambda|} \int \prod_x e^{-v(r_x)} r_x\, dr_x \prod_{b \in \Lambda^*} I_{n(b)}\big(c_b r_{b_-} r_{b_+}\big) \ \ge 0,$$

and $w(-n) = w(n)$ because $I_{-k} = I_k$. At $A = 0$ all phases equal 1, so $\sum_n w(n) = W_c(0)$. □

4.2. **Gaussian domination on boxes.**

**Proposition 4.2.** *Let $\Omega$ be a box and $g \in C^2(\Omega)$ real. Write $d = d_\Omega$, $d^* = d^*_\Omega$, and let $\Pi$ be the orthogonal projection of $C^2(\Omega)$ onto* $\operatorname{Ran} d$. *Then*

$$\langle e^{F(g)}\rangle_\Omega \le \exp\Big(\tfrac12\langle d^*g, (d^*d+\mu_0^2)^{-1}d^*g\rangle_\varepsilon\Big) \le e^{\frac12\|\Pi g\|^2_{2,\varepsilon}} \le e^{\frac12\|g\|^2_{2,\varepsilon}}.$$

*Consequently, for $t \in \mathbb{R}$, real $g_a \in C^2(\Omega)$ and $z \in \mathbb{C}^k$,*

$$\langle e^{tF(g)}\rangle_\Omega \le e^{\frac12 t^2\|g\|^2_{2,\varepsilon}}, \qquad \Big|\Big\langle \exp\sum_a z_a F(g_a)\Big\rangle_\Omega\Big| \le \exp\Big(\frac12\Big\|\sum_a \operatorname{Re}(z_a)g_a\Big\|^2_{2,\varepsilon}\Big).$$

*These bounds depend neither on the box nor on $e, \lambda, m^2, \delta m^2, \mu_0$.*

*Proof.* By Lemma 2.6(a) we may compute in the Proca state. Its Gaussian has density proportional to $\exp(-\frac12\langle A, (d^*d+\mu_0^2)A\rangle_\varepsilon)$, so with $C := (d^*d+\mu_0^2)^{-1}$ we have $\int e^{\langle u,A\rangle_\varepsilon}\gamma(dA) = e^{\frac12\langle u,Cu\rangle_\varepsilon}$ for complex $u$, with the bilinear extension of $\langle\cdot,\cdot\rangle_\varepsilon$. Put $f := d^*g$, so that $F(g) = \langle f, A\rangle_\varepsilon$, and $j_n := \varepsilon^{-2}n$, so that $e\varepsilon\sum_b n(b)A(b) = e\langle j_n, A\rangle_\varepsilon$. By Lemma 4.1 and Fubini's theorem, which applies since $\sum_n w(n)\int e^{\langle f,A\rangle_\varepsilon}d\gamma < \infty$,

$$\begin{aligned}\int e^{\langle f,A\rangle_\varepsilon} W(A)\,\gamma(dA) &= \sum_n w(n)\int e^{\langle f+ie j_n,A\rangle_\varepsilon}\gamma(dA)\\ &= e^{\frac12\langle f,Cf\rangle_\varepsilon}\sum_n w(n)\,e^{-\frac{e^2}{2}\langle j_n,Cj_n\rangle_\varepsilon}\cos\big(e\langle j_n, Cf\rangle_\varepsilon\big),\end{aligned}$$

where $W = W_c$ with $c \equiv \varepsilon$. The sine terms cancel in pairs because $w(n) = w(-n)$. The cosine is at most 1, and $\sum_n w(n)e^{-\frac{e^2}{2}\langle j_n,Cj_n\rangle_\varepsilon} = \int W\,d\gamma$ by the same identity at $f = 0$. Dividing by $\int W\,d\gamma$ gives the first inequality.

For the second, $\operatorname{Ran} d^* = (\operatorname{Ker} d)^\perp$ in $C^1(\Omega)$ is invariant under $d^*d$, and $d^*d$ is injective on it. On this subspace $d^*d + \mu_0^2 \ge d^*d > 0$, hence $(d^*d+\mu_0^2)^{-1} \le (d^*d)^{-1}$, where $(d^*d)^{-1}$ denotes the inverse on $\operatorname{Ran} d^*$. Since $f \in \operatorname{Ran} d^*$, $\langle f, Cf\rangle_\varepsilon \le \langle f, u\rangle_\varepsilon$ with $u := (d^*d)^{-1}f \in \operatorname{Ran} d^*$. Now $d^*(du - g) = 0$, so $du - g \in \operatorname{Ker} d^* = (\operatorname{Ran} d)^\perp$ and $du = \Pi g$. Hence $\langle f, u\rangle_\varepsilon = \langle d^*g, u\rangle_\varepsilon = \langle g, du\rangle_\varepsilon = \|\Pi g\|^2_{2,\varepsilon}$. The last inequality is $\|\Pi g\| \le \|g\|$.

The consequences follow by applying the bound to $tg$ and to $\sum_a \operatorname{Re}(z_a)g_a$, since the modulus of $\exp\sum_a z_aF(g_a)$ is $\exp F(\sum_a \operatorname{Re}(z_a)g_a)$. □

The argument is the box version of the corresponding torus argument of [v1, Lemma 7.3]. On the box only the facts that $\operatorname{Ran} d^*$ reduces $d^*d$ and that $d(d^*d)^{-1}d^*$ is the projection onto $\operatorname{Ran} d$ are used, and both hold on every finite cell complex.

4.3. **The infinite-volume state at fixed spacing.** Let $\mathcal{P}_\varepsilon$ be the set of positively oriented plaquettes of $\varepsilon\mathbb{Z}^3$, and let $E_\varepsilon = \mathbb{R}^{\mathcal{P}_\varepsilon}$ with the product $\sigma$-algebra. We write $F(p)$ for the coordinate functions, extend them to oriented plaquettes by $F(-p) = -F(p)$, and put $F(g) = \varepsilon^3\sum_p g(p)F(p)$ for finitely supported real $g$. For $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$, which acts on oriented cells as in Section 3, let $(\sigma_\sharp F)(p) := F(\sigma_*^{-1}p)$.

**Theorem 4.3.** *Let $\varepsilon > 0$ and the parameters be fixed as at the beginning of Section 2.*

(a) *Let $g_1, \dots, g_k$ be finitely supported real 2-cochains. For every exhausting sequence of boxes $(\Omega_n)$, the functions $z \mapsto \langle\exp\sum_a z_aF(g_a)\rangle_{\Omega_n}$ converge locally uniformly on $\mathbb{C}^k$ to an entire function, and the limit does not depend on the sequence.*

(b) *There is a unique probability measure $\omega_\varepsilon$ on $E_\varepsilon$ such that $\omega_\varepsilon(\exp\sum_a z_aF(g_a))$ equals the limit in (a) for all such data. It is carried by closed cochains: $(dF)(c) = 0$ for every cube $c$, $\omega_\varepsilon$-almost surely.*

(c) $\omega_\varepsilon$ *is invariant under* $\sigma_\sharp$ *for every* $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$.

(d) *For real* $g$ *with* $\sum_p |g(p)| < \infty$ *the series* $F(g)$ *converges absolutely* $\omega_\varepsilon$*-almost surely, and* $\omega_\varepsilon(e^{F(g)}) \le e^{\frac{1}{2}\|g\|_{2,\varepsilon}^2}$. *In particular* $\omega_\varepsilon(F(g)^2) \le \|g\|_{2,\varepsilon}^2$.

(e) *Let* $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, $\rho = \rho_{n,c}$, *and let* $G$ *be a bounded Borel function of finitely many* $F(p)$ *with* $p$ *lying in* $\{n \cdot x \ge c\}$. *Then* $\omega_\varepsilon\big(\overline{G \circ \rho_\sharp}\, G\big) \ge 0$.

(f) *For every finitely supported real* $g$ *and every nested exhausting sequence* $(C_J)$ *of boxes, in particular for King's cubes,*

$$\omega_\varepsilon(e^{iF(g)}) = \lim_J \langle e^{iF(g)}\rangle_{C_J}^{\mathrm{ND}}, \qquad \omega_\varepsilon(e^{F(g)}) = \lim_J \langle e^{F(g)}\rangle_{C_J}^{\mathrm{ND}}.$$

*Proof.* (a) For each box with all $\operatorname{supp} g_a \subset \Omega^{**}$, the function $\Phi_\Omega(z) = \langle \exp \sum z_a F(g_a)\rangle_\Omega$ is entire, by differentiation under the integral sign, and by Proposition 4.2 the family $(\Phi_\Omega)$ is locally bounded on $\mathbb{C}^k$ uniformly in $\Omega$. On $i\mathbb{R}^k$ the values are $\langle e^{iF(\sum t_a g_a)}\rangle_\Omega$, which converge along every exhausting sequence to a limit independent of the sequence, by Proposition 2.14(b). By Montel's theorem every subsequence of $(\Phi_{\Omega_n})$ has a locally uniformly convergent subsequence. All limits are entire and agree on $i\mathbb{R}^k$, hence everywhere, because an entire function on $\mathbb{C}^k$ that vanishes on $i\mathbb{R}^k$ vanishes identically (its Taylor coefficients at 0 vanish). Hence the whole sequence converges, and the limit is independent of the sequence.

(b) For a finite set $P \subset \mathcal{P}_\varepsilon$ the characteristic functions of the laws of $(F(p))_{p\in P}$ under $\langle\cdot\rangle_{\Omega_n}$ converge pointwise to a function continuous at 0. By Lévy's continuity theorem these laws converge weakly. The limits are consistent, so Kolmogorov's theorem gives a unique $\omega_\varepsilon$ with these finite-dimensional laws. For real $z$ the functions $\exp \sum z_a F(g_a)$ are uniformly integrable along the sequence by Proposition 4.2, applied with $2z$. Hence $\omega_\varepsilon(\exp \sum z_a F(g_a))$ equals the limit for real $z$, and by the same argument applied to the modulus for all complex $z$. For a cube $c$ and a box containing all its corners, $(dF)(c) = (ddA)(c) = 0$ identically, so the law of $(dF)(c)$ is $\delta_0$ under all $\langle\cdot\rangle_{\Omega_n}$ with $n$ large, hence under $\omega_\varepsilon$.

(c) Let $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$. It maps boxes onto boxes and preserves the induced complexes. Define $\hat\sigma A(b) = A(\sigma_* b)$ and $\hat\sigma\varphi(x) = \varphi(\sigma x)$. All terms of $S_\Omega$ are covariant: $v$ is the same at all sites, and $d^*_{\sigma\Omega}$ corresponds to $d^*_\Omega$. Hence $S_\Omega \circ \hat\sigma = S_{\sigma\Omega}$ as functions of the variables of $\sigma\Omega$, and the change of variables $\hat\sigma$, a signed permutation, gives $\langle G\circ\hat\sigma\rangle_{\sigma\Omega} = \langle G\rangle_\Omega$. For $G$ a function of the curvature, $G \circ \hat\sigma$ is $G$ composed with $F \mapsto (F(\sigma_* p))_p = (\sigma^{-1})_\sharp F$. Since $(\sigma\Omega_n)$ is again exhausting, (a) gives $\omega_\varepsilon(G \circ (\sigma^{-1})_\sharp) = \lim_n \langle G\circ\hat\sigma\rangle_{\sigma\Omega_n} = \lim_n \langle G\rangle_{\Omega_n} = \omega_\varepsilon(G)$ for all exponentials $G$, and these determine the law.

(d) For finitely supported $g$ the bound follows from Proposition 4.2 and (b). Applied to $\pm tg$ it gives $\omega_\varepsilon(\cosh tF(g)) \le e^{\frac{1}{2}t^2\|g\|^2}$, and $\cosh(ty) - 1 \ge \frac{1}{2}t^2y^2$, so $\omega_\varepsilon(F(g)^2) \le \lim_{t\to 0} 2(e^{t^2\|g\|^2/2} - 1)/t^2 = \|g\|_{2,\varepsilon}^2$. In particular $\omega_\varepsilon|F(p)| \le \varepsilon^{-3/2}$ for each $p$. For general $g$ with $\sum|g| < \infty$ the series $\sum_p |g(p)F(p)|$ therefore has finite expectation, hence converges almost surely. Let $g_m$ be the restriction of $g$ to plaquettes with $|c_p| \le m$. Then $F(g_m) \to F(g)$ almost surely, and Fatou's lemma gives $\omega_\varepsilon(e^{F(g)}) \le \liminf_m e^{\frac{1}{2}\|g_m\|^2} = e^{\frac{1}{2}\|g\|^2}$. The second-moment bound follows as before.

(e) First let $G = \phi(F(p_1), \dots, F(p_k))$ with $\phi$ bounded and continuous. By Lemma 3.4 there is an exhausting sequence of boxes $\Omega_n$ with $\rho\Omega_n = \Omega_n$, each containing the corners of $p_1, \dots, p_k$. As a function of $A$, $G$ is gauge invariant and belongs to $\mathfrak{A}_+$ for $\Lambda = \Omega_n$, and $\theta(G) = \overline{G\circ\rho_\sharp}$ because $\hat\rho(dA)(p) = (dA)(\rho_* p)$ and $\rho^{-1} = \rho$. By Corollary 3.3, $\langle \overline{G\circ\rho_\sharp}\, G\rangle_{\Omega_n} \ge 0$. The integrand is a bounded continuous function of finitely many curvatures, so (b) gives $\omega_\varepsilon(\overline{G\circ\rho_\sharp} G) \ge 0$. A bounded Borel function of $F(p_1), \dots, F(p_k)$ is an $L^2(\omega_\varepsilon)$-limit of such $\phi$, and $G \mapsto \omega_\varepsilon(\overline{G\circ\rho_\sharp} G)$ is continuous in $L^2(\omega_\varepsilon)$ because $\omega_\varepsilon$ is $\rho_\sharp$-invariant, by (c).

(f) This is Proposition 2.14(c) together with (b). □

## 5. Continuum limit points of the curvature field

5.1. **Test forms, laws and positivity.** Let $\mathcal{T} = \mathcal{S}(\mathbb{R}^3;\Lambda^2)$ be the space of real Schwartz two-forms $f = \sum_{\mu<\nu} f_{\mu\nu}\,dx^\mu \wedge dx^\nu$, with $f(x)(u,v) = \sum_{\mu<\nu} f_{\mu\nu}(x)(u_\mu v_\nu - u_\nu v_\mu)$ and

$$\langle f, h\rangle = \sum_{\mu<\nu} \int_{\mathbb{R}^3} f_{\mu\nu} h_{\mu\nu}\,dx, \qquad \|f\|_2^2 = \langle f, f\rangle.$$

Let $E(3) = \mathbb{R}^3 \rtimes O(3)$ act on $\mathbb{R}^3$ by $gx = Rx + a$. The pull-back $g^*f$ is $(g^*f)(x)(u,v) = f(gx)(Ru, Rv)$, and the push-forward is $g \cdot f := (g^{-1})^* f$; both preserve $\|\cdot\|_2$. Let $X = \mathcal{T}'$ be the space of continuous linear functionals on $\mathcal{T}$, with the $\sigma$-algebra $\mathcal{B}$ generated by the maps $F \mapsto F(f)$, $f \in \mathcal{T}$. A *law* is a probability measure on $(X, \mathcal{B})$. The group acts on $X$ by $(g \cdot F)(f) := F(g^* f)$, which is the push-forward of forms when $F$ is a smooth form, and on functions on $X$ by $\alpha_g \Phi := \Phi \circ g^{-1}$; thus $\alpha_g e^{iF(f)} = e^{iF(g\cdot f)}$. A law $m$ is $g$-invariant if $m(\alpha_g \Phi) = m(\Phi)$ for bounded measurable $\Phi$. We write $\tau_a$ for the translation $x \mapsto x + a$ and $\alpha_a = \alpha_{\tau_a}$. For an open set $U \subset \mathbb{R}^3$, $\Sigma(U)$ is the $\sigma$-algebra generated by the $F(f)$ with $f \in \mathcal{T}$ of compact support in $U$.

For a unit vector $\hat{n}$ and $c \in \mathbb{R}$ let $\pi = \{\hat{n}\cdot x = c\}$, $\pi^+ = \{\hat{n}\cdot x > c\}$, and let $r_\pi \in E(3)$ be the reflection in $\pi$. For $n \in \mathcal{N}$ we write $\pi_{n,c} = \{n \cdot x = c\}$, as before.

**Definition 5.1.** Let $m$ be a law.

(a) $m$ satisfies *Gaussian domination* (G) if $m(e^{F(f)}) \le e^{\frac{1}{2}\|f\|_2^2}$ for all $f \in \mathcal{T}$.

(b) $m$ is *carried by closed forms* if $F(d^*w) = 0$ $m$-almost surely for every $w \in \mathcal{S}(\mathbb{R}^3)$. Here $w$ is identified with the three-form $w\,dx^1 \wedge dx^2 \wedge dx^3$, and $d^*w \in \mathcal{T}$ is defined by $\langle d^*w, f\rangle = \int w\,(\partial_1 f_{23} - \partial_2 f_{13} + \partial_3 f_{12})\,dx$, that is, $(d^*w)_{23} = -\partial_1 w$, $(d^*w)_{13} = \partial_2 w$, $(d^*w)_{12} = -\partial_3 w$.

(c) $m$ is *reflection positive* at the plane $\pi$ if it is $r_\pi$-invariant and $m\big(\overline{\Phi \circ r_\pi}\,\Phi\big) \ge 0$ for every bounded $\Sigma(\pi^+)$-measurable $\Phi$. We write $\Theta_\pi \Phi := \overline{\Phi \circ r_\pi}$.

If $m$ is $r_\pi$-invariant, $\Phi \mapsto m(\Theta_\pi \Phi\,\Phi)$ is continuous on $L^2(\Sigma(\pi^+), m)$. Reflection positivity then extends to all $\Phi \in L^2(\Sigma(\pi^+), m)$ and, by Lemma B.2, it suffices to check it on bounded continuous cylinder functions $\phi(F(f_1), \dots, F(f_k))$ with $f_i$ of compact support in $\pi^+$.

*Hermite–Sobolev spaces.* Let $h_k$, $k \in \mathbb{N}$, be the real Hermite functions on $\mathbb{R}$, the eigenfunctions of $-\frac{d^2}{dx^2} + x^2$ with eigenvalues $2k+1$. For $\alpha \in \mathbb{N}^3$ and $\mu < \nu$, the forms $h_{\alpha,\mu\nu} = h_{\alpha_1} \otimes h_{\alpha_2} \otimes h_{\alpha_3}\,dx^\mu \wedge dx^\nu$ form an orthonormal basis of $L^2(\mathbb{R}^3;\Lambda^2)$ of eigenvectors of $-\Delta + |x|^2$, with eigenvalues $\lambda_\alpha = 2|\alpha| + 3$. For $s \in \mathbb{R}$ and $F \in X$ put

$$\|F\|_s^2 = \sum_{\alpha,\mu<\nu} \lambda_\alpha^s\, F(h_{\alpha,\mu\nu})^2 \in [0,\infty],$$

and $H^s = \{F : \|F\|_s < \infty\}$ for $s \le 0$; for $f \in \mathcal{T}$, $\|f\|_s$ is defined in the same way with $\langle h_{\alpha,\mu\nu}, f\rangle$. We use the following standard facts [RS, Sect. V.3]. The norms $\|\cdot\|_s$, $s \ge 0$, define the topology of $\mathcal{T}$, and $X = \bigcup_{r\ge 0} H^{-r}$. Each $H^{-r}$ is a separable Hilbert space, a measurable subset of $X$, and its Borel $\sigma$-algebra is the trace of $\mathcal{B}$. For $r' > r$ the embedding $H^{-r} \subset H^{-r'}$ is compact. Since $\#\{\alpha : \lambda_\alpha \le t\} \le Ct^3$, $\sum_\alpha \lambda_\alpha^{-t} < \infty$ for $t > 3$. Finally, $\|x^\beta \partial^\gamma f\|_{L^2} \le C\|f\|_{|\beta|+|\gamma|}$, so by the Sobolev embedding $H^2(\mathbb{R}^3) \subset C_b(\mathbb{R}^3)$,

$$\sup_x (1 + |x|^2)\,|f(x)| \le C\|f\|_4 \qquad (f \in \mathcal{T}). \tag{5.1}$$

5.2. **The lattice field in the continuum.** For a continuous two-form $f$ on $\mathbb{R}^3$ and $\varepsilon > 0$ define the 2-cochain $\iota_\varepsilon f$ by

$$(\iota_\varepsilon f)(p) = f(c_p)(u_p, v_p),$$

where $c_p$ is the centre and $(u_p, v_p)$ are the edge vectors of the oriented plaquette $p$. Thus $(\iota_\varepsilon f)(x;\mu,\nu) = f_{\mu\nu}(x + \frac{\varepsilon}{2}(e_\mu + e_\nu))$, and $\iota_\varepsilon f(-p) = -\iota_\varepsilon f(p)$. Similarly, for $w \in \mathcal{S}(\mathbb{R}^3)$ let $\iota'_\varepsilon w$ be the 3-cochain with value $w(c_c)$ on each positively oriented cube $c$ with centre $c_c$.

**Lemma 5.2.** *Let $f \in \mathcal{T}$ and $w \in \mathcal{S}(\mathbb{R}^3)$.*

(a) $\sum_p |\iota_\varepsilon f(p)| < \infty$. *Moreover $\|\iota_\varepsilon f\|_{2,\varepsilon} \le C\|f\|_4$ for $0 < \varepsilon \le 1$, with $C$ independent of $\varepsilon$ and $f$, and $\|\iota_\varepsilon f\|_{2,\varepsilon} \to \|f\|_2$ as $\varepsilon \to 0$.*

(b) *If $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$, then $\iota_\varepsilon(\sigma^* f)(q) = \iota_\varepsilon f(\sigma_* q)$ for every oriented plaquette $q$.*

(c) *$\|\iota_\varepsilon(d^* w) - d^*_\varepsilon \iota'_\varepsilon w\|_{2,\varepsilon} \to 0$ as $\varepsilon \to 0$, where $d^*_\varepsilon$ is the $\langle\cdot,\cdot\rangle_\varepsilon$-adjoint of the coboundary from 2-cochains to 3-cochains.*

*Proof.* (a) By (5.1), $|\iota_\varepsilon f(p)| \le C\|f\|_4(1 + |c_p|^2)^{-1}$, and $f$ decays faster than any power, so the first assertion holds. Also $\|\iota_\varepsilon f\|^2_{2,\varepsilon} \le C^2\|f\|^2_4\, \varepsilon^3 \sum_p (1 + |c_p|^2)^{-2}$, and the last sum is a Riemann sum of the integrable function $3(1 + |x|^2)^{-2}$ over three shifted copies of $\varepsilon\mathbb{Z}^3$, bounded uniformly for $\varepsilon \le 1$. The convergence $\|\iota_\varepsilon f\|^2_{2,\varepsilon} = \sum_{\mu<\nu} \varepsilon^3 \sum_x |f_{\mu\nu}(x + \frac{\varepsilon}{2}(e_\mu + e_\nu))|^2 \to \sum_{\mu<\nu} \int |f_{\mu\nu}|^2$ holds for Riemann sums of continuous functions dominated by $C(1 + |x|^2)^{-2}$. (b) Write $\sigma x = Rx + a$. Then $R$ is a signed permutation matrix, $\sigma_* q$ has centre $\sigma c_q$ and edge vectors $(Ru_q, Rv_q)$. Hence $\iota_\varepsilon f(\sigma_* q) = f(\sigma c_q)(Ru_q, Rv_q) = (\sigma^* f)(c_q)(u_q, v_q)$. (c) The lattice coboundary on 2-cochains has the same incidence signs as the exterior derivative on cubes. Hence, for the plaquette $p = (x;\mu,\nu)$ with $\{\mu,\nu,k\} = \{1,2,3\}$, $(d^*_\varepsilon \iota'_\varepsilon w)(p) = \pm\varepsilon^{-1}\big(w(c_p + \frac{\varepsilon}{2}e_k) - w(c_p - \frac{\varepsilon}{2}e_k)\big)$ and $(d^* w)_{\mu\nu}(c_p) = \pm\partial_k w(c_p)$, with the same sign. By Taylor's theorem the difference is bounded by $\varepsilon^2 \sup_{|t|\le\varepsilon} |\partial^3_k w(c_p + te_k)|$. Summing the squares with weight $\varepsilon^3$ gives $O(\varepsilon^4)$. □

Let $\omega_\varepsilon$ be the infinite-volume law of Theorem 4.3. By Theorem 4.3(d), $\omega_\varepsilon|F(p)| \le \varepsilon^{-3/2}$, so the event $E^0 = \{\sum_p (1 + |c_p|)^{-4}|F(p)| < \infty\}$ has full measure. On $E^0$ the formula

$$F_\varepsilon(f) := F(\iota_\varepsilon f) = \varepsilon^3 \sum_p (\iota_\varepsilon f)(p)\, F(p) \qquad (f \in \mathcal{T}) \tag{5.2}$$

converges absolutely and defines an element of $X$, by (5.1) applied to $(1 + |x|^2)f$. On the complement put $F_\varepsilon = 0$. Let $\mu_\varepsilon$ be the law of $F_\varepsilon$ under $\omega_\varepsilon$.

**Lemma 5.3.** *Let $\varepsilon > 0$, and let $F_\varepsilon$ and $\mu_\varepsilon$ be as above.*

(a) *For $f \in \mathcal{T}$: $\omega_\varepsilon(e^{F_\varepsilon(f)}) \le e^{\frac{1}{2}\|\iota_\varepsilon f\|^2_{2,\varepsilon}}$ and $\omega_\varepsilon(F_\varepsilon(f)^2) \le \|\iota_\varepsilon f\|^2_{2,\varepsilon}$.*

(b) *For $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$, regarded as an element of $E(3)$, $\sigma \cdot F_\varepsilon = (\sigma_\sharp F)_\varepsilon$ on $E^0 \cap \sigma_\sharp^{-1} E^0$. Hence $\mu_\varepsilon$ is $\sigma$-invariant.*

*Proof.* (a) is Theorem 4.3(d) with $g = \iota_\varepsilon f$, which is absolutely summable by Lemma 5.2(a). For (b), by Lemma 5.2(b) and the substitution $p = \sigma_* q$,

$$(\sigma \cdot F_\varepsilon)(f) = F_\varepsilon(\sigma^* f) = \varepsilon^3 \sum_q \iota_\varepsilon f(\sigma_* q) F(q) = \varepsilon^3 \sum_p \iota_\varepsilon f(p) F(\sigma_*^{-1} p) = (\sigma_\sharp F)_\varepsilon(f).$$

Since $\omega_\varepsilon$ is $\sigma_\sharp$-invariant (Theorem 4.3(c)), the law of $\sigma \cdot F_\varepsilon$ is that of $F_\varepsilon$. □

### 5.3. **Tightness and the properties of limit points.**

**Theorem 5.4.** *Let $L \ge 2$, $\varepsilon_K = L^{-K}$, and for each $K$ let $\omega_K := \omega_{\varepsilon_K}$ be the law of Theorem 4.3 for arbitrary parameters $\mu_{0,K} > 0$, $e_K$, $\lambda_K > 0$, $m^2_K$, $\delta m^2_K$. Let $\mu_K := \mu_{\varepsilon_K}$.*

(a) *For every $r > 7$, $\mu_K(H^{-r}) = 1$ and the sequence $(\mu_K)$ is tight on $H^{-r}$.*

(b) *Let $\mu$ be the limit, in the weak topology of laws on $H^{-r}$, of a subsequence of $(\mu_K)$, regarded as a law on $X$. Then $\mu$ satisfies (G); for real $f_1, \dots, f_m \in L^2(\mathbb{R}^3; \Lambda^2)$,*

$$|\mu(F(f_1) \cdots F(f_m))| \le 2 \cdot 2^{m/2} (m!)^{1/2} \prod_{i=1}^{m} \|f_i\|_2,$$

*with $F(f)$ for $f \in L^2$ as in Lemma B.1; $\mu$ is carried by closed forms; $\mu$ is invariant under $\mathbb{R}^3 \rtimes O_h$; and $\mu$ is reflection positive at every plane $\pi_{n,c}$ with $n \in \mathcal{N}$ and $c \in \mathbb{R}$.*

*Proof.* (a) Fix $r_0 \in (7, r)$. By Lemma 5.3(a) and Lemma 5.2(a), for $\varepsilon \le 1$,

$$\int \|F\|_{-r_0}^2 \, d\mu_\varepsilon = \sum_{\alpha, \mu < \nu} \lambda_\alpha^{-r_0} \, \omega_\varepsilon\big(F_\varepsilon(h_{\alpha,\mu\nu})^2\big) \le C^2 \sum_{\alpha,\mu<\nu} \lambda_\alpha^{-r_0} \|h_{\alpha,\mu\nu}\|_4^2 = 3C^2 \sum_\alpha \lambda_\alpha^{4-r_0} < \infty,$$

since $r_0 - 4 > 3$. Hence $\mu_K(H^{-r_0}) = 1$, and by Chebyshev's inequality $\mu_K(\|F\|_{-r_0} > M) \le C' M^{-2}$ uniformly in $K$. Closed balls of $H^{-r_0}$ are compact in $H^{-r}$.

(b) Write $\mu_K \to \mu$ along the subsequence. *(G).* For $f \in \mathcal{T}$ the function $F \mapsto e^{F(f)}$ is continuous and nonnegative on $H^{-r}$. Hence $\mu(e^{F(f)}) \le \liminf_K \mu_K(e^{F(f)}) \le \liminf_K e^{\frac{1}{2}\|\iota_{\varepsilon_K} f\|^2_{2,\varepsilon_K}} = e^{\frac{1}{2}\|f\|_2^2}$, by Lemma 5.3(a) and Lemma 5.2(a). The moment bound follows from (G) by Lemma B.1.

*Closedness.* Let $w \in \mathcal{S}(\mathbb{R}^3)$. The lattice Bianchi identity $dF = 0$ (Theorem 4.3(b)) gives $\langle d_\varepsilon^* \iota'_\varepsilon w, F\rangle_\varepsilon = \langle \iota'_\varepsilon w, dF\rangle_\varepsilon = 0$, the sums being absolutely convergent. Hence

$$F_\varepsilon(d^* w) = \langle \iota_\varepsilon d^* w - d_\varepsilon^* \iota'_\varepsilon w, F\rangle_\varepsilon,$$

and by Theorem 4.3(d) and Lemma 5.2(c), $\omega_\varepsilon(F_\varepsilon(d^*w)^2) \to 0$. Since $F \mapsto \min(F(d^*w)^2, 1)$ is bounded and continuous on $H^{-r}$, $\mu(\min(F(d^*w)^2, 1)) = 0$.

*Invariance.* Let $\chi_K(f) = \mu_K(e^{iF(f)})$ and $\chi(f) = \mu(e^{iF(f)})$, so that $\chi_K \to \chi$ pointwise on $\mathcal{T}$. If $\sigma \in \varepsilon_{K_0}\mathbb{Z}^3 \rtimes O_h$, then $\sigma \in \varepsilon_K \mathbb{Z}^3 \rtimes O_h$ for $K \ge K_0$, because $\varepsilon_{K_0} = L^{K-K_0}\varepsilon_K$. By Lemma 5.3(b), $\chi_K(\sigma^* f) = \chi_K(f)$ for $K \ge K_0$, hence $\chi(\sigma^* f) = \chi(f)$. This holds for $\sigma$ in the group generated by $O_h$ and the translations by $D = \bigcup_K \varepsilon_K \mathbb{Z}^3$. For $a \in \mathbb{R}^3$ choose $a_k \in D$ with $a_k \to a$. By (G) and Lemma B.1,

$$|\chi(\tau_{a_k}^* f) - \chi(\tau_a^* f)| \le \mu|F(\tau_{a_k}^* f - \tau_a^* f)| \le \|\tau_{a_k}^* f - \tau_a^* f\|_2 \to 0,$$

so $\chi$ is invariant under $\mathbb{R}^3 \rtimes O_h$. The characteristic functional determines a law on $X$, because it determines the finite-dimensional distributions of $(F(f_1), \dots, F(f_k))$. Hence $\mu$ is invariant under $\mathbb{R}^3 \rtimes O_h$.

*Reflection positivity.* Let $n \in \mathcal{N}$ and $c \in \bigcup_K \varepsilon_K \mathbb{Z}$, and let $r_\pi$ be the reflection in $\pi = \pi_{n,c}$. By the invariance just proved, $\mu$ is $r_\pi$-invariant. Let $\Phi = \phi(F(f_1), \dots, F(f_k))$ with $\phi \in C_b(\mathbb{R}^k)$ and $f_i \in \mathcal{T}$ of compact support in $\pi^+$. For $K$ large, $c \in \varepsilon_K \mathbb{Z}$ and $\varepsilon_K$ is smaller than the distance from $\bigcup \operatorname{supp} f_i$ to $\pi$. Then every plaquette $p$ with $\iota_{\varepsilon_K} f_i(p) \ne 0$ has its centre in $\operatorname{supp} f_i$ and its corners within $\varepsilon_K$ of the centre, so it lies in $\{n \cdot x \ge c\}$. There are finitely many such plaquettes. Thus $\Phi(F_{\varepsilon_K}) = G(F)$ for a bounded Borel function $G$ of finitely many $F(p)$ with $p$ in $\{n \cdot x \ge c\}$. Moreover $G(\rho_\sharp F) = \Phi((\rho_\sharp F)_{\varepsilon_K}) = \Phi(r_\pi \cdot F_{\varepsilon_K})$ by Lemma 5.3(b), where $\rho = \rho_{n,c} = r_\pi$. Theorem 4.3(e) gives $\mu_K(\overline{\Phi \circ r_\pi}\, \Phi) \ge 0$ for $K$ large. The function $\overline{\Phi \circ r_\pi}\, \Phi = \overline{\phi(F(r_\pi^* f_1), \dots)}\, \phi(F(f_1), \dots)$ is bounded and continuous on $H^{-r}$, so $\mu(\overline{\Phi \circ r_\pi}\, \Phi) \ge 0$. By density (Lemma B.2) and $r_\pi$-invariance, $\mu$ is reflection positive at $\pi_{n,c}$. For arbitrary $c \in \mathbb{R}$, let $b = c\, n/|n|^2$. Then $\tau_b$ maps $\pi_{n,0}$ onto $\pi_{n,c}$ and conjugates the two reflections, and $\alpha_b$ maps $\Sigma(\pi_{n,0}^+)$-measurable functions to $\Sigma(\pi_{n,c}^+)$-measurable functions. Since $0 \in \bigcup_K \varepsilon_K \mathbb{Z}$ and $\mu$ is translation invariant, positivity at $\pi_{n,c}$ follows from positivity at $\pi_{n,0}$. □

*Remark* 5.5. The moment bound of Theorem 5.4(b) is a bound of the form [OS75, (E0"), (4.2), p. 287] for product test functions, with $\sigma_m = 2 \cdot 2^{m/2}(m!)^{1/2}$ of factorial growth and the norm $\|\cdot\|_2$. Here $\mathbb{R}^4$ is replaced by $\mathbb{R}^3$, and the fields carry tensor indices.

## 6. The spectral cone and full rotation invariance

6.1. **Reconstruction.** Let $m$ be a law which is invariant under translations and under the reflection $r_n$ in the plane $\{n \cdot x = 0\}$, for a unit vector $n$, and which is reflection positive at this plane. Proposition B.3 constructs the Osterwalder–Schrader reconstruction: a Hilbert space $\mathcal{H}_n$, a contraction $[\cdot]$ from $L^2(\Sigma(\{n \cdot x > 0\}), m)$ onto a dense subspace, the vacuum $\psi_0 = [1]$, a self-adjoint $H_n \geq 0$ with $e^{-tH_n}[\Phi] = [\alpha_{tn}\Phi]$, a strongly continuous unitary representation $V_n$ of $n^\perp$ with $V_n(y)[\Phi] = [\alpha_y \Phi]$, and the joint spectral measure $\mathsf{E}_n$ of $H_n$ and the generators of $V_n$ on $[0, \infty) \times \mathbb{R}^2$. These objects are determined by $m$ and $n$; the construction goes back to [OS73], see [GJ, Sect. 6.1]. For $n = e_j$ we use the coordinates $p = (p_i)_{i \neq j}$ of $\mathbb{R}^2 \cong e_j^\perp$ and write $\mathsf{E}_j$, $H_j$.

6.2. **Two results from a recent preprint.** We quote two results of a recent preprint [S]. In [S] a law is a probability measure on $\mathcal{S}'(\mathbb{R}^3; \Lambda^2)$ with the $\sigma$-algebra generated by the evaluations, (G) is [S, (1.2), p. 3], and "carried by $dF = 0$" is our closedness. The action of $E(3)$ on tests and distributions [S, Sect. 1.2, p. 2] is the one of Section 5.1; in [S, (18.1), p. 104] it is written for the Hodge dual $b = \star f$. Reflection positivity at a plane [S, (1.1), p. 2], together with invariance under the reflection, is Definition 5.1(c), and the reconstruction of [S, Sect. 17] is that of Proposition B.3. Throughout [S, Sect. 18] the law is assumed to satisfy (G), to be carried by closed forms, to be invariant under translations and under $O_h$, and to be reflection positive at the coordinate planes through the origin [S, p. 104].

**Theorem 6.1** ([S, Prop. 18.1, p. 104])**.** *Let $\mu$ be a law which satisfies (G), is carried by closed forms, is invariant under $\mathbb{R}^3 \rtimes O_h$, and is reflection positive at the planes $\{x_j = 0\}$, $j = 1, 2, 3$. Let $i \neq j$. If $\mu$ is reflection positive at the plane $\{x_j + x_i = 0\}$, then*

$$\mathsf{E}_j\big\{(\omega, p) : \omega < |p_i|\big\} = 0.$$

**Theorem 6.2** ([S, Thm. 18.21, p. 123], [S, Thm. 1.2, p. 4])**.** *Let $\mu$ be a law which satisfies (G), is carried by closed forms, is invariant under $\mathbb{R}^3 \rtimes O_h$, and is reflection positive at every plane*

$$x_j = c \quad (1 \leq j \leq 3), \qquad x_j - x_i = c, \qquad x_j + x_i = c \quad (1 \leq j < i \leq 3), \qquad c \in \mathbb{R}.$$

*Then $\mu$ is invariant under $E(3)$, all its moments on $L^2$ test forms obey the tensor transformation law*

$$\mu\big(F(g \cdot f_1) \cdots F(g \cdot f_m)\big) = \mu\big(F(f_1) \cdots F(f_m)\big) \qquad (g \in E(3),\ f_i \in L^2(\mathbb{R}^3; \Lambda^2)),$$

*and $\mu$ is reflection positive at every affine plane.*

The proofs in [S, Sect. 18] use auxiliary facts from [S, Sects. 15 and 17]: the consequences of (G), the density of observables supported in half-spaces, and the reconstruction. Appendix B states and proves the versions of these facts that enter the definitions of the objects in Theorems 6.1 and 6.2 and the arguments of Section 7.

6.3. **Application to the continuum limit points.**

**Corollary 6.3.** *Let $\mu$ be a law as in Theorem 5.4(b). Then:*

(a) $\mathsf{E}_j\{(\omega, p) : \omega < |p_i|\} = 0$ *for all $i \neq j$;*
(b) *$\mu$ is invariant under $E(3)$, its moments obey the tensor transformation law, and $\mu$ is reflection positive at every affine plane.*

*Proof.* By Theorem 5.4(b), $\mu$ satisfies (G), is carried by closed forms and is invariant under $\mathbb{R}^3 \rtimes O_h$. The planes $x_j = c$, $x_j - x_i = c$ and $x_j + x_i = c$ are the planes $\pi_{n,c}$ with $n = e_j$, $n = e_j - e_i$ and $n = e_j + e_i$; up to the sign of $n$, which does not change the plane, these normals belong to $\mathcal{N}$. So $\mu$ is reflection positive at all of them. Theorems 6.1 and 6.2 apply. □

## 7. Ergodic components and the cluster property

7.1. **Decomposition and inheritance.** Let $\mathcal{I} \subset \mathcal{B}$ be the $\sigma$-algebra of translation-invariant sets, $\mathcal{I} = \{B \in \mathcal{B} : \tau_a B = B \text{ for all } a \in \mathbb{R}^3\}$, where $\tau_a$ acts on $X$ as in Section 5.1. A translation-invariant law $m$ is ergodic if $m(B) \in \{0, 1\}$ for $B \in \mathcal{I}$.

**Theorem 7.1.** *Let $\mu$ be a law as in Theorem 5.4(b). There is a family $(\mu_\xi)_{\xi \in X}$ of laws such that $\xi \mapsto \mu_\xi(B)$ is $\mathcal{I}$-measurable for every $B \in \mathcal{B}$, $\mu(\Phi) = \int \mu_\xi(\Phi)\, \mu(d\xi)$ for bounded measurable $\Phi$, and for $\mu$-almost every $\xi$:*

(a) *$\mu_\xi$ is translation invariant and ergodic, and it is carried by closed forms;*
(b) *$\mu_\xi$ is invariant under $\mathbb{R}^3 \rtimes O_h$ and reflection positive at every plane $\pi_{n,c}$, $n \in \mathcal{N}$, $c \in \mathbb{R}$;*
(c) *$\mu_\xi$ is invariant under $E(3)$ and reflection positive at every affine plane.*

*Parts (a) and (b) use only Theorem 5.4; part (c) uses Corollary 6.3(b).*

*Proof.* The family is the one of Theorem A.4, which gives (a); the set of closed forms is the measurable set $\bigcap_{w \in W_0} \{F(d^* w) = 0\}$ for a countable dense $W_0 \subset \mathcal{S}(\mathbb{R}^3)$. Part (b) is Theorem A.7 with $N$ the set of the eighteen unit vectors $\pm n/|n|$, $n \in \mathcal{N}$, for which $G_N = \mathbb{R}^3 \rtimes O_h$. Part (c) is Theorem A.7 with $N = S^2$, applicable by Corollary 6.3(b). □

*Remark* 7.2. The decomposition does not transfer Gaussian domination or moment bounds to the components, and we make no such claim. The statements about the components in this section concern the measures $\mu_\xi$ and the Hilbert spaces $L^2(\mu_\xi)$ only.

7.2. **The spectral cone of the components.** Let $m$ be a translation-invariant law which is invariant and reflection positive at a plane $\{n \cdot x = 0\}$. We write $[\cdot]_m$, $\psi_0^m = [1]_m$, $H_n^m$, $V_n^m$ and $\mathsf{E}_n^m$ for the class map, the vacuum vector, the Hamiltonian, the representation of $n^\perp$ and the joint spectral measure of the reconstruction of $m$ at this plane (Proposition B.3). For $m = \mu$ we omit the index $m$, for $m = \mu_\xi$ we write $[\cdot]_\xi$, $\psi_0^\xi$ and $\mathsf{E}_n^\xi$, and the index $n$ is omitted when the plane is fixed. When no confusion is possible we write $H_n$, $V_n$ for the operators of $\mu_\xi$ as well.

**Lemma 7.3.** *Let $\mu$ be a translation-invariant law, $n$ a unit vector and $\pi = \{n \cdot x = 0\}$. Suppose that $\mu$ is invariant and reflection positive at $\pi$, and that the same holds for $\mu$-almost every $\mu_\xi$ in the decomposition of Theorem A.4. Let $\mathsf{E}$ and $\mathsf{E}^\xi$ be the joint spectral measures of the reconstructions of $\mu$ and of $\mu_\xi$ at $\pi$. If $S \subset [0, \infty) \times \mathbb{R}^2$ is a Borel set with $\mathsf{E}(S) = 0$, then $\mathsf{E}^\xi(S) = 0$ for $\mu$-almost every $\xi$.*

*Proof.* Let $\mathcal{A}_+$ be the countable algebra $\mathcal{A}(D_{\pi^+})$ of Lemma A.1(b); it is dense in $L^2(\Sigma(\pi^+), m)$ for every law $m$. Fix $\Phi \in \mathcal{A}_+$. For a law $m$ which is invariant and reflection positive at $\pi$, let $\varrho_\Phi^m$ be the spectral measure of $[\Phi]$ for $\mathsf{E}^m$, a finite positive measure on $[0, \infty) \times \mathbb{R}^2$ with total mass $\|[\Phi]\|^2 \le \|\Phi\|_\infty^2$. On the $\mu$-null set of $\xi$ for which $\mu_\xi$ is not invariant and reflection positive at $\pi$ we put $\varrho_\Phi^{\mu_\xi} := 0$. By Proposition B.3,

$$m\big(\overline{\Phi \circ r_\pi}\; \alpha_{tn+y}\Phi\big) = \langle [\Phi], e^{-tH} V(y) [\Phi] \rangle = \int e^{-t\omega + ip\cdot y}\, \varrho_\Phi^m(d\omega, dp) \qquad (t \ge 0,\ y \in n^\perp). \tag{7.1}$$

Let $\mathcal{M}$ be the set of bounded Borel functions $h$ on $[0, \infty) \times \mathbb{R}^2$ for which $\xi \mapsto \int h\, d\varrho_\Phi^{\mu_\xi}$ is measurable and $\int \big( \int h\, d\varrho_\Phi^{\mu_\xi} \big) \mu(d\xi) = \int h\, d\varrho_\Phi^{\mu}$. By (7.1) and the decomposition, $\mathcal{M}$ contains the functions $e^{-t\omega + ip\cdot y}$ with $t > 0$, $y \in n^\perp$. It contains the constant 1, by letting $t \downarrow 0$ at $y = 0$

and using dominated convergence on both sides. The set $\mathcal{M}$ is a vector space, closed under complex conjugation and, by monotone convergence, under bounded monotone limits of real functions. The real and imaginary parts of the functions $e^{-t\omega+ip\cdot y}$, $t>0$, together with 1, span an algebra of bounded functions which generates the Borel $\sigma$-algebra of $[0,\infty)\times\mathbb{R}^2$. By the monotone class theorem $\mathcal{M}$ contains all bounded Borel functions, in particular $\mathbf{1}_S$. Hence $\int \varrho_\Phi^{\mu_\xi}(S)\,\mu(d\xi) = \varrho_\Phi^{\mu}(S) = \|\mathsf{E}(S)[\Phi]\|^2 = 0$, and $\|\mathsf{E}^\xi(S)[\Phi]_\xi\|^2 = \varrho_\Phi^{\mu_\xi}(S) = 0$ for almost every $\xi$. Since $\mathcal{A}_+$ is countable, for almost every $\xi$ this holds for all $\Phi\in\mathcal{A}_+$. The classes $[\Phi]_\xi$, $\Phi\in\mathcal{A}_+$, are dense in the reconstruction space of $\mu_\xi$, so $\mathsf{E}^\xi(S)=0$. □

### 7.3. **Uniqueness of the vacuum and clustering.**

**Theorem 7.4.** *Let $\mu$ be a law as in Theorem 5.4(b) and $(\mu_\xi)$ its decomposition. For $\mu$-almost every $\xi$:*

(a) *for $j=1,2,3$ the reconstruction of $\mu_\xi$ at $\{x_j=0\}$ satisfies* $\operatorname{Ker} H_j = \mathbb{C}\psi_0^\xi$*;*
(b) *for all $\Phi,\Psi\in L^2(\mu_\xi)$,*

$$\lim_{|a|\to\infty} \mu_\xi\big(\Phi\,\alpha_a\Psi\big) = \mu_\xi(\Phi)\,\mu_\xi(\Psi);$$

(c) *for every unit vector $n$ the reconstruction of $\mu_\xi$ at $\{n\cdot x=0\}$ satisfies* $\operatorname{Ker} H_n = \mathbb{C}\psi_0^\xi$.

*Parts (a) and (b) use Theorem 6.1 but not Theorem 6.2; part (c) uses Theorem 6.2.*

Parts (a)–(c) are the uniqueness of the vacuum and the cluster property [OS73, (E4)] (axiom OS4 in [GJ, Sect. 6.1]) for the components, stated for the measures $\mu_\xi$.

*Proof.* (a) By Theorem 7.1(b), almost every $\mu_\xi$ is invariant and reflection positive at $\{x_j=0\}$. By Corollary 6.3(a) and Lemma 7.3, applied to the countably many sets $\{\omega<|p_i|\}$, $i\neq j$, almost every $\mu_\xi$ satisfies $\mathsf{E}_j^\xi\{\omega<|p_i|\}=0$ for all $i\neq j$. Since $\mu_\xi$ is ergodic, Proposition B.4(b) gives $\operatorname{Ker} H_j = \mathbb{C}\psi_0^\xi$.

(b) Fix $\xi$ as in (a) and write $m=\mu_\xi$. First let $\Phi,\Psi$ be bounded and $\Sigma(B_R)$-measurable, where $B_R$ is the open ball of radius $R$ about 0. Let $a\in\mathbb{R}^3$ with $a_j=\max_k|a_k|\geq 4R$ for some $j$, and write $a=a_je_j+y$ with $y\perp e_j$. Put $\Phi_1:=\Theta_j\alpha_{-2Re_j}\Phi$ and $\Psi_1:=\alpha_{2Re_j}\Psi$, where $\Theta_j$ is the reflection operator of $\{x_j=0\}$. Both are bounded and $\Sigma(\{x_j>0\})$-measurable, and $\Theta_j\Phi_1=\alpha_{-2Re_j}\Phi$. By translation invariance and Proposition B.3,

$$m(\Phi\,\alpha_a\Psi) = m\big(\alpha_{-2Re_j}\Phi\cdot\alpha_{a-2Re_j}\Psi\big) = m\big(\Theta_j\Phi_1\cdot\alpha_{(a_j-4R)e_j+y}\Psi_1\big) = \int e^{-(a_j-4R)\omega+ip\cdot y}\,\varrho(d\omega,dp),$$

where $\varrho=\langle[\Phi_1],\mathsf{E}_j(\cdot)[\Psi_1]\rangle$ is a complex measure of total variation at most $\|\Phi\|_2\|\Psi\|_2$. By (a), $\mathsf{E}_j(\{0\}\times\mathbb{R}^2)=\mathbf{1}_{\{0\}}(H_j)$ is the orthogonal projection onto $\mathbb{C}\psi_0$, and $V_j(y)\psi_0=\psi_0$, so the spectral measure of $\psi_0$ is the point mass at $(0,0)$. Hence the restriction of $\varrho$ to $\{\omega=0\}$ is $\langle[\Phi_1],\psi_0\rangle\langle\psi_0,[\Psi_1]\rangle\delta_{(0,0)}$, and

$$\Big|m(\Phi\,\alpha_a\Psi) - \langle[\Phi_1],\psi_0\rangle\langle\psi_0,[\Psi_1]\rangle\Big| \leq \int_{\{\omega>0\}} e^{-(a_j-4R)\omega}\,|\varrho|(d\omega,dp) \longrightarrow 0$$

as $a_j\to\infty$, uniformly in $y$, by dominated convergence. Moreover $\langle[\Phi_1],\psi_0\rangle = m(\Theta_j\Phi_1)=m(\Phi)$ and $\langle\psi_0,[\Psi_1]\rangle = m(\Psi_1)=m(\Psi)$. If instead $-a_j=\max_k|a_k|$, then $m(\Phi\,\alpha_a\Psi)=m(\Psi\,\alpha_{-a}\Phi)$ by translation invariance, and the previous case applies. Every sequence with $|a|\to\infty$ splits into finitely many subsequences on which the index $j$ and the sign are fixed, and $\max_k|a_k|\geq|a|/\sqrt{3}$. This proves (b) for such $\Phi,\Psi$. In general, such functions are dense in $L^2(m)$ by Lemma B.2, and $|m(\Phi\,\alpha_a\Psi)|\leq\|\Phi\|_2\|\Psi\|_2$.

(c) Fix $\xi$ as in (b) for which, in addition, Theorem 7.1(c) holds. Then $m = \mu_\xi$ is invariant and reflection positive at $\{n \cdot x = 0\}$, and its reconstruction $\mathcal{H}_n$ is defined. For bounded $\Sigma(\{n \cdot x > 0\})$-measurable $\Phi, \Psi$, by (b) with $a = tn$,

$$\langle [\Phi], e^{-tH_n}[\Psi]\rangle = m(\Theta_n\Phi \cdot \alpha_{tn}\Psi) \longrightarrow m(\Theta_n\Phi)\, m(\Psi) = \langle [\Phi], \psi_0\rangle\langle \psi_0, [\Psi]\rangle \qquad (t \to \infty).$$

On the other hand $e^{-tH_n} \to \mathbf{1}_{\{0\}}(H_n)$ strongly. The classes of bounded functions are dense in $\mathcal{H}_n$, so $\mathbf{1}_{\{0\}}(H_n)$ is the orthogonal projection onto $\mathbb{C}\psi_0$, that is, $\operatorname{Ker} H_n = \mathbb{C}\psi_0$. □

7.4. **Proof of Theorem 1.1.** Part (a) is Theorem 4.3, with Proposition 2.14(c) for the identification with the Neumann/Dirichlet limit. Part (b) is Theorem 5.4. Part (c) is Corollary 6.3(b). Part (d) is Theorem 7.1 together with Theorem 7.4. □

## 8. A rotation criterion for scalar two-point functions

The Wick square $|\varphi|^2$, suitably renormalised, is the second natural gauge-invariant field of the model. We do not construct its continuum limit for the auxiliary states in the volume-first order (Section 10). This section gives an analytic criterion for rotation invariance of a scalar two-point function. It needs neither moment bounds of all orders nor the arguments of [S, Sect. 18]: cubic symmetry, a bound on the singularity at coinciding points, and positivity of spectral representations in an axis frame and a diagonal frame suffice. Such a criterion applies to the two-point function of the Wick square once that field is constructed with these properties.

Throughout, for $1 \le i < j \le 3$ let $k$ be the third index, and for an even real $\chi \in C_c^\infty(\mathbb{R})$ and a continuous function $G$ on $\mathbb{R}^3 \setminus \{0\}$ of polynomial growth at infinity put

$$G_\chi^{ij}(y) := \int_{\mathbb{R}}\int_{\mathbb{R}} G\big(y_1 e_i + y_2 e_j + (s-s')e_k\big)\, \chi(s)\chi(s')\, ds\, ds' \qquad (y \in \mathbb{R}^2 \setminus \{0\}). \tag{8.1}$$

**Proposition 8.1.** *Let $G : \mathbb{R}^3 \setminus \{0\} \to \mathbb{R}$ be continuous and suppose:*

(i) *$G \circ R = G$ for every $R \in O_h$;*

(ii) *there are $C, M$ and $a < 9$ with $|G(x)| \le C|x|^{-a}$ for $0 < |x| \le 1$ and $|G(x)| \le C|x|^M$ for $|x| \ge 1$;*

(iii) *for every $i < j$ and every even real $\chi \in C_c^\infty(\mathbb{R})$ there are positive Borel measures $\varrho, \varrho'$ on $\{(\omega, p) \in \mathbb{R}^2 : \omega \ge |p|\}$ such that*

$$G_\chi^{ij}(y) = \int e^{-\omega y_1 + ipy_2}\, \varrho(d\omega, dp) \quad (y_1 > 0), \qquad G_\chi^{ij}(y) = \int e^{-\omega y_+ + ipy_-}\, \varrho'(d\omega, dp) \quad (y_+ > 0),$$

*where $y_\pm = (y_1 \pm y_2)/\sqrt{2}$.*

*Then $G \circ R = G$ for every $R \in O(3)$.*

*Proof.* Increasing $a$ weakens (ii), so we may assume $1 < a < 9$, and we put $a' := a - 1 \in (0, 8)$. Fix $i < j$ and $\chi$, and write $G_\chi = G_\chi^{ij}$. Since $\chi$ is even, $(\chi * \chi)(t) = \int \chi(s)\chi(s-t)\, ds$, and $\chi * \chi$ is a bounded function of compact support; thus $G_\chi(y) = \int G(y_1 e_i + y_2 e_j + te_k)\, (\chi * \chi)(t)\, dt$. By (ii) and continuity, $|G_\chi(y)| \le C_\chi \int_{-T}^{T} (|y|^2 + t^2)^{-a/2} dt \le C'_\chi |y|^{-a'}$ for $0 < |y| \le 1$. Moreover $t \mapsto G_\chi(t, 0) = \int e^{-\omega t} d\varrho$ is nonincreasing on $(0, \infty)$, by (iii).

Fix $r > 0$ and let $h(\zeta) = G_\chi(r\cos\zeta, r\sin\zeta)$ for real $\zeta$. By (i), applied to the rotation by $\pi/2$ about $e_k$ and to the reflection $x_j \mapsto -x_j$, $h$ is $\pi/2$-periodic and even. For $\zeta = \alpha + i\beta$ with $|\alpha| < \pi/2$, the real part of $-\omega r\cos\zeta + ipr\sin\zeta$ is

$$-r\cos\alpha\,(\omega\cosh\beta + p\sinh\beta) \le -r\omega\cos\alpha\, e^{-|\beta|},$$

because $|p| \le \omega$ on the support of $\varrho$. Hence the first representation in (iii) converges absolutely and locally uniformly on the strip $|\alpha| < \pi/2$, and defines there a holomorphic extension $\tilde{h}$ of

$h$, with $|\tilde h(\alpha+i\beta)| \le G_\chi(r\cos\alpha\, e^{-|\beta|},0)$. For $|\alpha|\le\pi/4$ this gives $|\tilde h(\alpha+i\beta)|\le C_r e^{a'|\beta|}$ for all $\beta$, by the bound on $G_\chi$ near 0 and monotonicity. In the same way, since $y_+ = r\cos(\zeta-\pi/4)$ and $y_- = -r\sin(\zeta-\pi/4)$, the second representation gives a holomorphic extension on the strip $|\alpha-\pi/4|<\pi/2$. The two extensions agree on the real interval $(-\pi/4,\pi/2)$, hence on the intersection of the strips. By periodicity, the translates by $\frac{\pi}{2}\mathbb{Z}$ of the extension on $|\alpha|<\pi/2$ agree on overlaps and define an entire function, again denoted $\tilde h$, which is $\pi/2$-periodic and satisfies $|\tilde h(\alpha+i\beta)|\le C_r e^{a'|\beta|}$ for all $\alpha,\beta$.

Expand $\tilde h(\zeta)=\sum_{m\in\mathbb{Z}} c_m e^{4im\zeta}$ with $c_m=\frac{2}{\pi}\int_0^{\pi/2}\tilde h(\alpha+i\beta)e^{-4im(\alpha+i\beta)}d\alpha$, which does not depend on $\beta$. Then $|c_m|\le C_r e^{a'|\beta|+4m\beta}$ for all $\beta$. For $m\ge 2$ let $\beta\to-\infty$, and for $m\le -2$ let $\beta\to+\infty$; since $4|m|\ge 8>a'$, $c_m=0$. As $h$ is real and even, $c_{-1}=c_1\in\mathbb{R}$ and $c_0\in\mathbb{R}$, so $\tilde h(\zeta)=c_0+2c_1\cos 4\zeta$. At $\alpha=0$ the first representation gives

$$c_0+2c_1\cosh 4\beta=\tilde h(i\beta)=\int e^{-r(\omega\cosh\beta+p\sinh\beta)}\varrho(d\omega,dp)\ge 0,$$

and at $\alpha=\pi/4$ the second gives $c_0-2c_1\cosh 4\beta=\tilde h(\pi/4+i\beta)=\int e^{-r(\omega\cosh\beta-p\sinh\beta)}\varrho'(d\omega,dp)\ge 0$. Hence $2|c_1|\cosh 4\beta\le c_0$ for all $\beta$, so $c_1=0$, and $G_\chi$ is radial on $\mathbb{R}^2\setminus\{0\}$.

Thus $y\mapsto\int G(y_1e_i+y_2e_j+te_k)(\chi*\chi)(t)\,dt$ is radial for every even real $\chi\in C_c^\infty(\mathbb{R})$. Let $\eta_\delta\ge 0$ be an even mollifier and, for $s>0$, $\chi_\delta=\eta_\delta(\cdot-s/2)+\eta_\delta(\cdot+s/2)$. Then $\chi_\delta*\chi_\delta-2\eta_\delta*\eta_\delta=(\eta_\delta*\eta_\delta)(\cdot-s)+(\eta_\delta*\eta_\delta)(\cdot+s)$, which tends to $\delta_s+\delta_{-s}$ as $\delta\to 0$. By continuity of $G$ off the origin, $G(y_1e_i+y_2e_j+se_k)+G(y_1e_i+y_2e_j-se_k)=2G(y_1e_i+y_2e_j+se_k)$ is radial in $y\ne 0$; here we used (i) for the reflection $x_k\mapsto -x_k$. So $G$ is invariant under the rotations about the axis $e_k$. These rotations, for $k=1,2,3$, generate $SO(3)$, and $-I\in O_h$. □

**Proposition 8.2.** *Let $S_2$ be a symmetric bilinear form on real $C_c^\infty(\mathbb{R}^3)$ such that*

(i) *$S_2(h_1(\cdot-b),h_2(\cdot-b))=S_2(h_1,h_2)$ for $b\in\mathbb{R}^3$;*
(ii) *$|S_2(h_1,h_2)|\le C\,\mathcal{N}(h_1)\mathcal{N}(h_2)$ with $\mathcal{N}(h)=\max(\|h\|_{L^1},\|h\|_{L^\infty})$;*
(iii) *there is a continuous $G$ on $\mathbb{R}^3\setminus\{0\}$ with $G\circ R=G$ for all $R\in O(3)$ such that $S_2(h_1,h_2)=\iint h_1(x)h_2(y)G(x-y)\,dx\,dy$ whenever $\operatorname{supp}h_1\cap\operatorname{supp}h_2=\emptyset$.*

*Then $S_2(h_1\circ R^{-1},h_2\circ R^{-1})=S_2(h_1,h_2)$ for all $R\in O(3)$.*

*Proof.* For $R\in O(3)$ write $R_*h=h\circ R^{-1}$ and $D_R(h_1,h_2)=S_2(R_*h_1,R_*h_2)-S_2(h_1,h_2)$. Since $\mathcal{N}(R_*h)=\mathcal{N}(h)$ and $R_*$ commutes with translations up to $b\mapsto Rb$, the form $D_R$ is symmetric, translation invariant and bounded by $2C\mathcal{N}(h_1)\mathcal{N}(h_2)$. By (iii) and the invariance of $G$, it vanishes when the supports are disjoint. By the kernel theorem [Hö, Thm. 5.2.1] it has a distribution kernel on $\mathbb{R}^3\times\mathbb{R}^3$, supported on the diagonal by the last property and invariant under the diagonal translations. Hence $D_R(h_1,h_2)=\iint h_1(x)h_2(y)\,d_R(x-y)$ for a distribution $d_R$ supported at 0, that is $d_R=\sum_{|\beta|\le M}c_\beta\partial^\beta\delta$ [Hö, Thm. 2.3.4]. Since $D_R$ is real and symmetric, $d_R$ is real and even, so in Fourier variables $D_R(h_1,h_2)=(2\pi)^{-3}\int\overline{\hat h_1(\xi)}\hat h_2(\xi)P_R(\xi)\,d\xi$ with a real even polynomial $P_R$.

Fix a real $\phi\in C_c^\infty(\mathbb{R}^3)$, $\phi\ne 0$, and put $h_q(x)=\phi(x)\cos(q\cdot x)$, $q\in\mathbb{R}^3$. Then $\mathcal{N}(h_q)\le\mathcal{N}(\phi)$ and $\hat h_q=\frac12(\hat\phi(\cdot-q)+\hat\phi(\cdot+q))$. Since $\hat\phi$ is a Schwartz function and $\max(|\xi-q|,|\xi+q|)\ge|q|$, the cross terms in $|\hat h_q|^2$ contribute $o(1)$ as $|q|\to\infty$. Hence

$$D_R(h_q,h_q)=\frac14(2\pi)^{-3}\int|\hat\phi(\xi)|^2\big(P_R(\xi+q)+P_R(\xi-q)\big)\,d\xi+o(1).$$

The integral is a polynomial in $q$. If $P_R$ had degree $2l\ge 2$ with top-degree part $P_{2l}\ne 0$, its top-degree part would be $\frac12(2\pi)^{-3}\|\hat\phi\|_2^2P_{2l}(q)\ne 0$, and it would be unbounded. But $|D_R(h_q,h_q)|\le 2C\mathcal{N}(\phi)^2$, so $P_R$ is a constant $c(R)$ and $D_R(h_1,h_2)=c(R)\int h_1h_2$. From $D_{R_1R_2}(h_1,h_2)=$

$D_{R_1}(R_{2*}h_1, R_{2*}h_2) + D_{R_2}(h_1, h_2)$ we get $c(R_1R_2) = c(R_1) + c(R_2)$. Choosing $h$ with $\int h^2 \geq 1$ and $\mathcal{N}(h) \leq C''$ shows $|c(R)| \leq 2CC''^2$. A bounded homomorphism from a group into $(\mathbb{R}, +)$ vanishes, so $D_R = 0$. □

**Corollary 8.3.** *Let $S_2$ satisfy (i)–(iii) of Proposition 8.2, except that in (iii) the function $G$ is only assumed to satisfy (i)–(iii) of Proposition 8.1. Then $S_2$ is $O(3)$-invariant.*

*Proof.* By Proposition 8.1, $G$ is $O(3)$-invariant; apply Proposition 8.2. □

Bounds in terms of $\mathcal{N}(h)$ for Wick-square observables are of the type proved for the torus states of [v1] in [v1, Thm. 3.1(iii)]. No such bound is proved here for the auxiliary box states or their limits (Section 10, item 6). Hypothesis (iii) of Proposition 8.1 has the form of spectral representations at the planes $x_i = 0$ and $x_i + x_j = 0$ with the cone of Theorem 6.1. For a scalar field it would have to be derived from reflection positivity at those planes, which we do not do here.

**Example 8.4.** The condition $a < 9$ in Proposition 8.1 cannot be replaced by $a \leq 9$. Let $\alpha > 0$, let $s(q) = q_1^4 + q_2^4 + q_3^4$, and let

$$G(x) = \frac{1}{4\pi}\Big(\alpha + \big(\textstyle\sum_\mu \partial_\mu^4\big)^2\Big)^{\frac{1}{|x|}} \qquad (x \neq 0).$$

Then $G$ satisfies (i), (ii) with $a = 9$, and (iii), but $G$ is not $O(3)$-invariant. Indeed, $g(x) = (\sum_\mu \partial_\mu^4)^2|x|^{-1}$ is homogeneous of degree $-9$, invariant under $O_h$, and takes the values 124740, 280665/4 and 36960 at $e_1$, $(e_1+e_2)/\sqrt{2}$ and $(e_1+e_2+e_3)/\sqrt{3}$. This gives (i), (ii) and the failure of $O(3)$-invariance.

For (iii), take $(i, j, k) = (1, 2, 3)$; the other planes follow by symmetry. The function $G$ is the restriction to $\mathbb{R}^3 \setminus \{0\}$ of the tempered distribution with Fourier transform $P(q)/|q|^2$, $P = \alpha + s^2$, with the convention $\hat{u}(q) = \int e^{-iq\cdot x} u(x)\, dx$. For $q_\perp = (q_2, q_3) \neq 0$ let $w = |q_\perp|$. Since $P$ is even in $q_1$, $P(q) - P(iw, q_\perp)$ is divisible by $q_1^2 + w^2 = |q|^2$ in the ring of polynomials in $q$, and $P_0(q_\perp) := P(iw, q_\perp) = \alpha + (w^4 + q_2^4 + q_3^4)^2 \geq \alpha$ is a polynomial in $q_\perp$. Thus $P(q)/|q|^2 = P_1(q) + P_0(q_\perp)/|q|^2$ with a polynomial $P_1$, whose inverse Fourier transform is supported at 0. Computing the $q_1$-integral by residues, for $x_1 > 0$,

$$G(x) = (2\pi)^{-2} \int_{\mathbb{R}^2} e^{iq_\perp \cdot x_\perp}\, P_0(q_\perp)\, \frac{e^{-|q_\perp| x_1}}{2|q_\perp|}\, dq_\perp,$$

an absolutely convergent integral. Smearing in $x_3$ with $\chi * \chi$, whose Fourier transform is $\hat{\chi}^2 \geq 0$ since $\chi$ is even and real, gives the first representation in (iii). There $\varrho$ is the image of the positive measure $(2\pi)^{-2} P_0(q_\perp)\hat{\chi}(q_3)^2 (2|q_\perp|)^{-1} dq_\perp$ under $q_\perp \mapsto (|q_\perp|, q_2)$, which is carried by $\{\omega \geq |p|\}$. For the diagonal frame write $q_1 = (q_+ + q_-)/\sqrt{2}$, $q_2 = (q_+ - q_-)/\sqrt{2}$. Then $P$ is even in $q_+$, and with $w^2 = q_-^2 + q_3^2$,

$$s\big(\tfrac{iw+q_-}{\sqrt{2}}, \tfrac{iw-q_-}{\sqrt{2}}, q_3\big) = \tfrac{1}{2}w^4 - 3w^2 q_-^2 + \tfrac{1}{2}q_-^4 + q_3^4 \in \mathbb{R},$$

so the same computation gives $\varrho'$ with the positive density $\alpha + (\frac{1}{2}w^4 - 3w^2 q_-^2 + \frac{1}{2}q_-^4 + q_3^4)^2$ in place of $P_0$.

## 9. Comparison with the torus states: the two-point function

The companion paper [v1] constructs states of a closely related model on periodic cubic tori, with a counterterm scheme adapted to the tori, and takes the spacing and the torus to their limits jointly. By Remark 3.6, Theorem 3.2 does not apply to these tori. In this section we first show that, at a fixed spacing $\varepsilon$, the coefficient of the counterterm scheme of [v1] converges as the torus grows, to a limit $a_\natural(\varepsilon)$ given by the same graphs on $\varepsilon\mathbb{Z}^3$ (Proposition 9.2). We then

compare, at a fixed spacing, the infinite-volume limits of the torus states with the box states of Section 2 at the coefficient $a_\natural(\varepsilon)$. For all photon masses outside a countable set, the two have the same two-point function of the gauge field and of the curvature (Theorem 9.11). The proof combines the one-sided inequality between box and torus states given by the correlation inequalities of [BFS1], the convexity of the pressure in $\mu_0^2$, and a comparison of the pressures of a torus and of a box. For limits taken in the volume-first order it follows that the curvature two-point function of the torus limits is that of the limit points of Theorem 1.1(b), and hence is invariant under $E(3)$ (Corollary 9.14).

9.1. **The torus states and the counterterm.** Throughout this section $m > 0$, $\lambda > 0$, $e \neq 0$ and $\mu_* := m$ are fixed. The blocking factor $L \in \{2, 3\}$, the spacings $\varepsilon_K = L^{-K}\varepsilon_0$, $K \geq 0$, with $\varepsilon_0 = L^{-n_0}$, and the tori $T_j = (\mathbb{R}/\ell_j\mathbb{Z})^3$, $\ell_j = j!\,\ell_0$, with the lattice tori $T_{j,\varepsilon} = T_j \cap \varepsilon\mathbb{Z}^3$, are those of [v1, Sect. 2.1]. A *spacing of the family* is one of the $\varepsilon_K$; these are the spacings $L^{-K'}$, $K' = K + n_0$, of Theorems 1.1 and 5.4. For such $\varepsilon$ the number $N_j := \ell_j/\varepsilon$ of sites per side of $T_{j,\varepsilon}$ is an even integer [v1, Lemma 2.1], and we identify $T_{j,\varepsilon}$ with the box

$$\Lambda_j := [-\ell_j/2, \ell_j/2)^3 \cap \varepsilon\mathbb{Z}^3.$$

Under this identification the bonds of $T_{j,\varepsilon}$ are the bonds of $\Lambda_j^*$ together with the set $R_j$ of the $3N_j^2$ bonds $(x;\mu)$ of $\varepsilon\mathbb{Z}^3$ with $x \in \Lambda_j$ and $x_\mu = \ell_j/2 - \varepsilon$; on the torus such a bond joins $x$ to $x + \varepsilon e_\mu - \ell_j e_\mu$. A bond or plaquette of $\varepsilon\mathbb{Z}^3$ with all its vertices in $\Lambda_j$ is identified with the corresponding bond or plaquette of $T_{j,\varepsilon}$. In this way finitely supported functions on the bonds or plaquettes of $\varepsilon\mathbb{Z}^3$ are functions on those of $T_{j,\varepsilon}$ for all large $j$.

For a real function $f$ on a finite set of bonds we write $A(f) := \langle f, A\rangle_\varepsilon = \varepsilon^3 \sum_b f(b)A(b)$, and for a site $x$ we put $|A(x)|^2 := \sum_{\mu=1}^3 A(x;\mu)^2$. For a plaquette $p$ we write $F(p) = (dA)(p)$.

Let $\mu_0 > 0$ and $a \in \mathbb{R}$, let $\kappa = \mu_0\varepsilon$, and let $v$ be the function (2.3) with $m^2 + \delta m^2 = a$. On $\mathbb{R}^{3N_j^3} \times \mathbb{C}^{T_{j,\varepsilon}}$, the space of the bond variables of $T_{j,\varepsilon}$ and of the Higgs field, put

$$S^{\mathrm{T}}_{\mu_0,a}(A,\varphi) := \frac{\varepsilon}{2}\big(K_T(A) + \kappa^2|A|^2\big) + \sum_{x\in T_{j,\varepsilon}} v(|\varphi_x|) - \varepsilon\sum_b \mathrm{Re}\,\big(\overline{\varphi(b_-)}\, e^{ie\varepsilon A(b)}\varphi(b_+)\big),$$

where $|A|^2 = \sum_b A(b)^2$, the last sum runs over all $3N_j^3$ bonds of $T_{j,\varepsilon}$, the bonds of $R_j$ included, and

$$K_T(A) := \sum_{x\in T_{j,\varepsilon}} \sum_{\mu,\nu=1}^3 \big(A(x+\varepsilon e_\nu;\mu) - A(x;\mu)\big)^2.$$

This is the action $S_{\mu_0,a}$ of [v1, (2.1)–(2.2)]. Indeed, its gauge part is $\frac12\langle A, (-\Delta_\varepsilon + \mu_0^2)A\rangle_\varepsilon$, and $-\Delta_\varepsilon = d^*d + dd^*$ acts on each component $A(\cdot\,;\mu)$ as the lattice Laplacian [v1, Lemma 7.2]. In its matter part every site has six bonds, whose diagonal terms give the term $3\varepsilon r^2$ of $v$, as in Remark 2.2 (see also [v1, Prop. B.13]). Let $\mathbb{P}^a_{j,\mu_0}$ be the probability measure on $\mathbb{R}^{3N_j^3} \times \mathbb{C}^{T_{j,\varepsilon}}$ with density proportional to $e^{-S^{\mathrm{T}}_{\mu_0,a}}$ with respect to Lebesgue measure. It is invariant under the translations of $T_{j,\varepsilon}$ and under $(A,\varphi) \mapsto (-A, \overline{\varphi})$.

The coefficient of [v1] is given by the torus counterterm scheme [v1, Def. 2.2],

$$b_{T_j}(\varepsilon) := m^2 + \delta m^2_{K,T_j}(\varepsilon; m),$$

where $\delta m^2_{K,T_j}(\varepsilon; M)$ is the mass counterterm of [K1, (3.32)–(3.33), p. 659], computed on $T_{j,\varepsilon}$ with the scalar propagator $(-\Delta_\varepsilon + M^2)^{-1}$ and the photon propagator $(-\Delta_\varepsilon + \mu_*^2)^{-1}$. It is a polynomial in $(e^2, \lambda)$ with the monomials $\lambda, e^2, \lambda^2, e^2\lambda, e^4$. Since the photon propagator is taken at the reference mass $\mu_*$, the coefficient $b_{T_j}(\varepsilon)$ does not depend on $\mu_0$ [v1, Def. 2.2 and Rem. 2.3(ii)]. We write $b_j := b_{T_j}(\varepsilon)$ and $\mathbb{P}_{j,\mu_0} := \mathbb{P}^{b_j}_{j,\mu_0}$. For $\mu_0 \in (0, \mu_*]$ the restriction of $\mathbb{P}_{j,\mu_0}$

to the Borel functions of the curvatures $F(p)$ and of the $|\varphi_x|^2$ is the torus state $\langle\cdot\rangle_{\varepsilon,j,\mu_0}$ of [v1, Def. 2.7], by [v1, Lemma B.7]. The measures $\mathbb{P}_{j,\mu_0}$ are defined by the same formula for every $\mu_0 > 0$, and we use them for all $\mu_0 > 0$.

For a box $C$ and $a \in \mathbb{R}$ we write $\langle\cdot\rangle^{\mathrm{ND}}_{C;\mu_0,a}$ for the Neumann/Dirichlet state of Definition 2.4(c) with photon mass $\mu_0$ and $m^2 + \delta m^2 = a$.

9.2. **The limit of the torus counterterm.** We recall the explicit form of $b_{T_j}(\varepsilon)$ given in [v1, App. G], and define $a_\natural(\varepsilon)$ by the same expressions on $\varepsilon\mathbb{Z}^3$. Let $\varepsilon$ be a spacing of the family. Put $\mathrm{BZ}_\varepsilon := [-\pi/\varepsilon, \pi/\varepsilon)^3$ and $T_j^* := (2\pi/\ell_j)\mathbb{Z}^3 \cap \mathrm{BZ}_\varepsilon$, and for $k \in \mathbb{R}^3$ and $M > 0$ put

$$s_\mu(k) := \frac{2}{\varepsilon}\sin\frac{\varepsilon k_\mu}{2}, \qquad |s(k)|^2 := \sum_{\mu=1}^{3} s_\mu(k)^2, \qquad \hat G_M(k) := \big(|s(k)|^2 + M^2\big)^{-1}, \qquad \hat D := \hat G_{\mu_*}.$$

Let $\mathsf{X}$ be $T_{j,\varepsilon}$ or $\varepsilon\mathbb{Z}^3$. For a function $\phi$ on $\mathbb{R}^{3\mathsf{L}}$, $\mathsf{L} \geq 1$, put

$$\int_{T_{j,\varepsilon}} \phi := \ell_j^{-3\mathsf{L}} \sum_{k\in(T_j^*)^{\mathsf{L}}} \phi(k), \qquad \int_{\varepsilon\mathbb{Z}^3} \phi := (2\pi)^{-3\mathsf{L}} \int_{\mathrm{BZ}_\varepsilon^{\mathsf{L}}} \phi(k)\, dk,$$

and write $\int_{\mathsf{X},k}$ when the variable is named. For $M > 0$ and $x \in \varepsilon\mathbb{Z}^3$ let $G_{M,\mathsf{X}}(x) := \int_{\mathsf{X},k} e^{ik\cdot x}\hat G_M(k)$. This is the kernel of $(-\Delta_\varepsilon + M^2)^{-1}$ on $\mathsf{X}$ with respect to $\langle\cdot,\cdot\rangle_\varepsilon$; for $\mathsf{X} = T_{j,\varepsilon}$ it is an $\ell_j$-periodic function on $\varepsilon\mathbb{Z}^3$. Put $D_\mathsf{X}(x) := G_{\mu_*,\mathsf{X}}(x)$, $J_\mathsf{X} := m^2 \int_\mathsf{X} \hat D \hat G_m$ and

$$\Upsilon_\mathsf{X}(p) := -D_\mathsf{X}(0)\sum_\mu \big(1 - \cos\varepsilon p_\mu\big) - \int_{\mathsf{X},p'} \Big[\hat D(p-p')|s(p+p')|^2 - \hat D(p')|s(p')|^2\Big]\hat G_m(p').$$

Here and below $\mu$ runs over $1, 2, 3$. Define

$$c_\lambda(\mathsf{X}) := -16\, G_{m,\mathsf{X}}(0), \qquad c_{e^2}(\mathsf{X}) := -2D_\mathsf{X}(0) - J_\mathsf{X}, \qquad c_{\lambda^2}(\mathsf{X}) := 128\sum_{x\in\mathsf{X}} \varepsilon^3 G_{m,\mathsf{X}}(x)^3,$$

$$c_{e^2\lambda}(\mathsf{X}) := 16\int_\mathsf{X} \hat G_m^2\, \Upsilon_\mathsf{X},$$

and

$$\begin{aligned} c_{e^4}(\mathsf{X}) := {}& \tfrac{3}{4}\varepsilon^2 D_\mathsf{X}(0)^2 + m^2\int_\mathsf{X} \hat D\hat G_m^2\, \Upsilon_\mathsf{X} - \tfrac{\varepsilon^2}{2} D_\mathsf{X}(0)\big(D_\mathsf{X}(0) - J_\mathsf{X}\big) \\ & + \frac{1}{2}\sum_\mu \sum_{z\in\mathsf{X}} \varepsilon^3 \big[D_\mathsf{X}(z)^2 + D_\mathsf{X}(z+\varepsilon e_\mu)^2\big]\big[G_{m,\mathsf{X}}(z) + G_{m,\mathsf{X}}(z+\varepsilon e_\mu)\big] \\ & - \sum_\mu \int_{\mathsf{X},k}\int_{\mathsf{X},P} s_\mu(k)\big[s_\mu(k) + s_\mu(2P+k)\big]\hat G_m(k)\hat G_m(P)\hat D(k)\hat D(P-k) \\ & + \sum_\mu \int_{\mathsf{X},k} \hat D(k)^2\Big[2G_{m,\mathsf{X}}(\varepsilon e_\mu) - \int_{\mathsf{X},q} s_\mu(2q+k)^2 \hat G_m(q)\hat G_m(q+k)\Big]. \end{aligned}$$

For $\mathsf{X} = T_{j,\varepsilon}$ the sums over $z$ and $x$ run over a set of representatives of $\varepsilon\mathbb{Z}^3/\ell_j\mathbb{Z}^3$, for instance $\Lambda_j$. Every function of the momenta that occurs here is invariant under the translations by $(2\pi/\varepsilon)\mathbb{Z}^3$ in each momentum variable, so the sums over $(T_j^*)^{\mathsf{L}}$ do not depend on the choice of the representatives $T_j^*$ of the dual torus. For $T = T_{j,\varepsilon}$, in the notation of [v1, App. G], $c_\lambda(T) = -16C_T(m)$, $c_{e^2}(T) = -2D_T - J_T(m)$, $c_{\lambda^2}(T) = 128S_T(m)$, $c_{e^2\lambda}(T) = 16F_T(m)$ with

$e^2\Upsilon_T = \widetilde{\Pi}_T(\cdot\,;m)$ [v1, Lemma G.16], and $c_{e^4}(T) = Q_T(m)$, with its six terms written in the order of [v1, Cor. G.22]. By [v1, Def. 2.2 and Lemma G.1],

$$b_{T_j}(\varepsilon) = m^2 + \lambda c_\lambda(\mathsf{X}) + e^2 c_{e^2}(\mathsf{X}) + \lambda^2 c_{\lambda^2}(\mathsf{X}) + e^2\lambda\, c_{e^2\lambda}(\mathsf{X}) + e^4 c_{e^4}(\mathsf{X}) \qquad \text{with } \mathsf{X} = T_{j,\varepsilon}. \tag{9.1}$$

We define $a_\natural(\varepsilon)$ as the right side of (9.1) with $\mathsf{X} = \varepsilon\mathbb{Z}^3$. Its terms are finite: the momentum integrands are continuous, since all denominators are at least $\min(m^2, \mu_*^2) > 0$, and the sums over $\varepsilon\mathbb{Z}^3$ converge by Lemma 9.1(c) below. By [v1, Lemma G.16] for the orders $\lambda, e^2, \lambda^2, e^2\lambda$, and by [v1, Cor. G.22] and the paragraph on the lattice $\varepsilon\mathbb{Z}^3$ in its proof for the order $e^4$, the number $a_\natural(\varepsilon) - m^2$ is the mass counterterm given by the graphs of [K1, (3.32)–(3.33), p. 659] on $\varepsilon\mathbb{Z}^3$, with the propagators $(-\Delta_\varepsilon + m^2)^{-1}$ and $(-\Delta_\varepsilon + \mu_*^2)^{-1}$ on $\varepsilon\mathbb{Z}^3$. This interpretation is not used below. Like $b_{T_j}(\varepsilon)$, the coefficient $a_\natural(\varepsilon)$ does not depend on $\mu_0$. Since $\delta m^2 := a_\natural(\varepsilon) - m^2$ does not depend on the box, the results of Sections 2–5 apply to the box states with $m^2 + \delta m^2 = a_\natural(\varepsilon)$.

**Lemma 9.1.** *Let $\varepsilon$ be a spacing of the family.*

(a) *If $\phi$ is a continuous complex function on $\mathbb{R}^{3\mathsf{L}}$, $\mathsf{L} \geq 1$, then $\int_{T_{j,\varepsilon}} \phi \to \int_{\varepsilon\mathbb{Z}^3} \phi$ as $j \to \infty$.*

(b) *Let $M > 0$ and $G_M := G_{M,\varepsilon\mathbb{Z}^3}$. Then $G_M > 0$ and $\sum_{x\in\varepsilon\mathbb{Z}^3} \varepsilon^3 G_M(x) = M^{-2}$. Moreover $G_{M,T_{j,\varepsilon}}(x) = \sum_{n\in\mathbb{Z}^3} G_M(x + \ell_j n)$ and $G_{M,T_{j,\varepsilon}}(x) \leq \varepsilon^{-3}M^{-2}$ for $x \in \varepsilon\mathbb{Z}^3$, and the sum of $\varepsilon^3 G_{M,T_{j,\varepsilon}}$ over any set of representatives of $\varepsilon\mathbb{Z}^3/\ell_j\mathbb{Z}^3$ is $M^{-2}$.*

(c) *Let $M_1, M_2, M_3 \in \{m, \mu_*\}$ and $a_1, a_2, a_3 \in \varepsilon\mathbb{Z}^3$. Then the series $\sum_{z\in\varepsilon\mathbb{Z}^3} \varepsilon^3 \prod_{i=1}^3 G_{M_i}(z + a_i)$ converges, and*

$$\sum_{z\in T_{j,\varepsilon}} \varepsilon^3 \prod_{i=1}^3 G_{M_i,T_{j,\varepsilon}}(z + a_i) \to \sum_{z\in\varepsilon\mathbb{Z}^3} \varepsilon^3 \prod_{i=1}^3 G_{M_i}(z + a_i) \qquad (j \to \infty),$$

*where the sum on the left runs over a set of representatives of $\varepsilon\mathbb{Z}^3/\ell_j\mathbb{Z}^3$.*

*Proof.* (a) Since $N_j = \ell_j/\varepsilon$ is even, $T_j^* = (2\pi/\ell_j)\{-N_j/2, \dots, N_j/2 - 1\}^3$, and the cubes $k + [0, 2\pi/\ell_j)^3$, $k \in T_j^*$, partition $\mathrm{BZ}_\varepsilon$. Hence $\int_{T_{j,\varepsilon}} \phi = (2\pi)^{-3\mathsf{L}} \sum_{k\in(T_j^*)^\mathsf{L}} (2\pi/\ell_j)^{3\mathsf{L}} \phi(k)$ is a Riemann sum of $(2\pi)^{-3\mathsf{L}}\phi$ for a partition of $\mathrm{BZ}_\varepsilon^\mathsf{L}$ into cubes of side $2\pi/\ell_j \to 0$. Since $\phi$ is uniformly continuous on the closure of $\mathrm{BZ}_\varepsilon^\mathsf{L}$, these sums converge to $\int_{\varepsilon\mathbb{Z}^3} \phi$.

(b) Positivity of $G_M$ and the image formula are [v1, Lemma G.12(a),(b)]. For the sum, write $-\Delta_\varepsilon + M^2 = (6\varepsilon^{-2} + M^2)(1 - \theta P)$ on $\ell^2(\varepsilon\mathbb{Z}^3)$, where $\theta := (6 + \varepsilon^2 M^2)^{-1}$ and $(Pu)(x) := \sum_\mu \big(u(x + \varepsilon e_\mu) + u(x - \varepsilon e_\mu)\big)$. Since $\|P\| \leq 6$ and $6\theta < 1$, the Neumann series gives $G_M = \varepsilon^{-3}(6\varepsilon^{-2} + M^2)^{-1} \sum_{n\geq 0} \theta^n P^n \mathbf{1}_{\{0\}}$, a series of nonnegative functions. Since $\sum_x (P^n \mathbf{1}_{\{0\}})(x) = 6^n$, Tonelli's theorem gives $\sum_x \varepsilon^3 G_M(x) = (6\varepsilon^{-2} + M^2)^{-1}(1 - 6\theta)^{-1} = M^{-2}$. On the torus, $T_j^*$ has $N_j^3$ points and $\hat{G}_M \leq M^{-2}$, so $|G_{M,T_{j,\varepsilon}}(x)| \leq \int_{T_{j,\varepsilon}} \hat{G}_M \leq \varepsilon^{-3}M^{-2}$. Finally, the sum of $\varepsilon^3 e^{ik\cdot x}$ over a set of representatives of $\varepsilon\mathbb{Z}^3/\ell_j\mathbb{Z}^3$ is $\ell_j^3$ for $k = 0$ and 0 for the other $k \in T_j^*$, so the sum of $\varepsilon^3 G_{M,T_{j,\varepsilon}}$ over it is $\hat{G}_M(0) = M^{-2}$.

(c) Put $\mathsf{k}_i := G_{M_i}(\cdot + a_i)$, $\mathsf{k}_i^{(j)} := G_{M_i,T_{j,\varepsilon}}(\cdot + a_i)$ and $\beta := \varepsilon^{-3} \min(m, \mu_*)^{-2}$. By (b), $0 < \mathsf{k}_i \leq \mathsf{k}_i^{(j)} \leq \beta$, and $\prod_i \mathsf{k}_i \leq \beta^2 \mathsf{k}_1$ with $\mathsf{k}_1$ summable, so the series converges. Take $\Lambda_j$ as the set of representatives in the sum on the left. Since

$$\prod_i \mathsf{k}_i^{(j)} - \prod_i \mathsf{k}_i = \big(\mathsf{k}_1^{(j)} - \mathsf{k}_1\big)\mathsf{k}_2^{(j)}\mathsf{k}_3^{(j)} + \mathsf{k}_1\big(\mathsf{k}_2^{(j)} - \mathsf{k}_2\big)\mathsf{k}_3^{(j)} + \mathsf{k}_1\mathsf{k}_2\big(\mathsf{k}_3^{(j)} - \mathsf{k}_3\big),$$

and the three terms are nonnegative,

$$0 \leq \sum_{z\in\Lambda_j} \varepsilon^3 \Big(\prod_i \mathsf{k}_i^{(j)} - \prod_i \mathsf{k}_i\Big)(z) \leq \beta^2 \sum_{i=1}^3 \sum_{z\in\Lambda_j} \varepsilon^3 \big(\mathsf{k}_i^{(j)} - \mathsf{k}_i\big)(z) = \beta^2 \sum_{i=1}^3 \sum_{x\in\varepsilon\mathbb{Z}^3\setminus(\Lambda_j + a_i)} \varepsilon^3 G_{M_i}(x).$$

For the last equality note that $\Lambda_j + a_i$ is also a set of representatives, so by (b) the sums of $\varepsilon^3 G_{M_i,T_{j,\varepsilon}}$ over it and of $\varepsilon^3 G_{M_i}$ over $\varepsilon\mathbb{Z}^3$ are both $M_i^{-2}$. Every finite subset of $\varepsilon\mathbb{Z}^3$ lies in $\Lambda_j + a_i$ for all large $j$, so the right side tends to 0 because $G_{M_i}$ is summable. Moreover $\sum_{z\notin\Lambda_j} \varepsilon^3 \prod_i \mathsf{k}_i(z) \le \beta^2 \sum_{z\notin\Lambda_j} \varepsilon^3 \mathsf{k}_1(z) \to 0$. Together these give (c). □

**Proposition 9.2.** *Let $\varepsilon$ be a spacing of the family. Then each of the coefficients $c_\lambda$, $c_{e^2}$, $c_{\lambda^2}$, $c_{e^2\lambda}$, $c_{e^4}$ at $\mathsf{X} = T_{j,\varepsilon}$ converges, as $j\to\infty$, to its value at $\mathsf{X} = \varepsilon\mathbb{Z}^3$. In particular $b_{T_j}(\varepsilon) \to a_\natural(\varepsilon)$.*

*Proof.* The second statement follows from the first by (9.1) and the definition of $a_\natural(\varepsilon)$. The terms in the definitions of the coefficients are of two kinds.

*Momentum terms.* The numbers $G_{m,\mathsf{X}}(0)$, $G_{m,\mathsf{X}}(\varepsilon e_\mu)$, $D_\mathsf{X}(0)$, $J_\mathsf{X}$, $\int_\mathsf{X} \hat D^2$, $\int_{\mathsf{X},p} \hat G_m(p)^2 \sum_\mu (1-\cos\varepsilon p_\mu)$ and $\int_{\mathsf{X},p} \hat D(p)\hat G_m(p)^2 \sum_\mu (1-\cos\varepsilon p_\mu)$ are of the form $\int_\mathsf{X} \phi$ with $\phi$ continuous on $\mathbb{R}^3$; for instance $G_{m,\mathsf{X}}(\varepsilon e_\mu) = \int_{\mathsf{X},k} e^{i\varepsilon k_\mu} \hat G_m(k)$. The numbers

$$\int_{\mathsf{X},p}\int_{\mathsf{X},p'} \hat D(p)^n \hat G_m(p)^2 \Big[\hat D(p-p')|s(p+p')|^2 - \hat D(p')|s(p')|^2\Big] \hat G_m(p') \qquad (n=0,1),$$

$$\int_{\mathsf{X},k}\int_{\mathsf{X},P} s_\mu(k)\big[s_\mu(k) + s_\mu(2P+k)\big] \hat G_m(k)\hat G_m(P)\hat D(k)\hat D(P-k),$$

$$\int_{\mathsf{X},k}\int_{\mathsf{X},q} \hat D(k)^2 s_\mu(2q+k)^2 \hat G_m(q)\hat G_m(q+k)$$

are of the form $\int_\mathsf{X}\phi$ with $\phi$ continuous on $\mathbb{R}^6$. Continuity holds because all denominators are at least $\min(m^2,\mu_*^2) > 0$. Inserting the definition of $\Upsilon_\mathsf{X}$, the coefficients $c_\lambda$, $c_{e^2}$, $c_{e^2\lambda}$ and the terms of $c_{e^4}$ other than the sum over $z$ are finite sums of products of these numbers. For instance

$$c_{e^2\lambda}(\mathsf{X}) = -16 D_\mathsf{X}(0) \int_{\mathsf{X},p} \hat G_m(p)^2 \sum_\mu (1-\cos\varepsilon p_\mu) \\ - 16 \int_{\mathsf{X},p}\int_{\mathsf{X},p'} \hat G_m(p)^2 \Big[\hat D(p-p')|s(p+p')|^2 - \hat D(p')|s(p')|^2\Big] \hat G_m(p').$$

By Lemma 9.1(a), each of these numbers at $\mathsf{X} = T_{j,\varepsilon}$ converges to its value at $\mathsf{X} = \varepsilon\mathbb{Z}^3$, and hence so do the finite sums of products.

*Position terms.* After expanding the product of the two brackets, $c_{\lambda^2}(\mathsf{X})$ and the sum over $z$ in $c_{e^4}(\mathsf{X})$ are finite sums of sums $\sum_{z\in\mathsf{X}} \varepsilon^3 \prod_{i=1}^3 G_{M_i,\mathsf{X}}(z+a_i)$ with $M_i \in \{m,\mu_*\}$ and $a_i \in \{0,\varepsilon e_\mu\}$; for instance $D_\mathsf{X}(z)^2 G_{m,\mathsf{X}}(z+\varepsilon e_\mu)$ has $(M_1,M_2,M_3) = (\mu_*,\mu_*,m)$ and $(a_1,a_2,a_3) = (0,0,\varepsilon e_\mu)$. They converge by Lemma 9.1(c). □

*Remark* 9.3. (a) The proofs below use only that $\lim_j b_{T_j}(\varepsilon)$ exists and equals $a_\natural(\varepsilon)$; the interpretation of $a_\natural(\varepsilon)$ by graphs on $\varepsilon\mathbb{Z}^3$ plays no role.

(b) That the coefficient does not depend on $\mu_0$ is essential below: the action is then linear in $\mu_0^2$, so the pressure is convex in $\mu_0^2$ (Lemma 9.8). This is why the comparison is made with the box states at the coefficient $a_\natural(\varepsilon)$. For a coefficient that depends on $\mu_0$, for instance a counterterm computed with the photon propagator at the photon mass $\mu_0$ [v1, Rem. 2.3(ii)], only convexity jointly in $(\mu_0^2, a)$ is available, and the argument below does not give a countable exceptional set.

### 9.3. **Moment bounds and correlation inequalities.**

**Lemma 9.4.** *Let $\varepsilon > 0$, $\mu_0 > 0$ and $a \in \mathbb{R}$. Let $\mathbb{M}$ be either a box measure $\langle\cdot\rangle_{\gamma,\Lambda}$ of Definition 2.3, with $c_b \equiv \varepsilon$ and $m^2 + \delta m^2 = a$, or a measure $\mathbb{P}^a_{j,\mu_0}$. Write $r_x = |\varphi_x|$ and $a_- = \max(-a, 0)$.*

(a) *Suppose that* $\mathbb{M} = \mathbb{P}^a_{j,\mu_0}$, *or that* $\gamma = \gamma_Q$ *with* $Q \geq \kappa^2|\cdot|^2$. *Then for every real function* $f$ *on the bonds of* $\mathbb{M}$ *and every* $t \in \mathbb{R}$,

$$\mathbb{M}\big(e^{tA(f)}\big) \leq \exp\Big(\frac{t^2\|f\|^2_{2,\varepsilon}}{2\mu_0^2}\Big).$$

*Consequently* $\mathbb{M}(A(f)) = 0$, $\mathbb{M}(A(f)^2) \leq \|f\|^2_{2,\varepsilon}/\mu_0^2$, $\mathbb{M}(A(f)^4) \leq 16\|f\|^4_{2,\varepsilon}/\mu_0^4$, *and* $\mathbb{M}(A(b)^2) \leq (\mu_0^2\varepsilon^3)^{-1}$ *for every bond* $b$.

(b) *For every site* $x$,

$$\mathbb{M}(r_x^2) \leq \mathfrak{m}(a) := \frac{a_- + \big(a_-^2 + 32\lambda\varepsilon^{-3}\big)^{1/2}}{8\lambda}.$$

*Proof.* (a) Let $\Sigma$ be the covariance operator of the Gaussian part with respect to $\langle\cdot,\cdot\rangle_\varepsilon$. Thus $\int e^{\langle u,A\rangle_\varepsilon}\gamma(dA) = e^{\frac12\langle u,\Sigma u\rangle_\varepsilon}$ for complex $u$, where $\langle\cdot,\cdot\rangle_\varepsilon$ is extended bilinearly. On the torus $\Sigma = (-\Delta_\varepsilon + \mu_0^2)^{-1} \leq \mu_0^{-2}$. For $\gamma_Q$ with $Q \geq \kappa^2|\cdot|^2$ we have $\langle A, \Sigma^{-1}A\rangle_\varepsilon = \varepsilon Q(A) \geq \mu_0^2\|A\|^2_{2,\varepsilon}$, hence again $\Sigma \leq \mu_0^{-2}$. The matter integral $W(A) = \int e^{-S^{\mathrm{m}}}d\varphi$ has an expansion $W(A) = \sum_n w(n)\exp(ie\varepsilon\sum_b n(b)A(b))$ with $w(n) = w(-n) \geq 0$ and $\sum_n w(n) < \infty$: for box measures this is Lemma 4.1, and for tori it is [v1, Lemma 2.5], whose proof does not use the value of the coefficient. The computation in the proof of Proposition 4.2, with $f$ in place of $d^*g$, gives $\int e^{tA(f)}W\,d\gamma \leq e^{\frac12 t^2\langle f,\Sigma f\rangle_\varepsilon}\int W\,d\gamma$, hence the first bound. The moment bounds follow as in Lemma B.1(a), and $A(b) = A(\varepsilon^{-3}\mathbf{1}_b)$ with $\|\varepsilon^{-3}\mathbf{1}_b\|^2_{2,\varepsilon} = \varepsilon^{-3}$.

(b) Fix a site $x$ and regard $\varphi_x$ as a vector of $\mathbb{R}^2$. Let $S^{\mathrm{m}}$ be the matter part of the action of $\mathbb{M}$; the Gaussian part does not depend on $\varphi$. Since the density of $\mathbb{M}$ decays like $e^{-\lambda\varepsilon^3 r_x^4}$ in $\varphi_x$, integration by parts gives $\mathbb{M}(\varphi_x\cdot\nabla_{\varphi_x}S^{\mathrm{m}}) = 2$. Now $rv'(r) = 6\varepsilon r^2 + \varepsilon^3(ar^2 + 4\lambda r^4)$, and each hopping term containing $\varphi_x$ is real-linear in $\varphi_x$. Hence

$$\mathbb{M}\Big(6\varepsilon r_x^2 + \varepsilon^3\big(ar_x^2 + 4\lambda r_x^4\big) - \varepsilon\sum_{b\ni x} r_{b_-}r_{b_+}\cos u_b\Big) = 2,$$

where the sum runs over the $n_x \leq 6$ bonds $b$ of the hopping term at $x$, and $u_b = \vartheta_{b_+} - \vartheta_{b_-} + e\varepsilon A(b)$. Let $m_y := \mathbb{M}(r_y^2)$ and let $x$ be a site at which $m_x$ is maximal. Since $|\mathbb{M}(r_{b_-}r_{b_+}\cos u_b)| \leq \frac12(m_{b_-} + m_{b_+}) \leq m_x$ for $b \ni x$, the terms of order $\varepsilon$ have a nonnegative expectation. Together with $\mathbb{M}(r_x^4) \geq m_x^2$ and $am_x \geq -a_-m_x$ this gives $4\lambda m_x^2 - a_-m_x - 2\varepsilon^{-3} \leq 0$, that is, $m_x \leq \mathfrak{m}(a)$. □

**Lemma 9.5.** *Let* $X, X'$ *be real random variables with* $\mathbb{E}X^2 < \infty$ *and* $\mathbb{E}X'^4 < \infty$. *If* $\mathbb{E}\cos(tX) \leq \mathbb{E}\cos(tX')$ *for all* $t \in \mathbb{R}$, *then* $\mathbb{E}X^2 \geq \mathbb{E}X'^2$.

*Proof.* From $1 - \frac12 y^2 \leq \cos y \leq 1 - \frac12 y^2 + \frac{1}{24}y^4$ we get $1 - \frac12 t^2\mathbb{E}X^2 \leq \mathbb{E}\cos(tX) \leq \mathbb{E}\cos(tX') \leq 1 - \frac12 t^2\mathbb{E}X'^2 + \frac{1}{24}t^4\mathbb{E}X'^4$. Divide by $\frac12 t^2$ and let $t \to 0$. □

**Lemma 9.6.** *Let* $\varepsilon > 0$, $\mu_0 > 0$, $f$ *a finitely supported real function on* $(\varepsilon\mathbb{Z}^3)^*$ *and* $Y = e^{iA(f)}$. *All expectations of* $Y$ *below are real.*

(a) *If* $C \subset C'$ *are boxes with* $\operatorname{supp} f \subset C^*$ *and* $a \in \mathbb{R}$, *then* $\langle Y\rangle^{\mathrm{ND}}_{C;\mu_0,a} \leq \langle Y\rangle^{\mathrm{ND}}_{C';\mu_0,a}$.

(b) *If* $\varepsilon$ *is a spacing of the family,* $C \subset \Lambda_j$ *is a box with* $\operatorname{supp} f \subset C^*$ *and* $a_C \geq b_j$, *then* $\langle Y\rangle^{\mathrm{ND}}_{C;\mu_0,a_C} \leq \mathbb{P}_{j,\mu_0}(Y)$.

*Proof.* The expectations are real by the symmetry $(A,\varphi) \mapsto (-A,\overline{\varphi})$ of all the measures involved.

(a) The proof of Lemma 2.8 uses only that the observable depends on $A$ through $A|_{\Lambda^*}$, so it applies to $Y$. As in the proof of Lemma 2.9(a), $Q^{\mathrm{ND}}_{C'}(A) \geq Q^{\mathrm{ND}}_C(A|_{C^*}) + \kappa^2\sum_{b\in C'^*\setminus C^*}A(b)^2$.

(b) We use the correlation inequalities of [BFS1, Sect. 6] on the finite graph $T_{j,\varepsilon}$, in the form of [v1, Lemmas B.10–B.12]. They hold in the setting of [v1, Def. B.8], which contains the

measures on $\mathbb{R}^{3N_j^3} \times \mathbb{C}^{T_{j,\varepsilon}}$ with an arbitrary centred Gaussian on the bond variables, hopping coefficients $c_b \in [0,\varepsilon]$ on the bonds of $T_{j,\varepsilon}$ and single-site measures $re^{-v(r)}dr$ with $v$ of (2.3) for any coefficient (see [v1, Prop. B.13]).

*Step 1: the coefficient.* For $t \in [0,1]$ let $\langle\cdot\rangle_t := \mathbb{P}^{a(t)}_{j,\mu_0}$ with $a(t) = b_j + t(a_C - b_j)$. Differentiating under the integral sign, which is allowed because of the factor $e^{-\lambda\varepsilon^3 r^4}$ at every site,

$$\frac{d}{dt}\langle Y\rangle_t = -\frac{\varepsilon^3}{2}(a_C - b_j)\sum_{x\in T_{j,\varepsilon}} \langle Y; r_x^2\rangle_t, \qquad \langle X; Z\rangle := \langle XZ\rangle - \langle X\rangle\langle Z\rangle.$$

By the symmetry $(A,\vartheta) \mapsto (-A,-\vartheta)$, $\langle Y; r_x^2\rangle_t = \langle \cos A(f); r_x^2\rangle_t$, and this is nonnegative by [BFS1, Thm. 6.1, p. 276], in the form of [v1, Lemma B.10], applied to the functions $\cos A(f)$ and $r_x^2$; in the notation there the first is given by the polynomial 1, $n = 0$ and $\ell = A(f)$, and the second by the polynomial $r_x^2$, $m = 0$ and $\ell' = 0$. Hence $\langle Y\rangle_1 \le \langle Y\rangle_0 = \mathbb{P}_{j,\mu_0}(Y)$.

*Step 2: the hopping terms.* In $\langle\cdot\rangle_1$ put $c_b = 0$ for the bonds $b$ of $T_{j,\varepsilon}$ that are not in $C^*$. By [v1, Lemma B.12(b)] this does not increase $\langle Y\rangle$.

*Step 3: the gradient pairs.* Every pair of bonds summed in $K_C$ is a pair summed in $K_T$. Let $Q'$ be the form obtained from $K_T + \kappa^2|\cdot|^2$ by deleting the pairs summed in $K_T$ that contain a bond of $C^*$ but are not summed in $K_C$. Then $Q' \le K_T + \kappa^2|\cdot|^2$, and $Q'(A) = Q_C^{\mathrm{ND}}(A|_{C^*}) + Q''(A)$, where $Q''$ is a positive definite form in the variables on the bonds not in $C^*$. Replacing the Gaussian by $\gamma_{Q'}$ increases the covariance, so by [v1, Lemma B.12(a)] it does not increase $\langle Y\rangle$.

*Step 4: factorisation.* In the measure obtained, the Gaussian is the product of $\gamma_{Q_C^{\mathrm{ND}}}$ on $\mathbb{R}^{C^*}$ and of a Gaussian in the other bond variables. The matter weight is the product of the Dirichlet matter weight of $C$ with coefficient $a_C$, which depends on $A$ only through $A|_{C^*}$, and of single-site factors at the sites outside $C$; the single-site factors at the sites of $C$ are those of Remark 2.2. Since $Y$ depends only on $A|_{C^*}$, its expectation is $\langle Y\rangle^{\mathrm{ND}}_{C;\mu_0,a_C}$. □

## 9.4. **The Neumann/Dirichlet limit of the gauge field and the pressures.**

**Proposition 9.7.** *Let $\varepsilon > 0$, $\mu_0 > 0$ and $a \in \mathbb{R}$, and write $\langle\cdot\rangle_C := \langle\cdot\rangle^{\mathrm{ND}}_{C;\mu_0,a}$. Let $f, h$ be finitely supported real functions on $(\varepsilon\mathbb{Z}^3)^*$.*

(a) *There is a unique probability measure $\omega^{\mathrm{ND}} = \omega^{\mathrm{ND},a}_{\varepsilon,\mu_0}$ on $\mathbb{R}^{(\varepsilon\mathbb{Z}^3)^*}$ such that $\omega^{\mathrm{ND}}(e^{iA(f)}) = \sup_C \langle e^{iA(f)}\rangle_C$ for all such $f$, the supremum running over the boxes $C$ with $\operatorname{supp} f \subset C^*$. Moreover $\langle e^{iA(f)}\rangle_{C_n} \to \omega^{\mathrm{ND}}(e^{iA(f)})$ along every exhausting sequence of boxes $(C_n)$.*

(b) *$\omega^{\mathrm{ND}}$ is invariant under the translations by $\varepsilon\mathbb{Z}^3$ and under $A \mapsto -A$, and $\omega^{\mathrm{ND}}(e^{tA(f)}) \le \exp(t^2\|f\|^2_{2,\varepsilon}/(2\mu_0^2))$ for $t \in \mathbb{R}$.*

(c) *$\langle A(f)A(h)\rangle_{C_n} \to \omega^{\mathrm{ND}}(A(f)A(h))$ along every exhausting sequence of boxes, and $\langle A(f)^2\rangle_C \ge \langle A(f)^2\rangle_{C'} \ge \omega^{\mathrm{ND}}(A(f)^2)$ for boxes $C \subset C'$ with $\operatorname{supp} f \subset C^*$.*

(d) *The image of $\omega^{\mathrm{ND}}$ under $A \mapsto (F(p))_{p\in\mathcal{P}_\varepsilon}$ is the law $\omega_\varepsilon$ of Theorem 4.3 for the parameters $\mu_0, e, \lambda$ and $m^2 + \delta m^2 = a$.*

*Proof.* (a) By Lemma 9.6(a), $C \mapsto \langle e^{iA(f)}\rangle_C$ is nondecreasing, and it is real and at most 1. Let $s(f)$ be its supremum and $\delta > 0$, and choose $C_\delta$ with $\langle e^{iA(f)}\rangle_{C_\delta} > s(f) - \delta$. For large $n$, $C_n \supset C_\delta$, so $s(f) - \delta < \langle e^{iA(f)}\rangle_{C_n} \le s(f)$. Hence, for every finite set $B$ of bonds, the characteristic functions of the laws of $(A(b))_{b\in B}$ under $\langle\cdot\rangle_{C_n}$ converge pointwise. These laws are tight by Lemma 9.4(a), so they converge weakly by Lévy's continuity theorem. The limits are consistent, and Kolmogorov's theorem gives $\omega^{\mathrm{ND}}$. It is unique because its finite-dimensional laws are determined by their characteristic functions.

(b) The Neumann/Dirichlet state of the translate $C + y$, $y \in \varepsilon\mathbb{Z}^3$, is the image of that of $C$ under the translation, since $v$ is the same at every site and $K_{C+y}$ is the translate of $K_C$. Hence

the supremum in (a) is translation invariant. The characteristic functional is real, which is the invariance under $A \mapsto -A$. The bound follows from Lemma 9.4(a) and Fatou's lemma.

(c) By Lemma 9.4(a) the fourth moments of $A(f)$ and $A(h)$ under $\langle\cdot\rangle_C$ are bounded uniformly in $C$, so the weak convergence in (a) implies the convergence of second moments. The first inequality is Lemma 9.6(a) together with Lemma 9.5. The second follows by applying the first to $C'$ and the boxes of an exhausting sequence containing $C'$, and letting the box grow.

(d) Let $g$ be a finitely supported real 2-cochain and $C$ a box with $\operatorname{supp} g \subset C^{**}$. The bonds of the plaquettes of $C^{**}$ lie in $C^*$, so $F(g) = \langle g, d_C A\rangle_\varepsilon = A(d^*g)$, where $d^*g$ is the finitely supported 1-cochain obtained from the adjoint on $\varepsilon\mathbb{Z}^3$. Hence, for a nested exhausting sequence $(C_J)$, $\omega^{\mathrm{ND}}(e^{iF(g)}) = \lim_J \langle e^{iF(g)}\rangle^{\mathrm{ND}}_{C_J;\mu_0,a} = \omega_\varepsilon(e^{iF(g)})$, by (a) and Theorem 4.3(f). The finite-dimensional laws of the curvature under the two measures therefore coincide. □

For a spacing $\varepsilon$ of the family, $s > 0$ and $a \in \mathbb{R}$, let

$$Z_j^{\mathrm{T}}(s,a) := \int e^{-S^{\mathrm{T}}_{\sqrt{s},a}}\, dA\, d\varphi, \qquad Z_j^{\mathrm{ND}}(s,a) := \int e^{-\frac{\varepsilon}{2}(K_{\Lambda_j}(A)+s\varepsilon^2|A|^2)-S^{\mathrm{m}}_{\Lambda_j}(A,\varphi)}\, dA\, d\varphi,$$

with Lebesgue measure on $\mathbb{R}^{3N_j^3}\times\mathbb{C}^{T_{j,\varepsilon}}$ and on $\mathbb{R}^{\Lambda_j^*}\times\mathbb{C}^{\Lambda_j}$, where $S^{\mathrm{m}}_{\Lambda_j}$ is the matter action (2.4) with $c_b \equiv \varepsilon$ and $m^2 + \delta m^2 = a$. Thus $Z_j^{\mathrm{ND}}(s,a)$ is the normalisation of $\langle\cdot\rangle^{\mathrm{ND}}_{\Lambda_j;\sqrt{s},a}$. Put

$$\mathsf{p}_j^{\mathrm{T}}(s,a) := N_j^{-3}\log Z_j^{\mathrm{T}}(s,a), \qquad \mathsf{p}_j^{\mathrm{ND}}(s,a) := N_j^{-3}\log Z_j^{\mathrm{ND}}(s,a),$$

and

$$\mathsf{g}_j(s,a) := N_j^{-3}\sum_{b\in\Lambda_j^*}\langle A(b)^2\rangle^{\mathrm{ND}}_{\Lambda_j;\sqrt{s},a}.$$

**Lemma 9.8.** *Let $\varepsilon$ be a spacing of the family, $s > 0$ and $a, a' \in \mathbb{R}$.*

(a) *The functions $\mathsf{p}_j^{\mathrm{T}}(\cdot,a)$ and $\mathsf{p}_j^{\mathrm{ND}}(\cdot,a)$ are convex and continuously differentiable on $(0,\infty)$, with*

$$\partial_s \mathsf{p}_j^{\mathrm{T}}(s,a) = -\frac{\varepsilon^3}{2}\,\mathbb{P}^a_{j,\sqrt{s}}\big(|A(0)|^2\big), \qquad \partial_s \mathsf{p}_j^{\mathrm{ND}}(s,a) = -\frac{\varepsilon^3}{2}\,\mathsf{g}_j(s,a).$$

(b) $|\mathsf{p}_j^{\mathrm{T}}(s,a) - \mathsf{p}_j^{\mathrm{T}}(s,a')| \le \frac{\varepsilon^3}{2}|a-a'|\,\mathfrak{m}(\min(a,a'))$.

(c) $|\mathsf{p}_j^{\mathrm{T}}(s,a) - \mathsf{p}_j^{\mathrm{ND}}(s,a)| \le N_j^{-1}\Big(\frac{54}{s\varepsilon^2} + 3\varepsilon\,\mathfrak{m}(a) + \frac{3}{2}\Big|\log\frac{2\pi}{s\varepsilon^3}\Big|\Big)$.

*Proof.* (a) The actions are affine in $s$, with $\partial_s$ equal to $\frac{\varepsilon^3}{2}|A|^2$. Differentiation under the integral sign is allowed because of the Gaussian factor, and gives $\partial_s \log Z = -\frac{\varepsilon^3}{2}\langle|A|^2\rangle$ and $\partial_s^2 \log Z = \frac{\varepsilon^6}{4}\big(\langle|A|^4\rangle - \langle|A|^2\rangle^2\big) \ge 0$ for both partition functions. On the torus $|A|^2 = \sum_x |A(x)|^2$, and $\mathbb{P}^a_{j,\sqrt{s}}(|A(x)|^2)$ does not depend on $x$ by translation invariance. On the box, $\langle|A|^2\rangle^{\mathrm{ND}}_{\Lambda_j;\sqrt{s},a} = N_j^3\mathsf{g}_j(s,a)$.

(b) In the same way $\partial_a \log Z_j^{\mathrm{T}}(s,a) = -\frac{\varepsilon^3}{2}\sum_x \mathbb{P}^a_{j,\sqrt{s}}(r_x^2)$. By Lemma 9.4(b) this is at most $\frac{\varepsilon^3}{2}N_j^3\mathfrak{m}(a)$ in absolute value. Since $\mathfrak{m}$ is nonincreasing, the mean value theorem gives (b).

(c) Consider the box measure $\langle\cdot\rangle_{\mathrm{ref}} := \langle\cdot\rangle_{\gamma,\Lambda_j}$ with bond set $\Lambda_j^* \cup R_j$, where $R_j$ is regarded as a set of bonds of $\varepsilon\mathbb{Z}^3$ leaving $\Lambda_j$, and with $\gamma = \gamma_{Q_{\mathrm{ref}}}$, $Q_{\mathrm{ref}}(A) := Q^{\mathrm{ND}}_{\Lambda_j}(A|_{\Lambda_j^*}) + \kappa^2\sum_{b\in R_j}A(b)^2$, $\kappa^2 = s\varepsilon^2$. Through the identification of $\Lambda_j^*\cup R_j$ with the bonds of $T_{j,\varepsilon}$, it is a measure on the same space as $\mathbb{P}^a_{j,\sqrt{s}}$, with density proportional to $e^{-S_{\mathrm{ref}}}$, where $S_{\mathrm{ref}}(A,\varphi) := \frac{\varepsilon}{2}Q_{\mathrm{ref}}(A) + S^{\mathrm{m}}_{\Lambda_j}(A,\varphi)$. Its normalisation $Z_{\mathrm{ref}} = \int e^{-S_{\mathrm{ref}}}dA\,d\varphi$ satisfies $\log Z_{\mathrm{ref}} = \log Z_j^{\mathrm{ND}}(s,a) + \frac{3}{2}N_j^2\log(2\pi/(s\varepsilon^3))$,

since each bond of $R_j$ carries the factor $\int e^{-\frac{1}{2}s\varepsilon^3 A^2}dA = (2\pi/(s\varepsilon^3))^{1/2}$. Moreover $S^{\mathrm{T}}_{\sqrt{s},a} = S_{\mathrm{ref}} + W_A + W_\varphi$ with

$$W_A = \frac{\varepsilon}{2}\sum_{(b,b')\in P'}\big(A(b)-A(b')\big)^2 \geq 0, \qquad W_\varphi = -\varepsilon\sum_{b\in R_j} r_{b_-}r_{b_+}\cos u_b,$$

where $P'$ is the set of pairs summed in $K_T$ but not in $K_{\Lambda_j}$, and $u_b$ is as in the proof of Lemma 9.4. Each of the $3N_j^2$ bonds of $R_j$ belongs to at most six pairs of $K_T$. Every other pair of $K_T$ which is not a pair of $K_{\Lambda_j}$ has the form $((x;\mu),(x+\varepsilon e_\nu;\mu))$ with $x_\nu = \ell_j/2-\varepsilon$, and there are $9N_j^2$ pairs of this form. Hence $|P'|\leq 27N_j^2$. By Jensen's inequality,

$$-\langle W_A+W_\varphi\rangle_{\mathrm{ref}} \leq \log\frac{Z_j^{\mathrm{T}}(s,a)}{Z_{\mathrm{ref}}} \leq -\mathbb{P}^a_{j,\sqrt{s}}(W_A+W_\varphi).$$

Lemma 9.4 applies to both measures, since $Q_{\mathrm{ref}}\geq\kappa^2|\cdot|^2$. It gives $\mathbb{M}((A(b)-A(b'))^2)\leq 4(s\varepsilon^3)^{-1}$ and $|\mathbb{M}(r_{b_-}r_{b_+}\cos u_b)|\leq \mathfrak{m}(a)$ for both. Hence both expectations of $W_A+W_\varphi$ are at most $54N_j^2/(s\varepsilon^2)+3\varepsilon N_j^2\mathfrak{m}(a)$ in absolute value, and the claim follows after division by $N_j^3$. □

**Lemma 9.9.** *Let $\varepsilon$ be a spacing of the family and $a\in\mathbb{R}$. For $s>0$ put $\Gamma(s) := \omega^{\mathrm{ND},a}_{\varepsilon,\sqrt{s}}(|A(0)|^2)$.*

(a) *For every $s>0$, $\mathsf{g}_j(s,a)\to\Gamma(s)$ as $j\to\infty$, and $0\leq \mathsf{g}_j(s,a),\Gamma(s)\leq 3/(s\varepsilon^3)$.*
(b) *$\mathsf{g}_j(\cdot,a)$ and $\Gamma$ are nonincreasing on $(0,\infty)$. In particular $\Gamma$ has at most countably many points of discontinuity.*
(c) *For $s_0,s>0$, $\mathsf{p}_j^{\mathrm{ND}}(s,a)-\mathsf{p}_j^{\mathrm{ND}}(s_0,a)\to -\frac{\varepsilon^3}{2}\int_{s_0}^s\Gamma(u)\,du$ as $j\to\infty$.*

*Proof.* (a) The bounds follow from Lemma 9.4(a) and Proposition 9.7(b). Write $\langle\cdot\rangle_C := \langle\cdot\rangle^{\mathrm{ND}}_{C;\sqrt{s},a}$ and $B_n := [-n\varepsilon,n\varepsilon]^3\cap\varepsilon\mathbb{Z}^3$ for $n\in\mathbb{N}$. Let $x\in\Lambda_j$ with $x+B_n\subset\Lambda_j$, $n\geq 1$. Then the bonds $(x;\mu)$ lie in $(x+B_n)^*$ and in $\Lambda_j^*$, and by Proposition 9.7(b),(c) and translation covariance,

$$\omega^{\mathrm{ND}}\big(A(0;\mu)^2\big)\leq\langle A(x;\mu)^2\rangle_{\Lambda_j}\leq\langle A(x;\mu)^2\rangle_{x+B_n}=\langle A(0;\mu)^2\rangle_{B_n}.$$

The other sites of $\Lambda_j$ are at most $6nN_j^2$ in number, and each contributes at most $3/(s\varepsilon^3)$ to $N_j^3\mathsf{g}_j(s,a)$ and to $N_j^3\Gamma(s)=\sum_{x\in\Lambda_j}\sum_\mu\omega^{\mathrm{ND}}(A(0;\mu)^2)$. Hence

$$|\mathsf{g}_j(s,a)-\Gamma(s)|\leq 3\max_\mu\Big(\langle A(0;\mu)^2\rangle_{B_n}-\omega^{\mathrm{ND}}\big(A(0;\mu)^2\big)\Big)+\frac{18n}{N_j s\varepsilon^3}.$$

Let $j\to\infty$ and then $n\to\infty$, and use Proposition 9.7(c) for the sequence $(B_n)$.

(b) By Lemma 9.8(a), $\mathsf{g}_j(\cdot,a)=-\frac{2}{\varepsilon^3}\partial_s\mathsf{p}_j^{\mathrm{ND}}(\cdot,a)$ is nonincreasing. So is its pointwise limit $\Gamma$, and a monotone function has at most countably many discontinuities.

(c) By Lemma 9.8(a) the left side is $-\frac{\varepsilon^3}{2}\int_{s_0}^s\mathsf{g}_j(u,a)\,du$. Apply (a) and dominated convergence, with the bound $3/(u\varepsilon^3)$ on the interval between $s_0$ and $s$. □

For the coefficient $a=a_\natural(\varepsilon)$ we write $\omega^{\mathrm{ND}}_{\varepsilon,\mu_0}:=\omega^{\mathrm{ND},a_\natural(\varepsilon)}_{\varepsilon,\mu_0}$, $\Gamma_\varepsilon:=\Gamma$, and

$$\mathsf{E}_\varepsilon := \big\{\mu_0>0:\ \Gamma_\varepsilon \text{ is discontinuous at } \mu_0^2\big\}. \tag{9.2}$$

By Lemma 9.9(b) this set is countable. It is defined by the Neumann/Dirichlet box states alone. We also write $\omega_{\varepsilon,\mu_0}$ for the law $\omega_\varepsilon$ of Theorem 4.3 for the parameters $\mu_0,e,\lambda,\ m^2$ and $\delta m^2=a_\natural(\varepsilon)-m^2$; by Proposition 9.7(d) it is the image of $\omega^{\mathrm{ND}}_{\varepsilon,\mu_0}$ under $A\mapsto(F(p))_p$.

**Lemma 9.10** (Griffiths' lemma)**.** *Let $I$ be an open interval and $\varpi_j$, $j\in\mathbb{N}$, convex differentiable functions on $I$ which converge pointwise to a function $\varpi$. If $\varpi$ is differentiable at $s\in I$, then $\varpi_j'(s)\to\varpi'(s)$.*

*Proof.* For small $\eta > 0$, convexity gives $\eta^{-1}(\varpi_j(s)-\varpi_j(s-\eta)) \le \varpi_j'(s) \le \eta^{-1}(\varpi_j(s+\eta)-\varpi_j(s))$. Let $j \to \infty$ and then $\eta \downarrow 0$. □

9.5. **The two-point function at fixed spacing.** Let $\varepsilon$ be a spacing of the family and $\mu_0 > 0$. A *limit point* of $(\mathbb{P}_{j,\mu_0})_j$ is a probability measure $\omega$ on $\mathbb{R}^{(\varepsilon\mathbb{Z}^3)^*}$ such that, along a sequence $j_i \to \infty$,

$$\mathbb{P}_{j_i,\mu_0}\big(e^{iA(f)}\big) \to \omega\big(e^{iA(f)}\big) \qquad \text{for every finitely supported real } f \text{ on } (\varepsilon\mathbb{Z}^3)^*. \tag{9.3}$$

By Lemma 9.4(a) the laws of finitely many bond variables under $\mathbb{P}_{j,\mu_0}$ are tight, so by a diagonal argument and Kolmogorov's theorem every subsequence of $(\mathbb{P}_{j,\mu_0})_j$ has a further subsequence along which (9.3) holds for some $\omega$.

**Theorem 9.11.** *Let $\varepsilon$ be a spacing of the family, and let $\mathsf{E}_\varepsilon$ be the countable set* (9.2)*. Let $\mu_0 > 0$ and let $\omega$ be a limit point of $(\mathbb{P}_{j,\mu_0})_j$.*

(i) *$\omega(A(f)^2) \le \omega^{\mathrm{ND}}_{\varepsilon,\mu_0}(A(f)^2)$ for every finitely supported real $f$ on $(\varepsilon\mathbb{Z}^3)^*$.*

(ii) *If $\mu_0 \notin \mathsf{E}_\varepsilon$, then, for all bonds $b, b'$ and all plaquettes $p, q$ of $\varepsilon\mathbb{Z}^3$, as $j \to \infty$ along the whole sequence,*

$$\mathbb{P}_{j,\mu_0}\big(A(b)A(b')\big) \to \omega^{\mathrm{ND}}_{\varepsilon,\mu_0}\big(A(b)A(b')\big), \qquad \mathbb{P}_{j,\mu_0}\big(F(p)F(q)\big) \to \omega_{\varepsilon,\mu_0}\big(F(p)F(q)\big),$$

*the left sides being defined once $b, b', p, q$ lie in $\Lambda_j$. In particular $\omega(A(b)A(b')) = \omega^{\mathrm{ND}}_{\varepsilon,\mu_0}(A(b)A(b'))$ and $\omega(F(p)F(q)) = \omega_{\varepsilon,\mu_0}(F(p)F(q))$.*

For $\mu_0 \in (0, \mu_*]$, $\mathbb{P}_{j,\mu_0}(F(p)F(q)) = \langle F(p)F(q)\rangle_{\varepsilon,j,\mu_0}$ is the curvature two-point function of the torus states of [v1].

*Proof.* Write $\omega^{\mathrm{ND}} := \omega^{\mathrm{ND}}_{\varepsilon,\mu_0}$ and $a_\natural := a_\natural(\varepsilon)$, and let $(j_i)$ be the sequence in (9.3). By Lemma 9.4(a) and Fatou's lemma, $\omega(e^{tA(f)}) \le \exp(t^2\|f\|^2_{2,\varepsilon}/(2\mu_0^2))$; the fourth moments of $A(f)$ under $\mathbb{P}_{j,\mu_0}$ are bounded uniformly in $j$, so the first and second moments converge along $(j_i)$ to those under $\omega$, and $\omega(A(f)) = 0$. Since the $\mathbb{P}_{j,\mu_0}$ are invariant under the translations of $T_{j,\varepsilon}$, $\omega$ is invariant under the translations by $\varepsilon\mathbb{Z}^3$.

(i) Let $C$ be a box with $\operatorname{supp} f \subset C^*$, and put $a_j := a_\natural + |b_j - a_\natural| \ge b_j$. For large $j$, $C \subset \Lambda_j$, and Lemma 9.6(b) gives $\langle e^{itA(f)}\rangle^{\mathrm{ND}}_{C;\mu_0,a_j} \le \mathbb{P}_{j,\mu_0}(e^{itA(f)})$ for $t \in \mathbb{R}$. Let $j = j_i \to \infty$. The right side tends to $\omega(e^{itA(f)})$. By Proposition 9.2, $a_j \to a_\natural$, and the finite-dimensional integrals defining $\langle\cdot\rangle^{\mathrm{ND}}_{C;\mu_0,a}$ are continuous in $a$ by dominated convergence, since $v$ is increasing in $a$. Hence $\langle e^{itA(f)}\rangle^{\mathrm{ND}}_{C;\mu_0,a_\natural} \le \omega(e^{itA(f)})$, and the supremum over $C$ gives $\omega^{\mathrm{ND}}(e^{itA(f)}) \le \omega(e^{itA(f)})$, by Proposition 9.7(a). Both sides are expectations of $\cos(tA(f))$. Lemma 9.5, applied to $A(f)$ under $\omega^{\mathrm{ND}}$ and under $\omega$, gives (i).

(ii) Fix $s_0 > 0$ and put $\varpi_j(u) := \mathsf{p}^{\mathrm{T}}_j(u, b_j) - \mathsf{p}^{\mathrm{T}}_j(s_0, b_j)$ for $u > 0$. By Lemma 9.8(a), $\varpi_j$ is convex and differentiable, with $\varpi_j'(u) = -\frac{\varepsilon^3}{2}\mathbb{P}_{j,\sqrt{u}}(|A(0)|^2)$. By Lemma 9.8(b),(c) and Proposition 9.2, $\mathsf{p}^{\mathrm{T}}_j(u, b_j) - \mathsf{p}^{\mathrm{ND}}_j(u, a_\natural) \to 0$ for every $u > 0$; for (b) we use that $|b_j - a_\natural| \le 1$ for large $j$ and $\mathfrak{m}(\min(b_j, a_\natural)) \le \mathfrak{m}(a_\natural - 1)$. Together with Lemma 9.9(c), $\varpi_j(u) \to \varpi(u) := -\frac{\varepsilon^3}{2}\int_{s_0}^u \Gamma_\varepsilon(u')\,du'$ for every $u > 0$. Since $\Gamma_\varepsilon$ is monotone, $\varpi$ is differentiable at every point $u$ of continuity of $\Gamma_\varepsilon$, with $\varpi'(u) = -\frac{\varepsilon^3}{2}\Gamma_\varepsilon(u)$. If $\mu_0 \notin \mathsf{E}_\varepsilon$, then $s := \mu_0^2$ is such a point, and Lemma 9.10 gives

$$\mathbb{P}_{j,\mu_0}\big(|A(0)|^2\big) \to \Gamma_\varepsilon(s) = \omega^{\mathrm{ND}}\big(|A(0)|^2\big) \qquad (j \to \infty).$$

In particular $\omega(|A(0)|^2) = \omega^{\mathrm{ND}}(|A(0)|^2)$. Let $\Xi(f,h) := \omega^{\mathrm{ND}}(A(f)A(h)) - \omega(A(f)A(h))$, a symmetric bilinear form on the finitely supported real functions on $(\varepsilon\mathbb{Z}^3)^*$, which is positive semidefinite by (i). For a bond $b$ put $\hat\delta_b := \varepsilon^{-3}\mathbf{1}_b$, so that $A(\hat\delta_b) = A(b)$. By the translation invariance of $\omega$ and of $\omega^{\mathrm{ND}}$, $\Xi(\hat\delta_{(x;\mu)}, \hat\delta_{(x;\mu)}) = \Xi(\hat\delta_{(0;\mu)}, \hat\delta_{(0;\mu)}) \ge 0$, and the sum of these over

$\mu$ is $\omega^{\mathrm{ND}}(|A(0)|^2) - \omega(|A(0)|^2) = 0$. So $\Xi$ vanishes on the diagonal, and the Cauchy–Schwarz inequality $|\Xi(\hat\delta_b, \hat\delta_{b'})|^2 \le \Xi(\hat\delta_b, \hat\delta_b)\,\Xi(\hat\delta_{b'}, \hat\delta_{b'})$ gives $\omega(A(b)A(b')) = \omega^{\mathrm{ND}}(A(b)A(b'))$. Since $F(p) = \varepsilon^{-1}\sum_{b'\in\partial p} A(b')$ is a linear combination of bond variables, also $\omega(F(p)F(q)) = \omega^{\mathrm{ND}}(F(p)F(q))$, and this is $\omega_{\varepsilon,\mu_0}(F(p)F(q))$ by Proposition 9.7(d).

Finally, every subsequence of $(\mathbb{P}_{j,\mu_0})_j$ has a further subsequence along which (9.3) holds for some limit point $\omega$, and along which the second moments converge to those of $\omega$. These limits are $\omega^{\mathrm{ND}}(A(b)A(b'))$ and $\omega_{\varepsilon,\mu_0}(F(p)F(q))$, independently of the subsequence. Hence the convergence holds along the whole sequence. □

*Remark* 9.12. (a) The gauge-field part of (ii) compares the torus states with the Neumann/Dirichlet limit, whose Gaussian part, like that of the torus states, is a componentwise form. The auxiliary states of Definition 2.4(a) have a different gauge-fixing term, and they are compared only on gauge-invariant observables: on the curvature, $\omega_{\varepsilon,\mu_0}$ is the common limit of the auxiliary and of the Neumann/Dirichlet states (Theorem 4.3(f)). (b) The method gives only second moments. If the two-point functions agree, the inequality $\omega^{\mathrm{ND}}(e^{itA(f)}) \le \omega(e^{itA(f)})$ of the proof gives at order $t^4$ only $\omega^{\mathrm{ND}}(A(f)^4) \le \omega(A(f)^4)$. (c) For $\mu_0 \in \mathsf{E}_\varepsilon$ only part (i) is available.

9.6. **Volume-first continuum limits.** Let $\varepsilon$ be a spacing of the family and $\mu_0 \in (0, \mu_*]$. The torus states satisfy $\langle e^{F(g)}\rangle_{\varepsilon,j,\mu_0} \le e^{\frac12\|g\|^2_{2,\varepsilon}}$ for every real 2-cochain $g$ on $T_{j,\varepsilon}$ [v1, Lemma 7.3]. A *curvature limit point* of the torus states at $(\varepsilon, \mu_0)$ is a probability measure $\omega^{(\varepsilon)}$ on $E_\varepsilon = \mathbb{R}^{\mathcal{P}_\varepsilon}$ (Section 4) such that, along a sequence $j_i \to \infty$, $\langle e^{iF(g)}\rangle_{\varepsilon,j_i,\mu_0} \to \omega^{(\varepsilon)}(e^{iF(g)})$ for every finitely supported real 2-cochain $g$ on $\varepsilon\mathbb{Z}^3$. By the bound just quoted, the laws of finitely many curvatures are tight, so curvature limit points exist, by a diagonal argument and Kolmogorov's theorem. By Fatou's lemma $\omega^{(\varepsilon)}(e^{F(g)}) \le e^{\frac12\|g\|^2_{2,\varepsilon}}$ for finitely supported real $g$, and the proof of Theorem 4.3(d) applies to $\omega^{(\varepsilon)}$. Hence the continuum curvature field $F_\varepsilon$ of $\omega^{(\varepsilon)}$ is defined by (5.2), with $\omega^{(\varepsilon)}$ in place of $\omega_\varepsilon$, and Lemma 5.3(a) holds for it.

**Definition 9.13.** Let $\mu_0 \in (0, \mu_*]$. A *volume-first limit point* of the torus states of [v1] at photon mass $\mu_0$ is a law $\nu$ on $X$ obtained as follows. For each $K \ge 0$ let $\omega^{(K)}$ be a curvature limit point of the torus states at $(\varepsilon_K, \mu_0)$, and let $\nu_K$ be the law of the continuum curvature field $F_{\varepsilon_K}$ of $\omega^{(K)}$. Then $\nu$ is the weak limit on $H^{-r}$, for some $r > 7$, of a subsequence $(\nu_{K_i})$, regarded as a law on $X$.

Thus $\nu$ is a limit of the torus states of [v1] along a volume-first sequence: the torus grows at each fixed spacing, and then the spacing goes to zero. The limit states of [v1, Thm. 3.1(ii)] are taken jointly in the spacing and the torus [v1, Rem. 3.3(a)]; Definition 9.13 and Corollary 9.14 concern volume-first sequences only. By the proof of Theorem 5.4(a), which uses only Lemma 5.3(a) and Lemma 5.2(a), the sequence $(\nu_K)$ is tight on $H^{-r}$ for every $r > 7$, so volume-first limit points exist.

**Corollary 9.14.** *Put* $\mathsf{E} := \bigcup_{K\ge0} \mathsf{E}_{\varepsilon_K}$*, a countable set, and let* $\mu_0 \in (0, \mu_*] \setminus \mathsf{E}$*. Let* $\nu$ *be a volume-first limit point of the torus states at photon mass* $\mu_0$*, the limit of* $(\nu_{K_i})$ *on* $H^{-r}$ *as in Definition 9.13. For* $K \ge 0$ *let* $\mu_K$ *be the law of the continuum curvature field* $F_{\varepsilon_K}$ *of* $\omega_{\varepsilon_K,\mu_0}$*, the infinite-volume law of the auxiliary states at spacing* $\varepsilon_K$ *with the parameters* $\mu_0, e, \lambda$*,* $m^2$ *and* $\delta m^2 = a_{\natural}(\varepsilon_K) - m^2$*.*

(a) *For all* $f, h \in \mathcal{T}$ *the limit* $\lim_i \mu_{K_i}(F(f)F(h))$ *exists and equals* $\nu(F(f)F(h))$*. The sequence* $(\mu_{K_i})$ *has weak limit points on* $H^{-r}$*; each of them is a law* $\mu$ *as in Theorem 1.1(b), and* $\nu(F(f)F(h)) = \mu(F(f)F(h))$ *for all real* $f, h \in L^2(\mathbb{R}^3;\Lambda^2)$*.*

(b) $\nu\big(F(\gamma\cdot f)F(\gamma\cdot h)\big) = \nu\big(F(f)F(h)\big)$ *for all* $\gamma \in E(3)$ *and all real* $f, h \in L^2(\mathbb{R}^3;\Lambda^2)$*.*

*Proof.* Let $K \geq 0$, $\varepsilon = \varepsilon_K$, and let $(j_i)$ be the sequence that defines $\omega^{(K)}$.

*Step 1: plaquettes.* Since $\mu_0 \notin \mathsf{E}_\varepsilon$, Theorem 9.11(ii) gives $\langle F(p)F(q)\rangle_{\varepsilon,j,\mu_0} \to \omega_{\varepsilon,\mu_0}(F(p)F(q))$ as $j \to \infty$. Along $(j_i)$ the laws of finitely many curvatures converge weakly to those under $\omega^{(K)}$, and by [v1, Lemma 7.3] their fourth moments are bounded uniformly in $j$, as in Lemma B.1(a). Hence $\omega^{(K)}(F(p)F(q)) = \omega_{\varepsilon,\mu_0}(F(p)F(q))$ for all plaquettes $p, q$.

*Step 2: sampled test forms.* Let $\omega'$ be $\omega^{(K)}$ or $\omega_{\varepsilon,\mu_0}$. Then $\omega'(F(p)^2) \leq \varepsilon^{-3}$ for every $p$, by the bound $\omega'(F(g)^2) \leq \|g\|^2_{2,\varepsilon}$. If $g, g'$ are absolutely summable real 2-cochains, the partial sums of $F(g) = \varepsilon^3 \sum_p g(p)F(p)$ converge in $L^2(\omega')$, by Minkowski's inequality, and almost surely, so

$$\omega'\big(F(g)F(g')\big) = \varepsilon^6 \sum_{p,q} g(p)g'(q)\, \omega'\big(F(p)F(q)\big),$$

the double series converging absolutely. With $g = \iota_\varepsilon f$, $g' = \iota_\varepsilon h$ for $f, h \in \mathcal{T}$, which are absolutely summable by Lemma 5.2(a), Step 1 gives $\nu_K(F(f)F(h)) = \omega^{(K)}(F_\varepsilon(f)F_\varepsilon(h)) = \omega_{\varepsilon,\mu_0}(F_\varepsilon(f)F_\varepsilon(h)) = \mu_K(F(f)F(h))$.

*Step 3: the continuum limit.* For $f \in \mathcal{T}$ the map $F \mapsto F(f)$ is continuous on $H^{-r}$. By Lemma 5.3(a), for $\nu_K$ and for $\mu_K$, and by Lemma 5.2(a), $\nu_K(F(f)^4)$ and $\mu_K(F(f)^4)$ are at most $16C^4\|f\|_4^4$, uniformly in $K$, as in Lemma B.1(a). Hence weak convergence on $H^{-r}$ implies convergence of the expectations of $F(f)F(h)$. Along $(K_i)$ this gives $\nu(F(f)F(h)) = \lim_i \nu_{K_i}(F(f)F(h)) = \lim_i \mu_{K_i}(F(f)F(h))$, by Step 2. By Theorem 5.4(a), applied to the laws $\omega_{\varepsilon_K,\mu_0}$ at the spacings $\varepsilon_K = L^{-K'}$, $K' = K + n_0$ (with arbitrary parameters at the spacings $L^{-K'}$, $K' < n_0$), the sequence $(\mu_{K_i})$ is tight on $H^{-r}$. Each weak limit point $\mu$ along a further subsequence is a law as in Theorem 1.1(b), and, by the same uniform integrability, $\mu(F(f)F(h))$ is the limit of $\mu_{K_i}(F(f)F(h))$ along that subsequence, which is $\nu(F(f)F(h))$. Both $\nu$ and $\mu$ satisfy (G), by the proof of Theorem 5.4(b) and by Theorem 5.4(b), so $F(f)$ extends to real $f \in L^2(\mathbb{R}^3;\Lambda^2)$ as a contraction into $L^2$ (Lemma B.1(b)), and the identity extends by continuity.

*Step 4: Euclidean invariance.* Let $\mu$ be as in (a). By Theorem 1.1(c), $\mu(F(\gamma\cdot f)F(\gamma\cdot h)) = \mu(F(f)F(h))$ for $\gamma \in E(3)$ and real $f, h \in L^2(\mathbb{R}^3;\Lambda^2)$. Since $\gamma\cdot f, \gamma\cdot h \in L^2(\mathbb{R}^3;\Lambda^2)$, (a) applies to them, and (b) follows. □

*Remark* 9.15. (a) Part (b) uses Theorem 1.1(c), and hence the rotation theorem [S, Thm. 18.21] (Theorem 6.2). Part (a) and Theorem 9.11 do not use [S]. (b) Since $\mathsf{E}$ is countable, the countable sets $Q_0 \subset (0, \mu_*]$ accumulating at 0 of [v1, Thm. 3.1(ii)] can be chosen disjoint from $\mathsf{E}$. This choice would matter for the joint limit points of [v1] only if an estimate uniform in the spacing, as in Section 10, item 3, were proved; nothing is proved here for those limit points. (c) In the setting of Corollary 9.14, for $\mu_0 \in (0, \mu_*] \cap \mathsf{E}$, Theorem 9.11(i) still gives a one-sided bound. Every curvature limit point $\omega^{(K)}$ is the image under $A \mapsto (F(p))_p$ of a limit point $\omega$ of $(\mathbb{P}_{j,\mu_0})_j$ along a subsequence of its sequence, so $\omega^{(K)}(F(g)^2) = \omega(A(d^*g)^2) \leq \omega^{\mathrm{ND}}_{\varepsilon_K,\mu_0}(A(d^*g)^2) = \omega_{\varepsilon_K,\mu_0}(F(g)^2)$ for finitely supported real $g$. By Steps 2 and 3 of the proof, $\nu(F(f)^2) \leq \mu(F(f)^2)$ for $f \in \mathcal{T}$ and every weak limit point $\mu$ of $(\mu_{K_i})$ on $H^{-r}$.

## 10. Open problems

(1) *The external theorem.* Parts (c) and (d) of Theorem 1.1 rest on [S, Prop. 18.1, Thm. 18.21]. An independent verification of the argument of [S, Section 18] would be desirable.

(2) *The torus states of* [v1]*: the full law.* For $m > 0$, $0 < \lambda < 0.1m$ and $0 < e^2 < 0.1m$, the companion paper [v1] constructs continuum states, with cubic symmetry, of a closely related lattice model [v1, Thm. 3.1]. There the gauge field has the Feynman form on cubic tori; the bare mass is given by the torus counterterm scheme of [v1, Def. 2.2], in

which the counterterm is computed on each torus at a fixed reference photon mass $\mu_*$ and depends on the torus; the photon masses are $\mu_0 \in (0, \mu_*]$; the spacing and the torus go to their limits jointly; and the photon mass is then removed, $\mu_0 \downarrow 0$. By Remark 3.6, the argument of Theorem 3.2 does not apply to cubic tori: for an even number of sites per side, positivity at a diagonal plane fails for the algebra of the closed half-torus which contains the self-mapped layer. So Theorem 1.1 does not apply to those states directly. At a fixed spacing $\varepsilon$ and for $\mu_0$ outside the countable set $\mathsf{E}_\varepsilon$, Theorem 9.11 identifies the two-point functions of the gauge field and of the curvature of the torus limits with those of the box limits at the coefficient $a_\natural(\varepsilon)$. Equality of the full laws is open, already at a fixed spacing. The correlation inequalities give only $\omega^{\mathrm{ND}}_{\varepsilon,\mu_0}(e^{iA(f)}) \le \omega(e^{iA(f)})$ for the torus limits $\omega$, and, once the second moments agree, the terms of order $t^4$ of this inequality give only an inequality between fourth moments (Remark 9.12(b)). For $\mu_0 \in \mathsf{E}_\varepsilon$ only the inequality of Theorem 9.11(i) is known.

(3) *Joint limits.* Corollary 9.14 concerns the volume-first limit points of Definition 9.13. The limit states of [v1, Thm. 3.1(ii)] are joint limits in the spacing and the torus, and equality of two-point functions at each fixed spacing does not by itself give equality along joint sequences; the convergence in Theorem 9.11(ii) comes from Lemma 9.10 and has no rate. An estimate uniform in the spacing, for instance

$$\lim_{j\to\infty} \sup_{K\ge 0} \Big| \langle F(\iota_{\varepsilon_K} f) F(\iota_{\varepsilon_K} h)\rangle_{\varepsilon_K, j, \mu_0} - \omega_{\varepsilon_K,\mu_0}\big(F_{\varepsilon_K}(f) F_{\varepsilon_K}(h)\big) \Big| = 0$$

for real test two-forms $f, h$ of compact support, would give the conclusion of Corollary 9.14 for the joint limit points at $\mu_0 \notin \mathsf{E}$; it is open. The states of [v1] at zero photon mass, the limit $\mu_0 \downarrow 0$ and the state $\flat$ of [v1, Def. 2.7], are not reached, since the argument needs $\mu_0 > 0$.

(4) *King's continuum state.* The relation between the limit points $\mu$ and King's continuum state, in which the continuum limit is taken before the infinite-volume limit [K2, Thms. 2.1–2.2], is open. Theorem 4.3(f) identifies the two infinite-volume limits at fixed spacing; the question is whether the limits $\varepsilon \to 0$ and $\Omega \uparrow \mathbb{R}^3$ can be exchanged.

(5) *Uniqueness.* The continuum limit points are obtained along subsequences. Uniqueness of the limit, and its independence of the choice of $L$, are open.

(6) *The Wick square.* A continuum Wick square $|\varphi|^2$ for the auxiliary states in the volume-first order, with bounds of the type used in Section 8, is not constructed here. Given such a field, hypothesis (iii) of Proposition 8.1 has to be derived from reflection positivity at the axis and diagonal planes. Higher moments involving the Wick square are open beyond the two-point function. In particular, contact terms of order at least three with at least two Wick-square entries are not controlled by the bound $\mathcal{N}$ of Proposition 8.2.

(7) *Regularity of the components.* Gaussian domination and moment bounds for the ergodic components $\mu_\xi$, and hence a Wightman reconstruction for each component, are open.

(8) *Non-triviality.* All bounds used here are uniform in the parameters, and the point mass at $F = 0$ satisfies all conclusions of Theorem 1.1. Lower bounds which show that, for King's parameters, the limit points are not Gaussian (or not zero) are open.

(9) *The Higgs phase.* The present results hold for all bare parameters but do not distinguish phases. The relation of the limit points to a continuum theory in the Higgs phase, where the photon mass should be generated dynamically and the limit $\mu_0 \downarrow 0$ is natural, is open. The upper comparison Lemma 2.12 needs a margin of order $1/\mu_0$.

## Appendix A. Ergodic decomposition of reflection-positive laws

We use the notation of Section 5.1: $\mathcal{T} = \mathcal{S}(\mathbb{R}^3;\Lambda^2)$, $X = \mathcal{T}'$ with the $\sigma$-algebra $\mathcal{B}$, the action $(g\cdot F)(f) = F(g^*f)$ of $E(3)$, $\alpha_g\Phi = \Phi\circ g^{-1}$, and $\Sigma(U)$. Note that $\alpha_g$ maps $\Sigma(U)$-measurable functions to $\Sigma(gU)$-measurable ones, and that $r\tau_a r = \tau_{\bar r a}$ for a reflection $r$ with linear part $\bar r$. For a unit vector $n$ and $c\in\mathbb{R}$ we write $\pi_{n,c} = \{n\cdot x = c\}$, $\pi^+_{n,c} = \{n\cdot x > c\}$ and $r_{n,c}$ for the reflection; for $N = -N \subset S^2$ let $\mathcal{P}(N)$ be the set of planes $\pi_{n,c}$ with $n\in N$, $c\in\mathbb{R}$.

**Lemma A.1.** *The measurable space $(X,\mathcal{B})$ has the following properties.*

(a) *$(X,\mathcal{B})$ is Borel isomorphic to a Borel subset of the compact metric space $K = [0,1]^{\mathbb{N}}$.*

(b) *For a $\mathbb{Q}$-vector space $D\subset\mathcal{T}$ let $\mathcal{A}(D)$ be the $\mathbb{Q}[i]$-linear span of the functions $e^{iF(f)}$, $f\in D$. If $\mathcal{A}$ is an algebra of bounded measurable functions which contains $1$ and is closed under complex conjugation, then $\mathcal{A}$ is dense in $L^2(\sigma(\mathcal{A}),m)$ for every law $m$. There are countable $\mathbb{Q}$-vector spaces $D_0$ dense in $\mathcal{T}$ and, for every open $U$, $D_U\subset\{f\in\mathcal{T}:\operatorname{supp} f\subset U \text{ compact}\}$, such that $\sigma(\mathcal{A}(D_0)) = \mathcal{B}$ and $\sigma(\mathcal{A}(D_U)) = \Sigma(U)$; $\mathcal{A}(D_0)$ and $\mathcal{A}(D_U)$ are algebras of the stated kind.*

(c) *The map $(a,F)\mapsto\tau_a\cdot F$ is jointly measurable, and for $\Phi = \phi(F(f_1),\dots,F(f_k))$ with $\phi\in C_b(\mathbb{R}^k)$ the map $g\mapsto\Phi(g\cdot F)$ is continuous on $E(3)$ for every $F$.*

*Proof.* (a) Let $(h_l)$ be an enumeration of the Hermite basis of Section 5.1. The map $\iota(F) = (F(h_l))_l\in\mathbb{R}^{\mathbb{N}}$ is measurable and injective, because Hermite expansions converge in $\mathcal{T}$. Its inverse is measurable on the image, since $F(f) = \lim_M\sum_{l\le M}\langle h_l,f\rangle F(h_l)$. The image is $\bigcup_{r\in\mathbb{N}}\{t:\sum_l\lambda_l^{-r}t_l^2<\infty\}$, a Borel set, by the characterisation of $X = \bigcup_r H^{-r}$. Compose with a homeomorphism of $\mathbb{R}$ onto $(0,1)$ in each coordinate. (b) Let $\bar H$ be the $L^2(m)$-closure of $\mathcal{A}$, a closed complex subspace. It contains the polynomials in the real and imaginary parts of finitely many elements of $\mathcal{A}$, hence, by the Weierstrass approximation theorem on the compact range of these bounded functions, all continuous functions of them, and then all bounded Borel functions of them which are bounded pointwise limits of continuous ones. Hence it contains the indicators of the sets $\{\operatorname{Re}a_1<t_1,\dots,\operatorname{Im}a_k<t_k\}$, $a_i\in\mathcal{A}$, which form a $\pi$-system generating $\sigma(\mathcal{A})$. The sets $B$ with $\mathbf{1}_B\in\bar H$ form a $\lambda$-system. By the $\pi$–$\lambda$ theorem $\bar H$ contains all indicators of sets in $\sigma(\mathcal{A})$, hence all of $L^2(\sigma(\mathcal{A}),m)$. For $D_0$ take the $\mathbb{Q}$-span of a countable dense subset of $\mathcal{T}$; for $D_U$ take the $\mathbb{Q}$-span of countable subsets dense in the Fréchet spaces of test forms supported in $K_l$, where $K_l$ is a compact exhaustion of $U$. Every $F(f)$ is then a pointwise limit of $F(f_l)$ with $f_l$ in the corresponding space, and $F(f)$ is measurable with respect to $\sigma(e^{iF(tf)}:t\in\mathbb{Q})$. (c) For $\Phi = e^{iF(f)}$, $(a,F)\mapsto e^{iF(\tau_a^*f)}$ is continuous in $a$ and measurable in $F$, hence jointly measurable; the monotone class theorem extends this to all bounded measurable $\Phi$. Continuity in $g$ follows from the continuity of $g\mapsto g^*f_i$ from $E(3)$ to $\mathcal{T}$. □

**Lemma A.2.** *Let $\mu$ be a law and $\mathcal{I}'\subset\mathcal{B}$ a $\sigma$-algebra. There are laws $\mu_\xi$, $\xi\in X$, such that $\xi\mapsto\mu_\xi(B)$ is $\mathcal{I}'$-measurable and $\mu(W\mathbf{1}_B) = \int W(\xi)\mu_\xi(B)\,\mu(d\xi)$ for all $B\in\mathcal{B}$ and all bounded $\mathcal{I}'$-measurable $W$. They are unique up to a $\mu$-null set of $\xi$, and $\mu_\xi(hk) = h(\xi)\mu_\xi(k)$ for $\mu$-almost every $\xi$, for bounded $\mathcal{I}'$-measurable $h$ and bounded measurable $k$.*

*Proof.* By Lemma A.1(a) we regard $\mu$ as a law on $K$ carried by the Borel set $\iota X$. Let $\mathcal{D}$ be a countable $\mathbb{Q}[i]$-algebra containing 1, closed under complex conjugation and dense in $C(K)$ (for instance the polynomials in finitely many coordinates with coefficients in $\mathbb{Q}[i]$), and let $h_f$ be $\mathcal{I}'$-measurable versions of the conditional expectations $\mu(f\mid\mathcal{I}')$, $f\in\mathcal{D}$. Off an $\mathcal{I}'$-measurable null set the map $f\mapsto h_f(\xi)$ is $\mathbb{Q}[i]$-linear, commutes with conjugation, is nonnegative on the nonnegative elements of $\mathcal{D}$, and satisfies $h_1(\xi) = 1$; these are countably many conditions. Then $|h_f(\xi)|\le 2\sup|f|$, so the map extends by continuity to a positive normalised functional on $C(K)$, and the Riesz–Markov theorem gives a probability measure $\mu_\xi$ on $K$. On the null set put

$\mu_\xi = \mu$. The identity $\mu(Wf) = \int W \mu_\xi(f)\, d\mu$ passes from $\mathcal{D}$ to $C(K)$ by uniform limits, to open sets by monotone limits, and to all Borel sets by Dynkin's theorem. Since $\int \mu_\xi(K \setminus \iota X)\, d\mu = 0$, we may redefine $\mu_\xi = \mu$ on an $\mathcal{I}'$-measurable null set and assume that every $\mu_\xi$ is carried by $\iota X$. Two such kernels agree, for almost every $\xi$, on the countable family $\mathcal{A}(D_0)$, and hence as measures. The last identity holds because both sides are $\mathcal{I}'$-measurable and have equal integrals against every bounded $\mathcal{I}'$-measurable $W$. □

For a translation-invariant law $m$ and a box $B_R = [-R, R]^3$ put $M_R\Phi = |B_R|^{-1} \int_{B_R} \alpha_a \Phi \, da$, defined pointwise for bounded $\Phi$ by Lemma A.1(c).

**Lemma A.3.** *Let $m$ be translation invariant and $P_m$ the orthogonal projection of $L^2(m)$ onto $\{\Phi : \alpha_a\Phi = \Phi \ m\text{-a.e. for all } a\}$. Then $M_R\Phi \to P_m\Phi$ in $L^2(m)$ as $R \to \infty$. Every $\Phi$ with $\alpha_a\Phi = \Phi$ a.e. for all $a$ agrees a.e. with an $\mathcal{I}$-measurable function, and $P_m\Phi = m(\Phi \mid \mathcal{I})$. The law $m$ is ergodic if and only if $P_m\Phi = m(\Phi)$ for all $\Phi \in L^2(m)$.*

*Proof.* On invariant vectors $M_R$ acts as the identity. For $\Phi = \Psi - \alpha_b\Psi$ one has $\|M_R\Phi\|_2 \le |B_R \triangle (B_R + b)||B_R|^{-1}\|\Psi\|_2 \to 0$, and the closed span of such vectors is the orthogonal complement of the invariant vectors. Since $\|M_R\| \le 1$, $M_R \to P_m$ strongly. If $W$ is bounded and invariant a.e., then $\tilde W = \limsup_{R \in \mathbb{N}} M_R W$ is strictly invariant, because $|M_R W(\tau_b F) - M_R W(F)| \le 2\|W\|_\infty |B_R \triangle (B_R + b)|/|B_R|$. By Fubini, for almost every $F$ one has $W(\tau_{-a}F) = W(F)$ for almost every $a$, so $\tilde W = W$ a.e. Hence invariant vectors have $\mathcal{I}$-measurable versions, by truncation, and $\Phi - P_m\Phi \perp L^2(\mathcal{I}, m)$. The ergodicity criterion follows by considering indicators of sets in $\mathcal{I}$ and level sets of invariant functions. □

**Theorem A.4.** *Let $\mu$ be a translation-invariant law and $(\mu_\xi)$ the kernel of Lemma A.2 for $\mathcal{I}' = \mathcal{I}$. Then $\mu(W\Phi) = \int W(\xi)\mu_\xi(\Phi)\, \mu(d\xi)$ for bounded measurable $\Phi$ and bounded $\mathcal{I}$-measurable $W$. If $\mu(B) = 1$ then $\mu_\xi(B) = 1$ for almost every $\xi$. For every bounded $\mathcal{I}$-measurable $W$, $W = W(\xi)$ $\mu_\xi$-a.s. for almost every $\xi$. For $\mu$-almost every $\xi$ the law $\mu_\xi$ is translation invariant and ergodic.*

*Proof.* The first two assertions are immediate. By Lemma A.2, $\mu_\xi(|W - W(\xi)|^2) = \mu_\xi(W^2) - W(\xi)^2 = 0$ for almost every $\xi$. *Invariance.* For strictly invariant bounded $W$, $\mu(W\alpha_a\Phi) = \mu(\alpha_a(W\Phi)) = \mu(W\Phi)$, so $\mu_\xi(\alpha_a\Phi) = \mu_\xi(\Phi)$ for almost every $\xi$, for each $\Phi \in \mathcal{A}(D_0)$ and $a \in \mathbb{Q}^3$. By Lemma A.1(c) and dominated convergence this extends to all $a$, and by Lemma A.1(b) to all bounded $\Phi$. *Ergodicity.* Enumerate $\mathcal{A}(D_0) = \{\Phi_l\}$ and put $w_l(\xi) = \mu_\xi(\Phi_l)$, a version of $P_\mu\Phi_l$. By Lemma A.3 choose $R_k \uparrow \infty$ with $\|M_{R_k}\Phi_l - w_l\|^2_{L^2(\mu)} \le 2^{-k}$ for $l \le k$. Since $\int \|\cdot\|^2_{L^2(\mu_\xi)} \mu(d\xi) = \|\cdot\|^2_{L^2(\mu)}$, for almost every $\xi$ and every $l$ we get $M_{R_k}\Phi_l \to w_l$ in $L^2(\mu_\xi)$, and $w_l = w_l(\xi)$ $\mu_\xi$-a.s. By Lemma A.3 for $m = \mu_\xi$, $P_{\mu_\xi}\Phi_l = \mu_\xi(\Phi_l)$ for all $l$. By density and boundedness of $P_{\mu_\xi}$, $P_{\mu_\xi}\Phi = \mu_\xi(\Phi)$ for all $\Phi$, so $\mu_\xi$ is ergodic. □

**Lemma A.5.** *Let $\mu$ be translation invariant and reflection positive at $\pi = \pi_{n,c}$, and let $W$ be bounded with $\alpha_a W = W$ $\mu$-a.e. for all $a$. Then $W$ agrees a.e. with a $\Sigma(\pi^+)$-measurable function, and $W \circ r_\pi = W$ a.e.*

*Proof.* Given $\delta > 0$ choose $\Phi$ which is $\Sigma(K)$-measurable for a bounded open $K$, with $\|W - \Phi\|_2 < \delta$; this is possible by Lemma A.1(b), since $\bigcup_K \Sigma(K)$ generates $\mathcal{B}$. Then $\|W - \alpha_{tn}\Phi\|_2 = \|\alpha_{tn}(W - \Phi)\|_2 < \delta$, and $\alpha_{tn}\Phi$ is $\Sigma(\pi^+)$-measurable for large $t$. The space $L^2(\Sigma(\pi^+), \mu)$ is closed, so it contains $W$. The function $W \circ r_\pi$ is again invariant, since $r_\pi$ normalises the translations and preserves $\mu$, so it too is $\Sigma(\pi^+)$-measurable a.e. Hence $Z = W - W \circ r_\pi$ agrees a.e. with a bounded $\Sigma(\pi^+)$-measurable function, and $Z \circ r_\pi = -Z$. Reflection positivity gives $0 \le \mu(\overline{Z \circ r_\pi}\, Z) = -\mu(|Z|^2)$, so $Z = 0$. □

**Lemma A.6.** *Let $\mu$ be translation invariant and $g \in E(3)$ with $g_*\mu = \mu$. Then $g_*\mu_\xi = \mu_{g\xi}$ for almost every $\xi$. Moreover $g_*\mu_\xi = \mu_\xi$ for almost every $\xi$ if and only if $W \circ g = W$ a.e. for every bounded $\mathcal{I}$-measurable $W$.*

*Proof.* The map $g$ preserves $\mathcal{I}$, because $g\tau_a g^{-1}$ is a translation. The kernel $\xi \mapsto g_*\mu_{g^{-1}\xi}$ is $\mathcal{I}$-measurable and satisfies $\int W(\xi)\mu_{g^{-1}\xi}(g^{-1}B)\,\mu(d\xi) = \mu((W\circ g)\mathbf{1}_{g^{-1}B}) = \mu(W\mathbf{1}_B)$, so uniqueness in Lemma A.2 applies. If $g_*\mu_\xi = \mu_\xi$ a.e., then $\mu_\xi(|W\circ g - W(\xi)|^2) = \mu_\xi(|W - W(\xi)|^2) = 0$ for almost every $\xi$, by Theorem A.4, so $W \circ g = W$ a.e. Conversely, apply the hypothesis to $W = \mu_\cdot(\Phi)$, $\Phi \in \mathcal{A}(D_0)$: this gives $\mu_{g\xi}(\Phi) = \mu_\xi(\Phi)$ for almost every $\xi$. By the first part, $g_*\mu_\xi(\Phi) = \mu_\xi(\Phi)$ for almost every $\xi$ and all $\Phi$ in the countable family $\mathcal{A}(D_0)$, which determines laws. □

**Theorem A.7.** *Let $N = -N \subset S^2$ be closed, and let $\mu$ be translation invariant and reflection positive at every plane of $\mathcal{P}(N)$. Let $G_N$ be the closed subgroup of $E(3)$ generated by the translations and the reflections $r_{n,0}$, $n \in N$. Then for $\mu$-almost every $\xi$ the law $\mu_\xi$ is translation invariant, ergodic, $G_N$-invariant and reflection positive at every plane of $\mathcal{P}(N)$. For the eighteen unit vectors $\pm e_i$, $\pm(e_i \pm e_j)/\sqrt{2}$, $G_N = \mathbb{R}^3 \rtimes O_h$; for $N = S^2$, $G_N = E(3)$.*

*Proof.* Fix a countable dense $N_0 \subset N$. By Lemmas A.5 and A.6, almost every $\mu_\xi$ is $r_{n,0}$-invariant for every $n \in N_0$. For bounded invariant $W \geq 0$ and bounded $\Sigma(\pi^+)$-measurable $\Phi$, with $\pi = \pi_{n,0}$, $n \in N_0$, and $r = r_{n,0}$, Lemma A.5 applied to $\sqrt{W}$ gives

$$\mu\big(W\,\overline{\Phi\circ r}\,\Phi\big) = \mu\big(\overline{(\sqrt{W}\Phi)\circ r}\,\sqrt{W}\Phi\big) \geq 0.$$

Hence $\mu_\xi(\overline{\Phi\circ r}\Phi) \geq 0$ for almost every $\xi$, for each $\Phi$ in the countable algebra $\mathcal{A}(D_{\pi^+})$. Since $\Phi \mapsto m(\overline{\Phi\circ r}\Phi)$ is continuous on $L^2(m)$ for $r$-invariant $m$, and $\mathcal{A}(D_{\pi^+})$ is dense in $L^2(\Sigma(\pi^+), m)$, almost every $\mu_\xi$ is reflection positive at the planes $\pi_{n,0}$, $n \in N_0$.

The stabiliser $\{g \in E(3) : g_*\mu_\xi = \mu_\xi\}$ is a closed subgroup, because $g \mapsto \mu_\xi(\alpha_g\Phi)$ is continuous for $\Phi \in \mathcal{A}(D_0)$, by Lemma A.1(c) and dominated convergence, and $\mathcal{A}(D_0)$ determines laws. It contains the translations and the $r_{n,0}$ with $n \in N_0$, hence with $n \in N$, hence $G_N$. For $n \in N\setminus N_0$ take $n_k \to n$ in $N_0$, and let $\Phi = \phi(F(f_1),\dots,F(f_j))$ with $\phi \in C_b$ and $\operatorname{supp} f_i \subset \pi^+_{n,0}$ compact. Then $\operatorname{supp} f_i \subset \pi^+_{n_k,0}$ for large $k$, and $\Phi\circ r_{n_k,0} \to \Phi\circ r_{n,0}$ pointwise, so dominated convergence and density give positivity at $\pi_{n,0}$. Positivity at the translated planes follows from $r_{n,c} = \tau_{cn} r_{n,0} \tau_{-cn}$ and translation invariance.

The nine reflections $r_{n,0}$, $n \in \mathcal{N}$, are the sign changes of one coordinate and the maps $(x_i, x_j) \mapsto (x_j, x_i)$, $(x_i, x_j) \mapsto (-x_j, -x_i)$. They generate the group of signed permutation matrices, which is $O_h$. Every element of $O(3)$ is a product of at most three reflections, so the reflections $r_{n,0}$, $n \in S^2$, generate $O(3)$. □

*Remark* A.8. Invariance of a translation-invariant law under a connected group does not imply invariance of its ergodic components. Let $R$ be Haar distributed in $SO(3)$ and $\vartheta$ uniformly distributed in $[0, 2\pi)$, independent, and let $F_{R,\vartheta}(x) = (Re_1 \wedge Re_2)\cos(Re_1 \cdot x + \vartheta)$, a smooth closed two-form. Its law $\mu$ is invariant under translations and under $SO(3)$, since $g\cdot F_{R,\vartheta} = F_{gR,\vartheta}$ for rotations $g$. For fixed $R$, the law $\mu_R$ of $F_{R,\vartheta}$ is translation invariant and ergodic, because translations act transitively on $\vartheta$, and $\mu = \int \mu_R\, dR$. The kernel $\xi \mapsto \mu_{R(\xi)}$, where $\xi = F_{R(\xi),\vartheta}$, is a version of the kernel of Theorem A.4. Indeed, $\mu_R$ depends on $R$ only through its class modulo right multiplication by the four rotations $\operatorname{diag}(s_1, s_2, s_1 s_2)$, $s_i = \pm 1$, and a plane wave $F_{R,\vartheta}$ determines this class, since it determines its wave vector $Re_1$ and its polarisation $Re_1 \wedge Re_2$ up to sign. The set $P$ of these plane waves is compact in $H^{-s}$ for large $s$, as a continuous image of $SO(3) \times [0, 2\pi]$, and the class is a continuous function on $P$. Hence, for $B \in \mathcal{B}$, the function $\xi \mapsto \mu_{R(\xi)}(B)$, extended by $\mu(B)$ off $P$, is Borel and constant on the orbits of the translations,

hence $\mathcal{I}$-measurable. A bounded $\mathcal{I}$-measurable $W$ satisfies $W(F_{R,\vartheta}) = W(F_{R,0})$ for all $\vartheta$, so $\int W(\xi)\mu_{R(\xi)}(B)\,\mu(d\xi) = \int W(F_{R,0})\,\mu_R(B)\,dR = \mu(W\mathbf{1}_B)$, and the uniqueness in Lemma A.2 applies. No $\mu_R$ is $SO(3)$-invariant: $\mu_R$ is carried by plane waves with wave vector $\pm Re_1$. In Theorem A.7, invariance of the components comes from the reflections, through Lemma A.5.

## Appendix B. Gaussian-dominated and reflection-positive laws

This appendix collects the facts about laws on $X$ used in Sections 5–7.

**Lemma B.1.** *Let $m$ be a law satisfying (G), and for real $f \in \mathcal{T}$ write $Y = F(f)$ and $s = \|f\|_2^2$.*

(a) *$m(Y) = 0$, $m(Y^2) \le s$, $m(e^{t|Y|}) \le 2e^{t^2 s/2}$, and $m(|Y|^p) \le 2\cdot 2^{p/2}\Gamma(\frac{p}{2}+1)s^{p/2} \le 2\cdot 2^{p/2}(p!)^{1/2}s^{p/2}$ for $p \in \mathbb{N}$.*

(b) *The map $f \mapsto F(f)$ extends uniquely to a linear contraction from $L^2(\mathbb{R}^3;\Lambda^2)$ (real) to $L^2(m)$. The bounds of (a), and (G), hold for $f \in L^2$. If $f = 0$ a.e. outside an open set $U$, then $F(f)$ agrees a.e. with a $\Sigma(U)$-measurable function.*

(c) *$|m(F(f_1)\cdots F(f_p))| \le 2\cdot 2^{p/2}(p!)^{1/2}\prod_i \|f_i\|_2$ for real $f_i \in L^2$.*

(d) *If $g \in E(3)$ and $g_*m = m$, then $\alpha_g F(f) = F(g\cdot f)$ in $L^2(m)$ for $f \in L^2$.*

(e) *Two laws satisfying (G) whose moments $m(F(f_1)\cdots F(f_p))$ coincide for all $p$ and $f_i \in \mathcal{T}$ are equal.*

*Proof.* (a) Jensen's inequality gives $e^{tm(Y)} \le e^{t^2 s/2}$ for all $t$, so $m(Y) = 0$. From $\cosh(tY) - 1 \ge \frac{1}{2}t^2Y^2$ we get $m(Y^2) \le 2(e^{t^2s/2}-1)/t^2 \to s$. Next $m(e^{t|Y|}) \le m(e^{tY}) + m(e^{-tY})$, and Chernoff's bound gives $m(|Y| > u) \le 2e^{-u^2/2s}$. Hence $m(|Y|^p) = \int_0^\infty pu^{p-1}m(|Y| > u)\,du \le 2p\int_0^\infty u^{p-1}e^{-u^2/2s}du = 2\cdot 2^{p/2}\Gamma(\frac{p}{2}+1)s^{p/2}$. Log-convexity of $\Gamma$ gives $\Gamma(\frac{p}{2}+1)^2 \le \Gamma(1)\Gamma(p+1) = p!$. (b) By (a), $f \mapsto F(f)$ is a linear contraction from $\mathcal{T}$ with the $L^2$-norm to $L^2(m)$; extend by density. For $f_l \to f$ in $L^2$ with $f_l \in \mathcal{T}$, $F(f_l) \to F(f)$ almost surely along a subsequence, and Fatou's lemma preserves (G) and the bounds. Test forms supported in $U$ are dense in $L^2(U)$, and $L^2(\Sigma(U), m)$ is closed. (c) follows from (a), (b) and Hölder's inequality. (d) holds for $f \in \mathcal{T}$, and both sides are continuous in $f \in L^2$, since $\alpha_g$ is unitary on $L^2(m)$ and $\|g\cdot f\|_2 = \|f\|_2$. (e) For real $f \in \mathcal{T}$ the series $\sum_p \frac{i^p}{p!}m(F(f)^p)$ converges to $m(e^{iF(f)})$, by dominated convergence with the bound $e^{|F(f)|} \le e^{F(f)} + e^{-F(f)}$. So the moments determine the characteristic functional, hence the finite-dimensional distributions of $(F(f_1),\dots,F(f_k))$, hence the law. □

**Lemma B.2.** *Let $m$ be a law. For every open $U$, the bounded continuous cylinder functions $\phi(F(f_1),\dots,F(f_k))$ with $f_i \in \mathcal{T}$ of compact support in $U$ are dense in $L^2(\Sigma(U), m)$. The functions which are bounded and $\Sigma(B)$-measurable for some bounded open $B$ are dense in $L^2(X, \mathcal{B}, m)$. If $m$ is translation invariant, $a \mapsto \alpha_a$ is a strongly continuous unitary representation on $L^2(m)$.*

*Proof.* The first assertion follows from Lemma A.1(b), since $\mathcal{A}(D_U)$ consists of such functions. Every test form in $D_0$ is a limit in $\mathcal{T}$ of test forms of compact support, so $\bigcup_B \Sigma(B)$ generates $\mathcal{B}$, and the second assertion follows in the same way. For the third, $\alpha_a\Phi \to \Phi$ pointwise and boundedly as $a \to 0$ for bounded continuous cylinder functions $\Phi$, by Lemma A.1(c); these are dense, and the $\alpha_a$ are unitary. □

**Proposition B.3.** *Let $n$ be a unit vector, $r$ the reflection in $\pi = \{n\cdot x = 0\}$, and $m$ a law which is translation invariant, $r$-invariant and reflection positive at $\pi$. Let $L^2_+ = L^2(\Sigma(\pi^+), m)$ and $b(\Phi,\Psi) = m(\overline{\Phi\circ r}\,\Psi)$ for $\Phi, \Psi \in L^2_+$.*

(a) *$b$ is a positive semidefinite Hermitian form with $|b(\Phi,\Psi)| \le \|\Phi\|_2\|\Psi\|_2$. Let $\mathcal{H}_n$ be the completion of $L^2_+$ modulo the null space of $b$, $[\cdot] : L^2_+ \to \mathcal{H}_n$ the class map, and $\psi_0 = [1]$. Then $[\cdot]$ is a contraction with dense range, $\|\psi_0\| = 1$, and $\langle[\Phi],[\Psi]\rangle = b(\Phi,\Psi)$.*

(b) $T(t)[\Phi] := [\alpha_{tn}\Phi]$, $t \geq 0$, *defines a strongly continuous semigroup of self-adjoint contractions, so* $T(t) = e^{-tH_n}$ *with a self-adjoint* $H_n \geq 0$, *and* $H_n\psi_0 = 0$.
(c) *For* $y \in n^\perp$, $V(y)[\Phi] := [\alpha_y\Phi]$ *defines a strongly continuous unitary representation of* $n^\perp$ *which commutes with* $T$ *and fixes* $\psi_0$.
(d) *Choosing an orthonormal basis of* $n^\perp$, *there is a projection-valued measure* $\mathsf{E}_n$ *on* $[0,\infty)\times\mathbb{R}^2$ *with* $T(t)V(y) = \int e^{-t\omega+ip\cdot y}\,\mathsf{E}_n(d\omega,dp)$, *and* $b(\Phi,\alpha_{tn+y}\Psi) = \langle[\Phi],T(t)V(y)[\Psi]\rangle$.

*Proof.* (a) $\overline{b(\Psi,\Phi)} = m((\Psi\circ r)\overline{\Phi}) = m(\Psi\,\overline{\Phi\circ r}) = b(\Phi,\Psi)$ by $r$-invariance. Positivity is reflection positivity, extended from bounded functions to $L^2_+$ by continuity, and the bound is Cauchy–Schwarz. The remaining assertions follow from the construction and $b(1,1) = 1$. (b) For $t \geq 0$, $\alpha_{tn}$ maps $L^2_+$ into itself. Since $r\tau_{tn}r = \tau_{-tn}$ and $m$ is invariant, $b(\alpha_{tn}\Phi,\Psi) = b(\Phi,\alpha_{tn}\Psi)$, and $b(\alpha_{tn}\Phi,\alpha_{tn}\Phi) = b(\Phi,\alpha_{2tn}\Phi)$. Iterating the Cauchy–Schwarz inequality for $b$ gives

$$b(\alpha_{tn}\Phi,\alpha_{tn}\Phi) \leq b(\Phi,\Phi)^{1-2^{-k}}\,b(\alpha_{2^ktn}\Phi,\alpha_{2^ktn}\Phi)^{2^{-k}} \leq b(\Phi,\Phi)^{1-2^{-k}}\|\Phi\|_2^{2^{1-k}} \to b(\Phi,\Phi).$$

Hence $T(t)$ is well defined on classes, is a contraction and is symmetric. The semigroup law is clear. Strong continuity follows from $\|T(t)[\Phi]-[\Phi]\| \leq \|\alpha_{tn}\Phi-\Phi\|_2$ and Lemma B.2. A strongly continuous semigroup of self-adjoint contractions has the form $e^{-tH_n}$ with $H_n \geq 0$ self-adjoint [GJ, Thm. 6.1.3, pp. 94–95], and $T(t)\psi_0 = \psi_0$. (c) For $y \perp n$, $r\tau_y r = \tau_y$, so $b(\alpha_y\Phi,\alpha_y\Psi) = b(\Phi,\Psi)$; $\alpha_y$ maps $L^2_+$ onto itself with inverse $\alpha_{-y}$. Strong continuity follows as in (b). (d) By Stone's theorem for $\mathbb{R}^2$, $V(y) = \int e^{ip\cdot y}\mathsf{F}(dp)$ for a projection-valued measure $\mathsf{F}$ commuting with the spectral measure of $H_n$, since $V$ commutes with every $e^{-tH_n}$. Their product measure is $\mathsf{E}_n$. The last formula is the definition of $T$ and $V$. □

**Proposition B.4.** *In the situation of Proposition B.3, suppose in addition that* $m$ *is ergodic.*
(a) *The vectors fixed by all* $T(t)$ *and all* $V(y)$ *form the space* $\mathbb{C}\psi_0$.
(b) *If* $\mathsf{E}_n\{(\omega,p) : \omega < |p_i|\} = 0$ *for* $i = 1,2$, *then* $\operatorname{Ker} H_n = \mathbb{C}\psi_0$.

*Proof.* (a) Let $Q$ be the orthogonal projection onto the joint fixed space. The averages $\frac{1}{R}\int_0^R T(s)\,ds$ converge strongly to $\mathbf{1}_{\{0\}}(H_n)$, by the spectral theorem. The averages of $V(y)$ over $y \in [-R,R]^2$ converge strongly to the projection onto the fixed vectors of $V$, by the mean ergodic theorem. The two families commute, so the product $M_R$ of the averages converges strongly to $Q$. The class map is bounded and commutes with Bochner integrals, so $M_R[\Phi] = [M'_R\Phi]$, where $M'_R\Phi$ is the average of $\alpha_a\Phi$ over the box $B'_R = \{sn+y : 0 \leq s \leq R,\ y \in [-R,R]^2\}$. These boxes satisfy $|B'_R\triangle(B'_R+b)|/|B'_R| \to 0$, so the proof of Lemma A.3 applies to them. With ergodicity it gives $M'_R\Phi \to m(\Phi)$ in $L^2(m)$, hence $Q[\Phi] = m(\Phi)\psi_0 = \langle\psi_0,[\Phi]\rangle\psi_0$ for $\Phi \in L^2_+$. Since the classes are dense, $Q$ is the projection onto $\mathbb{C}\psi_0$. (b) If $H_n\eta = 0$, the spectral measure of $\eta$ is carried by $\{\omega = 0\}$, hence by the hypothesis by $\{(0,0)\}$. So $V(y)\eta = \eta$ for all $y$, and $\eta \in \mathbb{C}\psi_0$ by (a). □

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